AMENDMENT Reference No AMD-745828-5

Grant Agreement number: 745828 — Chemical building blocks from versatile MSW biorefinery (PERCAL)

The parties agree to amend the Grant Agreement as follows ('**Amendment**'):

1 . Removal of a beneficiary whose participation was terminated

The participation of the following beneficiary is terminated:

- SO.F.TER. SRL (SOFTER) - on the day after the notification of termination This

implies the **following changes** to the Grant Agreement:

- The end date of the participation is added, for the beneficiary, in the **Preamble**:

  SO.F.TER. SRL (SOFTER) SRL, 233925, established in VIA MASTRO GIORGIO 1, FORLI 47122, Italy, IT02019080403 - until termination date

In accordance with Article 50 of the Grant Agreement, the beneficiary's obligations continue to apply after termination.

If the estimated budget in Annex 2 is changed, the maximum amount referred to in Articles 44.1.1(b) (and in the declaration on joint and several liability (Annex 3a)) is that of the estimated budget before this amendment.

2 . Addition of a new beneficiary

The following new beneficiary is added:

- YPAREX BV (YPR) — as from 1 March 2018

This implies the **following changes** to the Grant Agreement:

- The new beneficiary and the 'accession date' is added to the **Preamble**:

  *"**YPAREX BV (YPR)** BV, 14049543 , established in GOOLGATENWEG 10, ENSCHEDE 7521BG , Netherlands, NL850232430B01 — as from 1 March 2018"*

3. Change of Annex 1 (description of the action)

Annex 1 is changed and replaced by the Annex 1 attached to this Amendment.

4 . Changes of Annex 2 (estimated budget of the action)

Annex 2 is changed and replaced by the Annex 2 attached to this Amendment.

All other provisions of the Grant Agreement and its Annexes remain unchanged.

This Amendment **enters into force** on the day of the last signature.

This Amendment **takes effect** on the date on which the amendment enters into force, except where a different date has been agreed by the parties (for one or more changes).

Please inform the other members of the consortium of the Amendment.

SIGNATURES

For the coordinator

For the JU

Enclosures:

Annex 2
Annex 1

ANNEX 1 (part A)

Bio-based Industries Research and Innovation action

NUMBER — 745828 — PERCAL

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1.1. The project summary

Project Number [1]	745828	Project Acronym [2]	PERCAL

One form per project

General information

Project title [3]	Chemical building blocks from versatile MSW biorefinery
Starting date [4]	01/07/2017
Duration in months [5]	36
Call (part) identifier [6]	H2020-BBI-JTI-2016
Topic	BBI-2016-R08 Emerging technologies for conversion of the organic content of Municipal Solid Waste and improving waste-to-chemicals value chains
Fixed EC Keywords	
Free keywords	Bioethanol, Municipal Solid Waste, Succinic acid, Lactic acid, surfactants, adhesives, solvents, polyester polyols, fermentation, membrane electrolysis, enzymatic hydrolysis

Abstract [7]

PERCAL will exploit Municipal Solid Waste (MSW) as feedstock to develop intermediate chemical products at high yield and low impurity level with huge industrial interest. These will be complementary to the bioethanol (current PERSEO technology), to achieve a cascade valorisation of the MSW components, i.e.:

- Lactic acid (LA) to produce: 1) Eco-friendly ethyl lactate solvents by reactive distillation from lactic acid &bio-ethanol to be used in cleaning products and inks and 2) hot-melt adhesives for cardboard and other non-food applications in combination with maleic anhydride by reactive extrusion.
- Succinic acid (SA) as an intermediate building blocks to production of polyols for the polyurethane industry.
- Biosurfactants by chemical and/or microbiological modification of protein and lipid fraction from remaining fractionof MSW fermentation.

IIn order to minimize the MSW heterogeneous composition (due to seasonal and origin variability driven by local economic, social and climate conditions) limitations as a fermentation feedstock, four main innovations will be proposed: i) New enzymatic cocktails to maximize hydrolysis of fermentable organic matter with low inhibitors production, ii) the use of high yield, specific and robust strains for each selected acid, iii) the extraction of fermentation by-products acting as inhibitors to succinic acid production via novel membrane electrolysis employing an integrated continuous fermentation coupled with simultaneous organic acid removal in comparison with SA sequential fermentation followed by its separation using an electrodialysis-based and iv) optimize simultaneous saccharification and fermentation for lactic acid production followed by a downstream separation process based on membrane electrodialysis. To maximize the yield and purity of target organic acids, continuous and single step fermentation process will be prioritized in order to allow their integration in the PERSEO plant.

1.2. List of Beneficiaries

Project Number [1]	745828	Project Acronym [2]	PERCAL

List of Beneficiaries

No	Name	Short name	Country	Project entry date[8]	Project exit date
1	INDUSTRIAS MECANICAS ALCUDIA SA	IMECAL	Spain		
2	AIMPLAS - ASOCIACION DE INVESTIGACION DE MATERIALES PLASTICOS Y CONEXAS	AIMPLAS	Spain		
3	LEIBNIZ-INSTITUT FUER AGRARTECHNIK UND BIOOEKONOMIE EV	ATB	Germany		
4	AGRICULTURAL UNIVERSITY OF ATHENS	AUA	Greece		
5	UNIVERSITEIT GENT	UGent	Belgium		
6	FUNDACION CENER-CIEMAT	CENER	Spain		
7	TBW RESEARCH GESMBH	TBWR	Austria		
8	VISUM LIMITED	VISUM	Ireland		
9	SO.F.TER. SRL	SOFTER	Italy		the day after the notification of termination
10	HAYAT KIMYA SANAYI ANONIM SIRKETI	HAYAT	Turkey		
11	COVESTRO DEUTSCHLAND AG	COVESTRO	Germany		
12	EXERGY LTD	EXERGY	United Kingdom		
13	YPAREX BV	YPR	Netherlands	01/03/2018	

1.3. Workplan Tables - Detailed implementation

1.3.1. WT1 List of work packages

WP Number[9]	WP Title	Lead beneficiary[10]	Personmonths[11]	Start month[12]	End month[13]
WP1	Characterization, handling and preprocessing of MSW from different localizations and seasons. Final products requirements definition	1 - IMECAL		1	18

WP2	Microbial conversions of MSW to lactic acid.	3 - ATB		2	32
WP3	Microbial conversions of MSW to succinic acid.	5 - UGent		3	34
WP4	Valorisation of fermentation byproducts	6 - CENER		8	34
WP5	Green chemical routes to produce solvents, adhesives & polymers from developed acids.	11 - COVESTRO		10	34
WP6	Study of adaptation and monitoring of PERSEO plant for the production of ethanol, LA & SA with suitable quality to produce the selected final products.	8 - VISUM		12	36
WP7	Environmental, regulatory & economic analyses.	4 - AUA		1	36
WP8	Communication, Dissemination, Exploitation, Business plan and Technology Transfer	12 - EXERGY		1	36
WP9	Project Management	1 - IMECAL		1	36
WP10	Ethics requirements	1 - IMECAL		1	36

1.3.2. WT2 list of deliverables

Deliverable Number[14]	**Deliverable Title**	**WP number[9]**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D1.1	Selected final products and intermediates requirements.	WP1	10 - HAYAT	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D1.2	Variability of biogenic fraction of MSW.	WP1	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	8
D1.3	Report on hydrolysis improvements.	WP1	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission	16

					Services)	
D1.4	Report on selected feedstocks.	WP1	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	14
D1.5	Report on validation of fermentation processes	WP1	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D2.1	Lactic acid fermentation process	WP2	3 - ATB	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D2.2	Down-stream processing	WP2	7 - TBWR	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D2.3	Scaling of the full process.	WP2	3 - ATB	Report	Confidential, only for members of the consortium (including the Commission Services)	32

Deliverable Number[14]	**Deliverable Title**	**WP number[9]**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D3.1	Selection of optimal bacterial strain using different MSW hydrolysates.	WP3	4 - AUA	Report	Confidential, only for members of the consortium (including the Commission Services)	18

D3.2	Report on the integrated SA production with electrolysis membrane extraction.	WP3	5 - UGent	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D3.3	Report on production and purification of SA using electrolysis or electrodialysis based processes	WP3	4 - AUA	Report	Confidential, only for members of the consortium (including the Commission Services)	28
D3.4	Report on process scale-up	WP3	5 - UGent	Report	Public	32
D4.1	Characterization of the fermentation residue.	WP4	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	20
D4.2	Pre-treatments for revalorization of the protein fraction.	WP4	6 - CENER	Report	Confidential, only for members of the consortium (including the Commission Services)	26
D4.3	Pre-treatments for revalorization of the lipid fraction.	WP4	6 - CENER	Report	Confidential, only for members of the consortium (including the Commission Services)	26
D4.4	Biosurfactant production from lipids and proteins.	WP4	10 - HAYAT	Report	Confidential, only for members of the consortium (including the Commission Services)	34
D5.1	Production of Ethyl lactate at laboratory level.	WP5	7 - TBWR	Report	Confidential, only for members of the consortium (including the	24

Deliverable Number[14]	**Deliverable Title**	**WP number[9]**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**

					Commission Services)	
D5.2	Ethyl-Lactate scale up and industrial validation.	WP5	10 - HAYAT	Report	Public	34
D5.3	Production of HMA at laboratory level.	WP5	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D5.4	HMA scale up and industrial validation	WP5	13 - YPR	Report	Public	34
D5.5	Production of polyols at laboratory level.	WP5	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D5.6	Polyols scale up and industrial validation	WP5	11 - COVESTRO	Report	Public	34
D6.1	Adaptation of PERSEO plant for the versatile production of chemical intermediates & identification of key parameters	WP6	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D6.2	Monitoring technologies and testing results.	WP6	8 - VISUM	Report	Confidential, only for members of the consortium (including the Commission Services)	32
D6.3	Integration of the production in a biotechnological plant.	WP6	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D7.1	Draft report on economic analysis	WP7	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D7.2	Simulation of the system and M&E balances	WP7	12 - EXERGY	Report	Confidential, only for members of the	36

					consortium (including the	

Deliverable Number[14]	Deliverable Title	WP number[9]	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
					Commission Services)	
D7.3	Report on Life Cycle Assessment (LCA).	WP7	4 - AUA	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D7.4	Report on Social Life Cycle Assessment (SLCA).	WP7	12 - EXERGY	Report	Public	36
D7.5	Report on regulatory analysis and safety aspects	WP7	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D7.6	Final report on economic analysis	WP7	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D8.1	Project web site	WP8	2 - AIMPLAS	Report	Public	3
D8.2	Initial Communication and dissemination Plan	WP8	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D8.3	Draft communication activities & dissemination Plan updates	WP8	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	18

D8.4	IPR policy implementation in line with Exploitation Plan	WP8	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D8.5	Report of the IPR management activities	WP8	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	36

Deliverable Number[14]	**Deliverable Title**	**WP number[9]**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D8.6	Final Exploitation Plan including Business Plans of key results	WP8	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D8.7	Technology Transfer Activities, including training to SMEs, other activities and workshops from M1 to M18	WP8	12 - EXERGY	Report	Public	18
D8.8	Guide to explore the conversion routes of MSW in high added value products.	WP8	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	34
D8.10	Technology Transfer Activities, including training to SMEs, other activities and workshops from M18 to M36	WP8	12 - EXERGY	Report	Public	36
D8.11	Final communication activities & dissemination Plan updates	WP8	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	36

D9.1	Risk indicators and Risk assessment	WP9	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	9
D10.1	NEC - Requirement No. 1	WP10	1 - IMECAL	Ethics	Confidential, only for members of the consortium (including the Commission Services)	1

Num.	Work Packages and Related Tasks	Partners Num.	Deliverables	Project Months																	
				2	**4**	**6**	**8**	**10**	**12**	**14**	**16**	**18**	**20**	**22**	**24**	**26**	**28**	**30**	**32**	**34**	**36**
WP1	**Final Requirements. MSW Feedstocks.**	**All (1)**	**D1.1-D1.5**				**H**		**H**			**H**									
T1.1	Final products & intermediates requirements.	1,2,9,(10), 11-12	D1.1																		
T1.2	Study of variability of organic MSW.	All(1)	D1.2																		
T1.3	Characterization of MSW samples.	1,3,4,(6)	D-1.4																		
T1.4	Pre-treatment and enzymatic hydrolysis improvement from PERSEO process	(1),6	D1.3																		
T1.5	Validation of fermentation processes.	1,3,4(6)	D1.5																		
WP2	**Lactic acid from MSW. Downstream.**	**1,(3),4,7**	**D2.1-D2.3**											**H**				**H**			
T2.1.	Screening & selection of bacterial strains.	(3)																			
T2.2	Fermentation processes into lactic acid.	(3),4	D2.1																		
T2.3	Downstream processing for recovery of LA	(3),7	D2.2																		
T2.4	Up-scaling of the entire process	1,(3)	D2.3																		
WP3	**Succinic acid from MSW. Downstream.**		**D3.1-D3.4**									**H**			**H**						
T3.1	Evaluation of MSW as feedstock.	(4),6	D3.1																		
T3.2	Optimisation of integrated bioprocess for SA production and in situ separation.	4,(5)	D3.2																		
T3.3	Succinic acid purification.	(4),5	D3.3																		
T3.4	Process scale-up.	4,(5)	D3.4																		
T3.5	Optimisation of bioprocess based on electrodialysis downstream separation system.	4,(5)																			

WP4	**Valorisation of fermentation by-products**		**D4.1-D4.4**															**H**		**H**	
T4.1	Characterization of the fermentation residue of the ethanol/LA and SA processes.	(1),6	D4.1																		
T4.2	Hydrolysis of protein fraction	1,(6)	D4.2																		
T4.3	Extraction & hydrolysis of the lipid fraction	1,(6)	D4.3																		
T4.4	Synthesis of the amino-lipid biosurfactants.	1,(6)																			
T4.5	Biosurfactants in relevant ind. applications.	(10)	D4.4																		
WP5	**Green chemical routes to produce HMA, solvent & polyols & Industrial Validation.**	**2,7,9-11**	**D5.1-D5.6**																	**H_3**	

1.3.3. WT3 Work package descriptions

Work package number [9]	WP1	**Lead beneficiary** [10]	1 - IMECAL
Work package title	Characterization, handling and pre-processing of MSW from different localizations and seasons. Final products requirements definition		
Start month	1	**End month**	18

Objectives

- Definition of intermediates requirements and related standard to effectively produce the targeted final products.
- Characterization of representative organic MSW samples to be used as substrates in the project.
- Improvement of enzymes cocktails to maximize organic hydrolysis for MSW compositions. Target to produce up to85% of potential fermentable sugars contained in the organic MSW.

Description of work and role of partners

WP1 - Characterization, handling and pre-processing of MSW from different localizations and seasons. Final products requirements definition [Months: 1-18]
IMECAL, AIMPLAS, ATB, AUA, UGent, CENER, TBWR, VISUM, HAYAT, COVESTRO, EXERGY
Task 1.1. Final products and intermediates requirements Task Leader: HAYAT; Other partners: YPR, COVESTRO, IMECAL, EXERGY and AIMPLAS.
Based on a review of the current petrol based or renewable selected products and intermediates properties, environmental requirements and especially economic limitations, a list of specific requirements will be defined for each one. In a further evaluation step these requirements will be summarised to specific requirement-lists that serve as a basis for future development. The aim here is to group requirements so that more general target areas can be defined that allow the use of material developments in more than one application. The report will also include the specific standards required for each case-study selected.

Task 1.2. Study of variability of organic MSW. Task Leader: IMECAL Other partners: all.
The main challenge to use the organic fraction of MSW as feedstock for bio-based industries lies in its high complexity and variability in composition. This heterogeneous composition depends on three main parameters:
- The waste collection system. The organic fraction of MSW contains less amount of inserts in function of the MSWcollection method employed (door-to-door, municipal bring points…)
- Geographical location. Differences in climate conditions and nutrition culture of EU-countries affects the specificcontents of food and therefore its chemical composition. Bio-waste coming from rural or urban areas also exhibits different compositional characteristics.
- Seasonality. Climate variations during the year have also shown an important impact in the food content and bio-waste composition.

In this task inputs regarding MSW composition variation from all the partners will be shared and bibliographic data will be analysed in order to better know the bio-waste composition and variability across Europe, PERCAL has partners and support companies of 12 different countries, including Mediterranean, Atlantic, continental and warm and cold countries. This analysis will serve as a basis to select ten MSW samples that will be characterized in task 1.3.

Task 1.3. Characterization of MSW samples. Task Leader: IMECAL Other partners: ATB & AUA.
In this task the selected substrates of task 1.2 will be fully analysed following the NREL methodology for lignocellulosic materials in order to know their potential for the intermediates production.
- The glucan and hemicellulose content, which are the polysaccharides that contain the fermentable sugars, will bedetermined by acid hydrolysis and subsequent HPLC analysis.
- Humidity degree will be measured by dying in oven at 105ºC and the ash content by kiln calcination at 575ºC.
- Fats and waxes will be analysed by means of Soxhlet extraction with chloroform.
- Soluble extractives will be analysed by water incubation.
- Pectin and Lignin content will be determined by extraction with EDTA or with acetic acid and NaClO2.
- Protein content will be determined by specific staining and spectrophotometry against bovine serum albumin(bicinchoninic acid assay) as well as by Total Nitrogen analysis (Kjeldahl method).

This analysis will serve as a basis to select three representative MSW samples that will be used as feedstocks in the project.

Task 1.4. Pre-treatment and enzymatic hydrolysis improvement from PERSEO bioethanol® process. Task Leader: IMECAL

The selected samples will be pre-treated by means of thermo-chemical processes in order to liberate glucan and hemicellulose content for subsequent enzymatic hydrolysis. Both treatments will be based on state-of-the art processes performed currently at PERSEO semi-industrial plant. Current saccharification processes are able to liberate around 65% of all potential fermentable sugars present in the organic fraction of MSW. New enzymes cocktails supplied by NOVO will be tested with the aim of producing more than 85% of all the potential fermentable sugars present, while minimising the generation of inhibitors.

The dosage of the enzymatic cocktail will be calculated based on the present protein content. Five dosages will be evaluated for each cocktail, in duplicate. Fermentable sugars will be analysed at the beginning of the enzymatic hydrolysis (t=0), at 24 and 48 h.

In order to diminish the inhibitory effects of some compounds (weak acids, furfural, hydroxymethyl furfural) towards the fermentative microorganisms in the pre-treatment strategy might be modulated depending on the intrinsic resistance of microbial strains to these hydrolysates. The amount inhibitors produced will be determined by liquid chromatography.

Task 1.5. Validation of fermentation processes. Task Leader: IMECAL; Other partners: AUA & ATB.

The hydrolysates generated in Task 1.3 with the three representative samples will be validated in terms of toxicity and potential carbon sources for the ethanol, LA and SA production. Tolerance tests will be performed with the yeast responsible of ethanol fermentation, and the different bacteria employed for the production of LA and SA. The tolerance of the selected strains will be evaluated in 96-well growth experiment assays in ten different substrate concentrations. Microbial growth will be measured spectrophotometrically. Flask scale fermentation reactions will be performed taking into account the tolerance of the different strains. Production of the target molecules will be tested following the indications of the different partners involved in fermentation. Production of the target compounds, together with other metabolites, will be followed by HPLC (RID detector). Interaction with task 1.3 will be required if inhibition issues arise. Detoxification or feed strategies will be proposed if required to cope with tolerance or inhibition problems.

Participation per Partner

Partner number and short name	**WP1 effort**
1 - IMECAL	
2 - AIMPLAS	
3 - ATB	
4 - AUA	
5 - UGent	
6 - CENER	
7 - TBWR	
8 - VISUM	
10 - HAYAT	
11 - COVESTRO	
12 - EXERGY	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D1.1	Selected final products and intermediates requirements.	10 - HAYAT	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D1.2	Variability of biogenic fraction of MSW.	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	8
D1.3	Report on hydrolysis improvements.	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	16
D1.4	Report on selected feedstocks.	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	14
D1.5	Report on validation of fermentation processes	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	18

Description of deliverables

D1.1. Report of selected final products and intermediates requirements.
D1.2. Report on variability of biogenic fraction of MSW.
D1.3. Report on hydrolysis improvements.
D1.4. Report on Selected feedstocks.
D1.5. Report on validation of fermentation processes

D1.1 : Selected final products and intermediates requirements. [6]

Selected final products and intermediates requirements.

D1.2 : Variability of biogenic fraction of MSW. [8]

Variability of biogenic fraction of MSW.

D1.3 : Report on hydrolysis improvements. [16]

Report on hydrolysis improvements.

D1.4 : Report on selected feedstocks. [14]

Report on selected feedstocks.

D1.5 : Report on validation of fermentation processes [18]

Report on validation of fermentation processes

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS1	Composition of MSW available	1 - IMECAL	14	Comparison with reported values in literature for similar feedstocks.
MS2	Improved enzymatic hydrolysis.	1 - IMECAL	16	Comparison of hydrolysis yields
MS3	Samples of hydrolysed MSW available for the different subsequent tasks	1 - IMECAL	18	Sufficient amounts of different fractions for lab tests within WP2, WP3 and WP4

Work package number [9]	WP2	**Lead beneficiary** [10]	3 - ATB
Work package title	Microbial conversions of MSW to lactic acid.		
Start month	2	**End month**	32

Objectives

This WP2 aims at bioconversion of the hydrolysed carbohydrates from organic municipal solid waste to lactic acid (LA). The utilisation and upgrading of organic municipal solid waste streams and the purification of lactic acid as a monomer for green solvents will be addressed. Biotechnological tools for the development of non-food industrial products and services and/or biogenic sources as feedstock are applied.

Description of work and role of partners

WP2 - Microbial conversions of MSW to lactic acid. [Months: 2-32]
ATB, IMECAL, AUA, TBWR, HAYAT
Lactic acid (LA) from organic municipal solid waste streams (MSW) feedstocks will be produced by ATB at laboratory and pilot-scale using sequential saccharification and fermentation approaches. In another valorization path, suitable raw material will be applied for lactic acid production via simultaneous saccharification and fermentation approaches using newly isolated strains. These complementary processes will be evaluated in parallel and assessed regarding optimum LA production that is suitable for subsequent processing.

Task 2.1. Screening and selection of new/alternative bacterial strains. Task Leader: ATB.
Screening and selection of new/alternative bacterial strains with respect to high product yields, sufficient substrate uptake, tolerance against eventually remaining impurities/inhibitors and product quality. One of the first targets will be the selection of the most efficient and robust LA producing microorganism among the ones mentioned in the state of the art or beyond. ATB will employ several screening algorithms to identify microorganisms that satisfy the performance criteria (e.g. high specific growth rate, high productivity, maximum sugar consumption, optical purity, high process stability etc.) with regard to industrial application. We will screen the most promising microorganisms in first simple fermentations in tubes, bottles, shake flasks and lab-scale bioreactors.

Task 2.2. Fermentation processes for converting biogenic fraction MSW into lactic acid. Task Leader: ATB Other partners: AUA.
Develop efficient fermentation processes for converting bio waste into lactic acid (L(+) isomer). Mixtures and/or single fractions of the (C6/C5) sugar solution from pre-treated fractions will be converted to lactic acid in order to provide compounds for further processing steps.
Depending on the steps before (WP1/sub-tasks for pre-treatment and/or hydrolysis) and the available analytical information the different materials, hydrolysates, sugars, mixtures etc. have to be characterized in view of their main components and possible impurities. If necessary additional enzymatic treatment could be adapted for a better release of sugars and/or nutrients. Finally a fermentation broth has to be designed which represents an optimal mixture of all necessary carbohydrates, macro- and micro-nutrients for the bacteria.
ATB will develop an efficient LA fermentation process through the optimization and elucidation of various parameters. Optimization of fermentation conditions will focus on initial sugar concentration, addition of commercial (e.g. yeast extract) or crude nutrient supplements (e.g. oilseed meal hydrolysates, grass juice), and addition of pH correcting agents (e.g. ammonia, caustic etc.). Further studies will focus on the evaluation of kinetic parameters including the inhibitory effect of by-products and/or impurities to microbial growth and product formation. Fermentation runs will be carried out in batch, fed-batch and repeated batch fermentation mode in order to compare productivity, yield, final LA concentration, cost aspects and environmental criteria. The adapted strain(s) may be able to sustain inhibitory conditions during continuous fermentation (optional).

Task 2.3. Downstream processing (DSP) for recovery of lactic acid Task Leader: ATB; Other partners: TBWR
ATB will adapt and/or develop LA purification considering process scalability, purity/quality, robustness, separation yield and cost-competitiveness. One criterion should be the comparison with already existing figures related to other fermentation feedstocks, whereas the down-stream procedure for eventually unknown MSW-related components and impurities needs to be adapted due to missing experiences. In the case that there are still residual sugars in the broth, a nanofiltration step could be integrated in order to recover unconsumed substrate to enhance the overall process productivity. All the necessary components (filtration, electrodialysis, ion exchange, evaporation etc.) are available on bench (20-50 litres) as well as pilot scale (800-1000 litres).

It is foreseen that the fermentation broth will also contain a variety of other compounds such as a significant amount of minerals and remaining nitrogen (most likely no protein but polypeptides or single amino acids. In addition other organic acids which have its origin from the waste (such as acetic acid, propionic acid or butanolic acid) are also expected as impurities. Within this task different DSP- configurations will be tested for the recovery of the key product of lactic acid. Partner TBWR has piloted a separation technology to recover LA and amino acid (AA) in a combined processing which is currently demonstrated at production scale. In case the feedstock origin can also deliver AA at significant level, a transfer of technology from grass silage juice to PERCAL intermediates will be tested and assessed. This task is closely connected to WP5 the esterification of lactic acid with ethanol to ethyl lactate and the production of hot-melt adhesives by reactive extrusion. There is potential to compromise on quality of LA (e.g. use a racemic mixture, some remaining ash etc…) if it is used as intermediate for ethyl lactate.

Task 2.4. Up-scaling of the entire process Task Leader: ATB; Other partners: IMECAL
Up-scaling of the entire process chain including downstream processing for the provision of product samples in high quality. The optimised process in lab-scale (1-3 L bioreactors) will be scaled up in pilot-scale at 50 L bioreactor system. Process operation will be mainly dependent on the fermentation mode that will be selected in previous tasks 2.2 and 2.3. The pilot facility for the production of lactic acid at ATB and the corresponding pre-treatment and down-stream unit operations will be adapted to the waste materials under investigation to compare the process performance with previous feedstocks. General parameters such as utilisation of carbon sources, possible limitations by lacking nutrients, response/stability of microorganism and bioreactor and downstream separation performance at higher scales will be elucidated. This task will also provide several product samples in certain qualities for further processing and useful data for economic and environmental assessment, respectively. From the evaluation view not only the quantitative estimation but also the qualitative (impurities, optical purity) validation of the "waste-based lactic acid" will be an important part of the project.
The overall bioprocess design for lactic acid production at the pilot scale will be proposed, including up- and down-stream processes. Overall process testing will include all accompanying steps, such as hydrolysis, filtration, fermentation, full downstream processing based on membrane processes (i.e. electrodialysis) achieving a highly pure concentrated lactic acid, and will rely on previous results generated within WP2. Moreover, the wastewater from the pilot scale down streaming is collected to investigate its reutilization in the preceding processes.

Participation per Partner

Partner number and short name	**WP2 effort**
1 - IMECAL	
3 - ATB	
4 - AUA	
7 - TBWR	
10 - HAYAT	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D2.1	Lactic acid fermentation process	3 - ATB	Report	Confidential, only for members of the consortium (including the Commission	18

				Services)	
D2.2	Down-stream processing	7 - TBWR	Report	Confidential, only for members of the	24

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
				consortium (including the Commission Services)	
D2.3	Scaling of the full process.	3 - ATB	Report	Confidential, only for members of the consortium (including the Commission Services)	32

Description of deliverables

D2.1 Report of lactic acid fermentation process (ATB)
D2.2 Report of down-stream processing (TBWR)
D2.3 Report of up scaling of the full process (ATB)

D2.1 : Lactic acid fermentation process [18]

Lactic acid fermentation process

D2.2 : Down-stream processing [24]

Down-stream processing

D2.3 : Scaling of the full process. [32]
Scaling of the full process.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS4	LA with purity higher than 97% for L-LA .	3 - ATB	22	Application of chiral HPLC to estimate the optical purity of all LA samples
MS5	Yield at pilot plant facilities similar to other feedstocks already tested.	3 - ATB	30	Max. 10% losses compared to current process using pure lignocellulosic biomass feedstock.

Work package number [9]	WP3	**Lead beneficiary [10]**	5 - UGent
Work package title	Microbial conversions of MSW to succinic acid.		
Start month	3	**End month**	34

Objectives

- Optimisation of fermentation media composition for industrially competitive succinic acid production using MSWhydrolysates as the major source of nutrients.
- Development and optimisation of an integrated bioprocess combining SA production with separation from thefermentation broth via electrolysis membrane extraction targeting maximisation of productivity and yield.
- Comparison of succinic acid separation and purification based on a electrolysis membrane extraction system or anelectrodialysis-based downstream separation system.
- Optimisation of continuous cultures for succinic acid production and separation using immobilised cultures and a lowcost MSW hydrolysate as fermentation feedstock.
- Scale-up of the most efficient succinic acid production process.

Description of work and role of partners

WP3 - Microbial conversions of MSW to succinic acid. [Months: 3-34]
UGent, IMECAL, ATB, AUA, CENER
Task 3.1. Evaluation of MSW as fermentation feedstock. Task Leader: AUA Other partners: CENER.

Subtask 3.1.1 – Selection of bacterial strain and optimisation of fermentation media: The primary target will be the selection of the most efficient and robust succinic acid producing bacterial strain when the different MSW hydrolysates produced in WP1 are used. AUA will screen Basfia succiniciproducens, Actinobacillus succinogenes and Escherichia coli AFP184 that have been previously identified by AUA as the most efficient bacterial strains depending on the feedstock employed. AUA will screen the bacterial strains in single or two-stage (only in the case of E. coli AFP184) fermentations in Duran bottles and bench-scale bioreactors with constant supply of CO2 in the succinic acid production stage.
The optimal bacterial strain will be employed in the optimisation of the fermentation medium. The hydrolysis of polysaccharides, proteins (the supply of crude N sources will reduce the supply of yeast extract) and phytic acid (source of phosphorus) contained in MSW will lead to the production of a nutrient-rich MSW hydrolysate. AUA will employ ICP-MS analysis in order to identify the content of minerals and thus minimise the supply of commercial salts. Preliminary fermentations in Duran bottles will be carried out in order to assess the inhibitory effect of the MSW hydrolysates (produced in WP1) on bacterial growth and succinic acid production as compared to control fermentations carried out with commercial sugars (using the same mixture of sugars present in MSW hydrolysates), yeast extract and corn steep liquor. In the case that lower efficiency is observed when the MSW hydrolysates are used, AUA will study the effect of potential inhibitors present in MSW hydrolysates (e.g. furfural, 5-hydroxy-methyl-furfural, acetic acid, phenolics etc) and the effect of commercial nutrient supplementation (e.g. yeast extract, corn steep liquor) using a 96-well microplate reader followed by design of experiments (DoE) carried out in Duran bottles. The DoE will focus on the optimisation of nutrient composition. AUA will optimise the fermentation medium regarding nutrient content (addition of yeast extract, salts, MgCO3 and corn steep liquor) and will propose pre-treatment strategies for the removal of potential inhibitors.

Subtask 3.1.2 – Optimisation of lab-scale fed-batch and continuous cultures: AUA will optimise fed-batch and continuous cultures using the selected bacterial strain and fermentation media identified in Task 3.1.1. Optimisation of fed-batch cultures will focus on initial sugar concentration, addition of MgCO3 or Na2CO3 and feeding strategies. Continuous cultures will be optimised regarding initial sugar concentration, dilution rate and bacterial cell immobilisation. Fermentation modes of operation will be compared regarding productivity, yield, final sucinic acid concentration, capital/operating cost requirements and environmental criteria (e.g. non-renewable energy utilisation, waste generation).

Task 3.2. Optimisation of integrated bioprocess for succinic acid production and in situ separation. Task Leader: UGENT Other partners: AUA.

Subtask 3.2.1 – Study of electrolysis membrane extraction using a synthetic system: A number of commercially available anion exchange membranes will be assessed according to electrochemical parameters such as applied current, resistance and potential across the electrochemical cell, and subsequent flux of succinic acid and critical co-products (e.g. acetic acid, lactic acid). After an appropriate membrane is chosen, the ideal operational parameters can then be assessed according to other process parameters, including extractant concentration, pH and conductivity. This will be critical towards minimising the input power per extracted kg of succinic acid, and equivalent masses of OH- generated to manage the fermentation.

Subtask 3.2.2 – Optimisation of the integrated system at lab-scale: AUA will use the data from subtask 3.2.1 for the development of the integrated system, which will consist of 7 L bioreactor coupled to a high-rate electrochemical cell performing the extraction of organic acids produced during fermentation via recycling of fermentation broth through the extraction cell. This novel system will be developed at AUA in lab-scale in close collaboration with visiting researchers from UGENT. AUA and UGENT will study the effect of organic acid extraction on productivity and yield during fermentation. The effect of the applied current will be elucidated along with the effect of the produced H2 on bacterial metabolism. The integrated system will be evaluated using either fed-batch or continuous cultures using in the second case either free cells or immobilised cultures. The nutrients that pass through the membrane together with organic acids will be identified in order to assess the potential effect on fermentation efficiency.

Task 3.3. Succinic acid purification. Task Leader: AUA Other partners: UGENT
Conventional purification of succinate salts from fermentation broths requires cell removal via filtration, colour and protein removal via activated carbon treatment, acidification of organic acids using resins, concentration of succinic acid and removal of acetic and formic acids via evaporation, crystalisation of succinic acid followed by drying. AUA has optimised the above process to yield succinic acid crystals with purity higher than 99.5% and recovery yield higher around 80% when fermentations were carried out in spent sulphite liquor based media. This downstream process will be used as base-case scenario. AUA will optimise the purification of succinic acid produced by the electrolysis extraction system. The optimal concentration of succinic acid in the extracted aqueous solution will be identified in order to minimise energy consumption during evaporation which is required in order to remove the acetic and formic acids that also pass through the extraction electrolysis cell. The composition of the aqueous solution after the extraction will be identified in order to develop the appropriate separation scheme. Any sufficiently small anion will cross the membrane (e.g. chloride, sulfate, phosphate, formate, and more), and effective recovery and recycle of these compounds will be investigated.

Task 3.4. Process scale-up Task Leader: UGENT; Other partners: AUA
UGENT will use the data from tasks 3.1-3.3 regarding the lab-scale operation of the fermentation and in situ membrane electrolysis extraction in order to move the operation over to a pilot scale unit of > 50 L. The aim will be to control the succinic acid production via fermentation exclusively with the applied current (i.e. the electrolysis of water to hydroxide), and develop a system to continuously extract the succinic acid crystals from the acidic extract, such as a cooling and settling unit. The process can be investigated under conditions for maximum production rate, versus maximum yield, versus minimal power input to develop a set of parameters to compare against existing processes in succinic acid production. Other operational parameters more unique to membrane electrolysis, (e.g. membrane resistance, electrode potential, electrode fouling, voltage losses) will be investigated in detail, to compare the membrane electrolysis lifetime and operation against the other existing technologies. The process scale up will mimic the fermentation, extraction and purification of succinic acid, with the goal of minimising product output according to added chemicals as a first goal (i.e. completely replacing chemical caustic and acid control with electrochemistry), and followed by minimising the power investment per kg of succinic acid product. At an anticipated production rate of 20 g/L/d, here we aim to produce 1 kg/d in continuous, steady state operation, which should allow sufficient product for assessment and benchmarking against existing processes.

Task 3.5. Optimisation of bioprocess based on electrodialysis downstream separation system. Task Leader: ATB Other partners: AUA
ATB and AUA will optimise the production and purification of succinic acid in 20 – 50 L scale experiments using the unit operations (bioreactor, ultra/nanofiltration, softening, bipolar membrane electrodialysis (BMED), cation/anion exchange resins, activated carbon, vacuum evaporation) available at ATB. Succinic acid separation will commence with cell removal using hollow fiber membranes. Softening is then carried out by a series of anion and cation exchange resins to remove various impurities and metals that could cause membrane fouling, pipe jam or pump erosion. BMED will be employed to convert the succinate (and other organic acid by-products) into the respective acid form. In the BMED configuration, which is composed of three compartments, both the anions of the organic acids and the sodium cations are transferred across the anion and cation exchange membranes, and then they are combined with H+ and OHrespectively, which are produced in the bipolar membrane intermediate layer, to form the organic acids and NaOH in the acid compartment and base compartment, respectively. The process parameters

to be evaluated and optimized are the initial organic acid feed, the voltage of BMED process as well as the conductivity and the pH. Another important factor is to reduce the water transport per molarity succinate treated. The most important hurdle in this process is to prevent membrane fouling. In order to do so, the adequate pretreatment methods should be carried out in conjunction to the following measures: a) reversing the polarity of electrodes, b) decrease operating current, c) improve the hydraulic conditions in stack compartments, d) add chemicals to prevent the formation of precipitates or microbial growth, e) extract products before saturation. After electrodialysis, the product stream is processed via cationic/anionic resin or activated carbon in order to remove colour impurities, residual carbon or/and nitrogen sources. The final purification steps involve vacuum evaporation, crystalisation and drying. This process will be carried at ATB in collaboration with researchers from AUA. The results will be compared with the results achieved in previous tasks by electrolysis membrane extraction.

Participation per Partner

Partner number and short name	WP3 effort
1 - IMECAL	
3 - ATB	
4 - AUA	
5 - UGent	
6 - CENER	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D3.1	Selection of optimal bacterial strain using different MSW hydrolysates.	4 - AUA	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D3.2	Report on the integrated SA production with electrolysis membrane extraction.	5 - UGent	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D3.3	Report on production and purification of SA using electrolysis or electrodialysis based processes	4 - AUA	Report	Confidential, only for members of the consortium (including the Commission Services)	28
D3.4	Report on process scaleup	5 - UGent	Report	Public	32

Description of deliverables

D3.1 Selection of optimal bacterial strain using different MSW hydrolysates
D3.2 Report on the integrated SA production with electrolysis membrane extraction
D3.3 Report on production and purification of SA using electrolysis or electrodialysis based processes
D3.4 Report on process scale-up

D3.1 : Selection of optimal bacterial strain using different MSW hydrolysates. [18]
Selection of optimal bacterial strain using different MSW hydrolysates.

D3.2 : Report on the integrated SA production with electrolysis membrane extraction. [24]
Report on the integrated SA production with electrolysis membrane extraction.

D3.3 : Report on production and purification of SA using electrolysis or electrodialysis based processes [28]
Report on production and purification of SA using electrolysis or electrodialysis based processes

D3.4 : Report on process scale-up [32]
Report on process scale-up

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS6	Production of succinic acid in fed-batch and continuous processes	4 - AUA	18	Report on lab-scale experiments and comparison with preliminary experiments with the integrated system.
MS7	Extraction of succinic acid directly from the extraction cell at 99% purity.	4 - AUA	24	Report on the purity of succinic acid crystals extracted directly from the concentrated solution produced via membrane electrolysis.

Work package number [9]	WP4	Lead beneficiary [10]	6 - CENER
Work package title	Valorisation of fermentation by-products		
Start month	8	End month	34

Objectives

The objective of WP4 is to develop novel biosurfactants by employing the lipid and protein fraction remaining after the fermentation processes to ethanol, LA or SA, of pre-treated and hydrolysed MSW.

Description of work and role of partners

WP4 - Valorisation of fermentation by-products [Months: 8-34]
CENER, IMECAL, HAYAT
Task 4.1 Characterization of the fermentation residue of the ethanol/LA and SA processes Task Leader: CENER Other partners: IMECAL
Fermentation residues from the fermentation of ethanol, SA and LA processes will be analyzed employing the characterization techniques described in Task 1.2. Those residues with the highest content in protein and lipids will be selected for the revalorization of these fractions within the following tasks. Depending on the composition the strategy for the lipid and protein fraction will be evaluated. Processes for the fractionation of lipids and proteins rich fractions are well established in the literature, but adaptations and optimizations to the raw material are needed.

Task 4.2 Hydrolysis of protein fraction Task Leader: CENER Other partners: IMECAL
Hydrolysis of proteins into amino acids is the first step of converting proteins into surfactants. It is necessary to work with concentrated biomass, therefore if its water content is very high concentration by centrifugation or filtration will be applied. In order to increase the yield of the enzymatic hydrolysis some pretreatment processes will be studied: extraction with ethanol to remove color, previous lipid extraction, milling, heating, additional enzymes. Test conditions will be selected from literature in term of solid loads and other variables. The enzymatic processes will be evaluated on these pre-treated samples. Enzymes working at neutral or basic pH will be preferred such as papain (pH 7.0), pancreatin (pH7.5), Alcalase® 2.5L (pH 8) and Flavourzyme®1000L (pH 7). Working temperature will be around 50-60ºC, and 3-4 hours of incubation. Optimal dosage described in literature will be tested. Finally, free aminoacids will be quantified by a colorimetric test based on nihidrin reaction and its aminogram will be analyzed by HPLC from samples treated with phenylthio isocyanate
Chemical hydrolysis methods (based on acid or alkali) will be revised from. Acid hydrolysis completely destroys tryptophan and partially damages tyrosine, serine and threonine. Alkaly hydrolysis can modify arginine, lysine, cystein, serine and threonine. Therefore, pure chemical hydrolysis will be avoided. The enzymatic approach will be the preferred one, although, combination with mild acid hydrolysis will be evaluated in order to maximize yield and minimize processing time. The protein hydrolyzate will be employed as a Nitrogen source for fermentation procedures and as synthons for biosurfactant production.

Task 4.3 Extraction and hydrolysis of the lipid fraction Task Leader: CENER Other partners: IMECAL
There are two main approaches for lipid valorisation: the first one is a first step of extraction, followed by the subsequent esterification or saponification reactions; the second approach is the direct treatment without separation. In the second case extreme chemical agents are required, and it can be harmful for the protein fraction, therefore, within the framework of this project the lipids will be extracted as the first step in revalorization process. Lipid extraction will be performed on the dried fermentation residue because water can affect the subsequent saponification reactions. It can be performed with a single solvent or with solvent mixtures. Extraction tests will be performed in a soxhlet system with organic solvents (dichloromethane, ethyl acetate: ethanol, or mainly ethyl lactate: ethanol) for 18 hours. After the extraction, solvent will be removed and the degree of extraction will be quantified for each case. The lipid fraction will be analyzed in order to know the content in acylglycerols (by GC-FID). Analysis of fatty acid composition will be performed by GC-FID or GC-MS of methanolized samples.

Task 4.4. Synthesis of the amino-lipid biosurfactant Task Leader: CENER Other partners: IMECAL
By the combination of chemical, enzymatic or a combination of both strategies, surfactants will be obtained by the condensation of amino acid and lipids. The N-acylation of the amino group by condensation with fatty acids from the saponified lipid fraction or directly from the acylglycerides, will be prepared by either chemical methods or by lipases (from Candida antarctica & Rhizomucor meihei). Aqueous or solvent reaction systems will be evaluated. For the chemical approach the activation of the acyl carbon of the fatty acids will be performed by the formation of acyl chlorides in strongly alkaline aqueous medium containing the aminoacids. Protection of Nitrogen-amino group will be

taken into account for some amino acids. A mixture of different molecules is expected, so therefore state of the art isolation and characterization techniques will be applied. Surface active properties of the surfactants will be tested such as surface tension at the critical micelle concentration, critical micelle concentration, maximum surface excess concentration at the air/aqueous solution interface and minimum area per surfactant molecule at the air/aqueous solution interphase.

Task 4.5. Validation of the biosurfactants in relevant industrial applications Task Leader: HAYAT.
HAYAT uses Linear Alkyl Benzene Sulphonic Acid (LABSA), it is an anionic surfactant widely used in the production of varieties of detergents and emulsifiers. HAYAT will evaluate the performance of the new surfactant from MSW in a standard formulation in one of their detergents following their own internal methodology and suitable EN/ISO/ASTM/ DIN standards such as ISO 4319, ASTM D2960-98, ASTM D2960-05, and DIN 44983. If required, additional additives or blends with LABSA or other commercial surfactants will be employed. Full validation and new products data-sheets will be provided at the end of the project.

Participation per Partner	
Partner number and short name	**WP4 effort**
1 - IMECAL	
6 - CENER	
10 - HAYAT	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D4.1	Characterization of the fermentation residue.	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	20
D4.2	Pre-treatments for revalorization of the protein fraction.	6 - CENER	Report	Confidential, only for members of the consortium (including the Commission Services)	26
D4.3	Pre-treatments for revalorization of the lipid fraction.	6 - CENER	Report	Confidential, only for members of the consortium (including the Commission Services)	26
D4.4	Biosurfactant production from lipids and proteins.	10 - HAYAT	Report	Confidential, only for members of the consortium (including the Commission Services)	34

Description of deliverables

D4.1 Report on characterization of the fermentation residue
D4.2 Report on pre-treatments for revalorization of the protein fraction
D4.3 Report on pre-treatments for revalorization of the lipid fraction D4.4 Report on biosurfactant production from lipids and proteins

D4.1 : Characterization of the fermentation residue. [20] Characterization of the fermentation residue.

D4.2 : Pre-treatments for revalorization of the protein fraction. [26] Pre-treatments for revalorization of the protein fraction.

D4.3 : Pre-treatments for revalorization of the lipid fraction. [26] Pre-treatments for revalorization of the lipid fraction.

D4.4 : Biosurfactant production from lipids and proteins. [34] Biosurfactant production from lipids and proteins.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS8	A method for the production of amino acid and lipid surfactants from MSW byproducts.	6 - CENER	32	Comparison of surface active properties with those described in literature.
MS9	Biosurfactants has fulfil the HAYAT internal standards.	10 - HAYAT	34	Comparison of bio-surfactant properties with commercial ones using DIN/EN/ISO standard.

Work package number [9]	WP5	**Lead beneficiary [10]**	11 - COVESTRO
Work package title	Green chemical routes to produce solvents, adhesives & polymers from developed acids.		
Start month	10	**End month**	34

Objectives

The main objectives of WP5 are the production and industrial validation (TRL5) of the follow products using the monomers developed in WP2 &3: 1) Ethyl lactate by reactive distillation of LA and ethanol, 2) Hot Melt Adhesives by reactive extrusion of LA, and 3) Polyester-Polyols by conversion of SA with diamines. Polyester Polyols will be validated by PU formation using Di-isocyanates.

Description of work and role of partners

WP5 - Green chemical routes to produce solvents, adhesives & polymers from developed acids. [Months: 10-34]
COVESTRO, AIMPLAS, TBWR, HAYAT, YPR
Task 5.1. Ethyl lactate as a solvent. Task Leader: TBWR, Other partners: HAYAT.
Lactic acid and ethanol will used for esterification to produce ethyl lactate. The main technological focus is to test and further develop a reactive distillation process for simultaneous esterification and separation of product in one column. TBWR is able to bring in a reactive distillation prototype into the project which was designed specifically for ethyl lactate within an ongoing national R&D project. (Column DN 50mm; height 4.5 meters).

Subtask 5.1.1 Esterification test in lab scale for validation of PERCAL intermediates. This task involves batch esterification test in lab and at bench scale to test different solid catalyst designs. Also potential degeneration of catalyst resigns need to be tested and evaluated which is related to PERCAL - LA quality and potential impurities.

Subtask 5.1.2 Design, validate and modify a chemical model for the esterification process. The esterification process and simultaneous product separation with a reactive distillation will be described by a multi parameter chemical model. The specific model will be used to simulate processing to get first results on the influence of the feedstock characteristic (e.g. concentration of LA and ethanol) and its interaction on esterification performance. The model is used as a tool to plan systematic experiments in 5.1.3 and can also be adjusted by results of the experiments.

Subtask 5.1.3 Upscale reactive distillation to produce ethyl lactate. Within this task an existing reactive distillation pilot prototype column will be used to upscale the esterification and separation process. This task involves a series of systematic trials to run the reactive distillation at various process parameters. The esterification test/process is assessed using analytical data of various streams of the reactive distillation. Results are used to adapt and fine tune the chemical model and refine the set of experiments. The goal is to identify optimized operation parameters for the production of ethyl lactate by means of reactive distillation. Also this task will deliver first sets of products as samples to be provided to industrial partners.

Subtask 5.1.4 Assessment of the esterification process and up-scaling. This task uses and analyses the results generated in Task 5.1 to refine key information and data on esterification process performance (efficiency, energy demand, product recovery rate, potential technical bottlenecks etc.) and also characterise the quality of ethyl lactate generated. The objective is to compile concrete data for the technical and economic analysis on the process and product to establish the basis for a sound evaluation of further upscaling of reactive distillation.

Subtask 5.1.5 Ethyl Lactate Validation. HAYAT will validate the Ethyl Lactate in two products of their current products using their internal methodology and the suitable EN/ISO/ASTM/DIN tests such as ISO 4319, ASTM D2960-98, ASTM D2960-05, and DIN 44983. In concrete, they studied the substitution of propylene glycol monomethyl ether in inks for printing and isopropanol in windows cleaners. Full validation and new products data-sheets will be provided at the end of the project.

Task 5.2. LA based hot melt adhesives by reactive extrusion. Task Leader: AIMPLAS Other partners: YPR.

Subtask 5.2.1 Production and purificaction of lactide from lacti acid. The procedure of lactide obtention from lactic acid consists basically of heating up lactic acid in a glass vessel under vacuum whilst steering and distilling off water in order to promote a polycondesation reaction to produce a lactic acid polycondesate. Once the polyester is formed an addition of a catalyst will lead us to transform the polycondensate into lactide. The dimer (lactide) will be purified from the crude by distillation and by solution crystallization. For the production of up several kilos, a 15L reactor of AIMPLAS's kilolab facilities will be employed.
Subtask 5.2.2 Production of Low molecular PLA and PLA-copolymers by reactive extrusion. Lactides will be used for their Ring Opening polymerization using a large L/D twin screw extruder as a continuous reactor. In order to optimize the reaction conditions in terms of reaction time, initiatior, ratio of copolymer, catalyst and temperature a pre-optimization employing a Haake minilab extruder will be carried out.
Once the different polymers and copolymers have been obtained by reactive extrusion and by solution procedures, those compounds will be characterized employing several techniques as Differential scanning calorimetry (DSC), thermogravimetry (TGA), Gel permeation chromatography (GPC) and rheological tests which are currently available in AIMPLAS. In addition the molecular weight of the polyester will be analyzed employing a new NMR method based in the correlation of diffusion coefficients acquired by proton NMR spectroscopy and the MW. The method lead us to explore and analyze with just some milligrams of sample properties as MW & polidispersity index
Once the best conditions has been optimized several tests will be done to adjust the reactive extrusion conditions to obtain a PLA grade with a low molecular weight suitable to be used in hot melt adhesives. Special attention will be done in the selection of type and amount of the catalyzer. The reaction will be designed to be done in the residence time of conventional extrusion process. The PLA will be characterized.

Subtask 5.2.3 HMA by PLA grafting. Initial tests to PLA graft will be done at lab scale in a discontinuous 1 L reactor using some reactive substances (glycolic acid, maleic acid, methacrylate or glycidyl acrylate) and suitable peroxides to create a chemical graft bonds between the matrix and selected monomers. The objective is to achieve a percentage of graft superior to 1% w/w. The percentage of co-monomer will be determinate by Fourier-transform infrared spectroscopy (FT-IR) and their thermal stability by differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA).The best results obtained in solution will be transferred to a reactive extrusion process employing a Haake minilab extruder.Once the reaction conditions have been optimized the process will be scale up using a twin esxtrusion processing parameters (screw design, residence time, feeding of each component, etc..) will be optimized. The best PLA grafted grade obtained will be stabilized and processing improved by adding by melt compounding the suitable additives to be used as a HMA (e.g., antioxidants, UV stabilizers, plasticizers, tackifiers, waxes, etc…).
Finally, in order to have a cost competitive product, we the possibility to produce the HMA in one compounding steps, using lactides and reactive co-monomers will be studied, i.e., lactide polymerization, reactive commoner grafting and adding and mix the rest additives for final HMA formulation.

Subtask 5.2.4 HMA scale-up and Validation. The production of HMA will be scaled-up and validated in the YPR industrial facilities. The company is producing several tons of petrol based HMA each year and they will be used the same methodology to validate the new renewable source. Other partners such as HAYAT and COVESTRO will use the lactic acid to sealing delivery box cardboard or labels prototypes and validated according their internal procedures.

Task 5.3. Butanediol from succinic acid using green chemistry routes. Task Leader: COVESTRO Other partners: AIMPLAS.
COVESTRO has a pilot plant equipped with several reactors of different sizes to produce different type of building blocks from several monomers. Using these facilities, the SA obtained in WP3 will be combined with several diamines to produce high molecular weight Polyester-Polyols. Processing parameters such as temperature, pressure, residence time and other chemical to control the final polyester chain length. Polyester Polyols will be validated by PU formation using Di-isocyanates using an internal methodology that use own defined tests and suitable EN/ISO standards.

Participation per Partner

Partner number and short name	**WP5 effort**
2 - AIMPLAS	
7 - TBWR	
10 - HAYAT	
11 - COVESTRO	
13 - YPR	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D5.1	Production of Ethyl lactate at laboratory level.	7 - TBWR	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D5.2	Ethyl-Lactate scale up and industrial validation.	10 - HAYAT	Report	Public	34
D5.3	Production of HMA at laboratory level.	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D5.4	HMA scale up and industrial validation	13 - YPR	Report	Public	34

D5.5	Production of polyols at laboratory level.	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	24
D5.6	Polyols scale up and industrial validation	11 - COVESTRO	Report	Public	34

Description of deliverables

D5.1 Report on production of Ethyl lactate at laboratory level.
D5.2 Report on Ethyl-Lactate scale up and industrial validation.
D5.3 Report on production of HMA at laboratory level.
D5.4 Report on HMA scale up and industrial validation.
D5.5 Report on production of polyols at laboratory level.
D5.6 Report on polyols scale up and industrial validation.

D5.1 : Production of Ethyl lactate at laboratory level. [24] Production of Ethyl lactate at laboratory level.

D5.2 : Ethyl-Lactate scale up and industrial validation. [34] Ethyl-Lactate scale up and industrial validation.

D5.3 : Production of HMA at laboratory level. [24] Production of HMA at laboratory level.

D5.4 : HMA scale up and industrial validation [34]
HMA scale up and industrial validation

D5.5 : Production of polyols at laboratory level. [24] Production of polyols at laboratory level.

D5.6 : Polyols scale up and industrial validation [34]
Polyols scale up and industrial validation

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS10	Simultaneous esterification & products separation (reactive distillation) for ethyl lactate established.	7 - TBWR	34	Continuous mode reactive distillation established and validated at scale 3-5 L/h.
MS11	Renewable HMA has fulfil the end-user requirements or YPR internal standards.	2 - AIMPLAS	34	Comparison of HMA main properties with commercial EVA and renewable source HMA using the suitable EN/ ISO.
MS12	Polyester-Polyols by conversion of SA with diamines have similar reactivity than using commercial bio-based SA (less than 20% time increase) and the PU formation obtained have at least 80% of performance than commercial ones.	11 - COVESTRO	34	Comparative analysis using internal and well-defined methodology of COVESTRO.

Work package number [9]	WP6	**Lead beneficiary** [10]	8 - VISUM
Work package title	Study of adaptation and monitoring of PERSEO plant for the production of ethanol, LA & SA with suitable quality to produce the selected final products.		
Start month	12	**End month**	36

Objectives

The main objectives of WP6 are to study the adaptability of a versatile biotechnological plant to obtain different chemical intermediates and to determine the most relevant parameters to control and monitor the PERSEO's process at industrial level.

Description of work and role of partners

WP6 - Study of adaptation and monitoring of PERSEO plant for the production of ethanol, LA & SA with suitable quality to produce the selected final products. [Months: 12-36]
VISUM, IMECAL, ATB, AUA
Description of work.
Process parameters of PERSEO Bioethanol® semi-industrial plant are currently controlled with a SCADA technology. New monitoring technologies developed by VISUM will improve the performance of the plant control by obtaining relevant parameter information through advanced sensors and adding an intelligent plant control platform with decision support models. The objectives of this WP consist of studying the adaptability of PERSEO Bioethanol® plant to produce lactic acid and succinic acid as well as the design and testing of new sensors and control platform for the monitoring of the process parameters of interest in a laboratory scale and the study of the scalability of the processes and control system at full plant level.

Task 6.1. Study of parameters of the pre-treatment and fermentation process for the production of bioethanol, lactic & succinic acid and the by-products stream. Task Leader: IMECAL Other partners: VISUM.
In order to determine the main parameters in the fermentative processes, current control parameters measured on line in PERSEO pilot plant during the simultaneous saccharification and fermentation processes for the production of bioethanol are pH, temperature, pressure and material flow. Additional parameters that would be useful to monitor in real time for the better control of the process progress are the viscosity and homogeneity of the feedstock, quantification of liberated sugars and quantification of generated ethanol, which are currently measured off line.
Moreover, taking into account the optimization for the microbial conversion of MSW to lactic and succinic acid (WP 2 and WP3) the adaptation of PERSEO plant will be studied for the production of both intermediates. In particular, the required conditions of the fermentation unit (temperature, retention time, pH control, addition of nutrients…) for the optimum production of the organic acids at semi-industrial scale will be evaluated, studying the main parameters to be controlled for the respective fermentation processes. Control of the substrate viscosity progress and quantification of liberated sugars will be common to the three different type of fermentations. Quantification in real time of lactic and succinic acid produced will be also evaluated.

Task 6.2. Study of parameters of the separation/purification processes for the production of bioethanol, lactic acid and succinic acid. Task Leader: IMECAL Other partners: VISUM
To determine the main parameters in the separation/fermentation processes. Each of the target intermediates require different separation and purification techniques. Bioethanol is purified by means of distillation. The range of temperature, pressure, flows and amount of ethanol at the top of distillation columns are all important parameters to be studied. In case of lactic acid, the integration of all the necessary chemical purification systems (filtration, electrodialysis, ion exchange, evaporation units etc.) for the achievement of the required quality will be evaluated and the relevant parameters to control will be identified. Similarly, the adaptation of the electrochemical separation described in WP3 at laboratory scale for the purification of SA will be also studied to be implemented in PERSEO plant.

Task 6.3. Monitoring of the processes parameters. Task Leader: VISUM Other partners: IMECAL.
Based on the parameters studied by IMECAL in Tasks 6.1 and 6.2, VISUM will build at least 1 monitoring system with adjusted sensors, possibly a sampling system, and a data acquisition and treatment platform to be able to track the Critical Control Parameters (CCPs) and the Critical Quality Attributes (CQAs). Potential sensors to be used include but are not limited to NIR or other optical technologies. This system will be applied at laboratory scale where the different processes for the production of bioethanol, lactic acid and succinic acid will be studied. These tests will allow to study the feasibility of monitoring the parameters listed as critical to quality of the processing of OMSW and will provide deep understanding of the processes in real time. VISUM will also develop the associated calibration models to analyse

the measurements of the sensors in order to obtain a tailored and fit for purpose monitoring system. These models will be developed if possible, from existing similar models or in case there are no suitable previous studies, new chemical analysis will be carried out to establish or validate the spectroscopic calibration models.

Task 6.4. Integration and adaptation of processes to a biotechnological plant configuration Task Leader: IMECAL; Other partners: VISUM
In this task, we plan to study the adaptation and monitoring of the intermediates production processes to a unified biotechnological plant. Based on the different processes studied in previous tasks T6.1 and T6.2, IMECAL will study their integration and adaptation to a MSW biotechnology pilot plant for the versatile production of bioethanol, lactic and succinic acid. VISUM will study the scaling of the monitoring system to a unified plant configuration in which all process steps of the production of bioethanol, lactic acid and succinic acid are performed continuously in a single biotechnological plant. To this end, a plant control system will be studied and evaluated and a configuration for controlling the processes of the unified plant will be reported in Deliverable 6.3. To this end a process modelling task will be required which will be done in conjunction with IMECAL.

Participation per Partner

Partner number and short name	**WP6 effort**
1 - IMECAL	
3 - ATB	
4 - AUA	
8 - VISUM	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type**[15]	**Dissemination level**[16]	**Due Date (in months)**[17]
D6.1	Adaptation of PERSEO plant for the versatile production of chemical intermediates & identification of key parameters	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D6.2	Monitoring technologies and testing results.	8 - VISUM	Report	Confidential, only for members of the consortium (including the Commission Services)	32
D6.3	Integration of the production in a biotechnological plant.	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	36

Description of deliverables

D6.1 Report on the adaptation of PERSEO plant for the versatile production of chemical intermediates and identification of key process parameters.

D6.2 Report on the monitoring technologies and testing results.
D6.3 Report on the integration of the production in a biotechnological plant configuration.

D6.1 : Adaptation of PERSEO plant for the versatile production of chemical intermediates & identification of key parameters [18]

Adaptation of PERSEO plant for the versatile production of chemical intermediates & identification of key parameters

D6.2 : Monitoring technologies and testing results. [32]

Monitoring technologies and testing results.

D6.3 : Integration of the production in a biotechnological plant. [36]

Integration of the production in a biotechnological plant.

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS13	Main process parameters identified.	1 - IMECAL	18	Study of a versatile plant for bioethanol, LA and SA production.
MS14	Monitoring system built and tested.	8 - VISUM	30	The sensors and control system will be available for lab validation and suitable results will be reported.

Work package number [9]	WP7	**Lead beneficiary** [10]	4 - AUA
Work package title	Environmental, regulatory & economic analyses.		
Start month	1	**End month**	36

Objectives

The objectives of WP7 are summarized in the follow points:
- Study the environmental suitability of the processes by a Life Cycle Assessment (LCA).
- To demonstrate the technical integration of the modelled processed through virtual simulation
- Assess the economic viability each the product stream through the development of a technical and economic evaluationto guide the decision points throughout the project and assess economic viability.
- Ensure that the new material and their processing technology complies with the relevant regulatory landscape.

Description of work and role of partners

WP7 - Environmental, regulatory & economic analyses. [Months: 1-36]
AUA, IMECAL, AIMPLAS, ATB, UGent, CENER, TBWR, VISUM, HAYAT, COVESTRO, EXERGY
Task 7.1 Evaluation of synergies with other projects involved in MSW valorisation. Task Leader: AIMPLAS, Other partners: IMECAL, EXERGY.
There are several projects aiming to make biofuels and/or bio-based products from MSW (FP6, FP7 and H2020). Some of them feed the waste to microorganism; some others first go via a hydrothermal liquefaction process or a gasification. In the second case, it is important to check that no chemical by-products will affect the downstream microorganism process. Synergies with current or past fuel related projects will be evaluated using the CORDIS database, personal contacts and information from the different partners will be employed. A stakeholder board, or an advisory board will be proposed in the first months of the project that include mainly companies and research centres from the biofuel sector (such as Lanzatech, Enerkem, French institute of Petroleum and New Energies, BP, Shell or Total). In terms of Board members, we plan to involve several related European associations.

Task 7.2. Modelling and simulation of the overall process. Mass and energy balances. Leader EXERGY, with the support of all partners
In order to evaluate the environmental impact of the project in terms of energy and mass streams, including the actual CO2 balance which aims to be positive, virtual simulations of the entire PERCAL process at the process scale and/or larger scale (as a case study) will be run in order to evaluate the different balances. The results will be complemented with the LCA outcomes in order to evaluate the real environmental impact of the project.
Additionally, simulations will be used for the optimisation of the process and the identification of potential hotspots. Results will be reported along with potential enhancements of the overall PERCAL process and will be used for the evaluation of the exergy analysis. The capacity for the up-scaled simulation will be decided during the project's execution.

Task7.3. Product Life-Cycle-Assessment. Leader AUA, with the support of all partners.
Life Cycle Assessment considers the entire life cycle of a product to assess the environmental impacts associated with all the stages of the product's life from-cradle-to-grave (i.e. from raw material extraction through material processing, manufacture, distribution, use, repair and maintenance, and disposal or recycling).
To ensure the consistency of the LCA studies performed in PERCAL, Task 7.3 will build upon existing and ongoing LCA activities on bio-based products and processes, and will take into account the guidelines of the ILCD handbook project , which has developed guidelines to make LCA studies more coherent and comparable. Furthermore, the LCA approach will be used to compare bio-based and non-biobased products within the same LCA framework, this comparative approach generates results which are more robust than a reliance on absolute values and will help in the marketing claims of the advantage of the new solutions vs. conventional synthetic counterparts.
The LCAs will be performed using dedicated software (e.g. Umberto NXT software) and the Ecoinvent database for a selected set of raw material/production process/product, taking into account use and disposal or re-use, i.e. 'from cradleto-grave'. The LCA will follow the ILCD guidelines and ISO 14040 methodologies. This will include the definition of the following aspects: goal, scope and system boundaries; life cycle inventory; life cycle impact assessment and interpretation of the results.
A dynamic approach with feedback to the process development activities will be implemented, using the outputs of the LCA analysis. The pilot tests and demonstrators will likewise provide feedback to the Life Cycle Inventory, with up-to-date field data.

The LCA will serve both as a characterisation tool for the performance of the developed processes, and as a means of identifying areas where process optimisation can make the largest environmental savings. This information will be fed back to the RTD development and scale-up activities so that effective process optimisations may be realised.

Task 7.4 Economic analysis. Lead partner: EXERGY, Other partners: All partners
An economic study is required to validate the economic feasibility of the improved materials and technologies developed in the project and take further decisions. Parallel to the environmental analysis, an economic analysis of the materials will be conducted using life cycle costing (LCC) and will be complemented with techno-economic assessment (TEA). LCC is a methodology for the systematic evaluation of the economic costs associated with the entire life cycle of a product or process, including not only direct costs but also external relevant costs and benefits. LCC also follows a life cycle approach, but it assesses the economic costs of the product or process instead of the environmental impacts as in LCA. The procedure used for the economic analysis will be similar to that used for the environmental analysis. The economic costs of current scenarios will be assessed and compared with the costs of alternative scenarios and current solutions. EXERGY will be responsible for conducting the LCC studies, which will take into account not only investment and operating costs but also other external relevant costs, such as environmental costs.

Task 7.5 Social analysis. Lead partner: EXERGY, Other partners: All partners
Social impacts must also be taken into account when developing new technologies. A social analysis is necessary to determine whether the products coming from MSW can be accepted by customers.
A social analysis of the materials and technologies will be conducted using social life cycle assessment (S-LCA), which will be performed in parallel to the environmental (LCA) and economic analyses (LCC). S-LCA is a methodology to assess the social and socio-economic aspects of materials or processes and their potential positive and negative impacts along their life cycle. These aspects assessed in S-LCA are those that may directly or indirectly affect stakeholders, including workers, local community, consumers, value chain actors and society. S-LCA also follows a life cycle approach, but the impacts in this case are related to socio-economic processes and comprise human rights, working conditions, health and safety, socio-economic repercussion, and so forth.
As in previous analyses, the procedure used for the social analysis will involve the assessment and comparison of social impacts between current scenarios and alternative scenarios, these latter being based on the new developments from this project.
EXERGY will be responsible for conducting the S-LCA studies and the data required for the development of the S-LCA studies will be provided by other project partners and by organise some discussions along the project with engaging researchers, industry, policymakers, NGOs and Trade Unions, as well as desk research on identifying potential issues.

Task 7.6: Process inefficiencies Analysis Leader: EXERGY.
This task will study process life cycles from a thermodynamic perspective in order to analyse inefficiencies at different stages in process and propose potential improvements. An exergy assessment methodology to determine specific exergy contents to all mass and energy in and outflows will be developed. Where necessary, computerized simulation tools (i.e. COMSOL or TRNSYS) will be utilized.
The potential for improvements will be assessed by the overall exergy efficiency, which is the real exergy over the exergy used in the process. This study will compare: (1) the efficiency of the processes before and after the implantation of PERCAL innovations and (2) the efficiency with regards to other similar biological processes.
Aside from the determination of the efficiency for the different heat transfers along the process, exergy analysis will establish the potential causes for the detected inefficiency and will decrease cost associated to operation and maintenance of the system by means of a suitable assignment of energy optimisation measures at different levels of the process.

Task 7.7. Regulatory and safety analysis. Task Leader: AIMPLAS, Other participants: All partners.
The verification that the products comply with the existing regulatory landscape for the use of the rest raw materials and for the use of the final products on the envisaged applications and fulfil EC Directives and specific national laws in the EU28, regulatory compliance will be regularly checked throughout the project . The assessment should be performed before the end of the project in order to enable any processing/product specification changes. Input from final users is required. A regulatory compliance report will be written to make sure that the products and technologies also fulfil all standards and legislation in force with regards to safety criteria.

Participation per Partner

Partner number and short name	WP7 effort

1 - IMECAL	
2 - AIMPLAS	
3 - ATB	
4 - AUA	
5 - UGent	
6 - CENER	
7 - TBWR	
8 - VISUM	
10 - HAYAT	
11 - COVESTRO	
12 - EXERGY	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type**[15]	**Dissemination level**[16]	**Due Date (in months)**[17]
D7.1	Draft report on economic analysis	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D7.2	Simulation of the system and M&E balances	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D7.3	Report on Life Cycle Assessment (LCA).	4 - AUA	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D7.4	Report on Social Life Cycle Assessment (SLCA).	12 - EXERGY	Report	Public	36

D7.5	Report on regulatory analysis and safety aspects	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D7.6	Final report on economic analysis	12 - EXERGY	Report	Confidential, only for members of the consortium (including	36

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
				the Commission Services)	

Description of deliverables

D7.1 Draft report on economic analysis
D7.2 Final report on simulation of the system and M&E balances
D7.3 Report on Life Cycle Assessment (LCA).
D7.4 Report on Social Life Cycle Assessment (S-LCA).
D7.5 Report on regulatory analysis and safety aspects.
D7.6 Final report on economic analysis

D7.1 : Draft report on economic analysis [18]

Draft report on economic analysis

D7.2 : Simulation of the system and M&E balances [36]

Simulation of the system and M&E balances

D7.3 : Report on Life Cycle Assessment (LCA). [36]

Report on Life Cycle Assessment (LCA).

D7.4 : Report on Social Life Cycle Assessment (S-LCA). [36]

Report on Social Life Cycle Assessment (S-LCA).

D7.5 : Report on regulatory analysis and safety aspects [36]

Report on regulatory analysis and safety aspects

D7.6 : Final report on economic analysis [36]

Final report on economic analysis

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS15	Environmental performance of the new materials and processes according to standards and EU legislation in force.	2 - AIMPLAS	36	Check the environmental regulatory assessments that affect the product.

MS16	Final cost of intermediates and products at least 15% cheaper than current biobased market analogues.	12 - EXERGY	34	Final costs of the products taking into account all process steps.
MS17	The new bio-based products and intermediates fulfil the related regulatory requirements.	1 - IMECAL	36	Check the regulatory assessments that affect the product.

Work package number [9]	WP8	**Lead beneficiary** [10]	12 - EXERGY
Work package title	Communication, Dissemination, Exploitation, Business plan and Technology Transfer		
Start month	1	**End month**	36

Objectives

This WP encompasses the development and delivery of the project Plan for Exploitation of the Results and Dissemination (PERD).WP8 comprises of six interlinked actions designed to ensure successful exploitation of the projects technical innovations: 1) Communication, 2) Dissemination, 3) Exploitation, 4) Business Models and Plans, 5) Intellectual Property Management and 6) Technology Transfer.

Description of work and role of partners

WP8 - Communication, Dissemination, Exploitation, Business plan and Technology Transfer [Months: 1-36]
EXERGY, IMECAL, AIMPLAS, ATB, AUA, UGent, CENER, TBWR, VISUM, HAYAT
Task 8.1 Communication. Task Leader: EXERGY with the support of all partners.
The purpose of the communication plan is to maximise awareness of project aims and outputs amongst a broad stakeholder base. Communication activities complement dissemination actions as they add public value to the achievements of the project by transforming the sometimes-complex scientific and technological results into media resources. The communication plan will identify channels for tailored, target-specific communication and will include guidelines for using and disseminating the knowledge gained during the project lifetime. This plan will identify the relevant trade media (e.g. bio-based chemicals and materials, sustainability, industrial biotechnology as well as household products, adhesives, paints, etc. and the relevant specialty chemicals publications).
Stakeholder mapping - At the start of the project, the Consortium will review and expand the list (email pool) of groups, projects, experts and other stakeholders, identified during the proposal phase, and develop a stakeholder mapping to use in its communication activities.
Messaging – WP8 will work on defining the key messages that need to be used in the communications materials. The messages, based on scientific information, will be concise, clear and easy to understand, and tailored to the target stakeholder group. The project communications tools will include: a) Project logo and marketing communications tools - A PERCAL logo along with the project's visual identity will be designed and produced. The logo and the project's visual identity will be rolled out to the different communications tools that will be produced to engage consumers; b) Project website and communication channels - At the beginning of the project, WP8 will set up the PERCAL website. The website (in English language) will be developed to provide an overview of the project objectives and programme of work (i.e project's workshop, events and other activities), to track its progress and present relevant results of individual work packages in near real time (public deliverables). Another channel for dissemination of the PERCAL objectives, organisation and results is through social media. WP8 will set-up and maintain a presence in social media networks (LinkedIn, Twitter, YouTube). The PERCAL social media presence will broaden the reach to the public and experts interested in following the project and offering feedback.

Task 8.2 Dissemination. Task Leader: AIMPLAS with the support of all partners.
The purpose is to support project exploitation through technical dissemination and industrial outreach. The PERCAL Consortium will use a variety of platforms tailored, when possible, to reach particular industry and research groups and maintain communication with them. Press releases will be drafted and disseminated and opportunities to publish through the trade media will be compiled. The consortium will also carefully select events organised by others to participate in and present the project results towards the end of it.
A dissemination plan (a Working Document among the partners) with tailored means to reach main target audiences will be developed and will designate specific points of contact for value chain participants, research communities, press, the general public and the European bioeconomy process industry that will be maintained during the course of the

project, so that the PERCAL consortium can always be contacted by relevant actors in the field and interested persons for any query on content and/or findings but also as opportunities for possible collaborations.

Task 8.3 Exploitation. Task Leader: EXERGY with the support of all partners.

The Exploitation Plan will be designed at the beginning of the project as a working document and updated constantly throughout the project's life cycle. The plan will be reviewed yearly with progress and identified issues recorded and exploitations actions updated. The Exploitation Plan will address the following aspects: 1) List of project results and identification of key exploitable results (KER i.e. most promising results), 2) the final exploitation plan will incorporate the information and knowledge developed in Task 8.4 and also an exploitation risk analysis, 3) an initial version of the exploitation plan will be drawn with the initial results identified in the proposal stage.

At the end of the project, a detailed exploitation plan will be finalised. Exploitable outputs will be identified and discussed at regular exploitation workshops;

Exploitation workshops - The whole consortium will be involved in exploitation workshops that will be led by EXERGY. The objective of the workshops will be the characterisation of the exploitable results and identification of new results as the project progresses. The workshops will serve as screening mechanism to identify the key exploitable results to be taken through to business model and plan development (Task 8.4). The exploitation workshops will be organised on a yearly basis, and whenever possible just before/after the General Assemblies. The outcomes of these workshops will feed into the Exploitation Plan.

Exploitation risk analysis will lead to a risk management plan, taking into account the internal and external factors which may prevent the successful exploitation of the project results by follows three steps: 1) the identification of exploitation risks. They may include technical and non-technical; the risk register will draw from the information and knowledge gathered in all work packages & tasks, 2) the assessment of these exploitation risks along two dimensions: probability of occurrence & potential impact. A matrix combining these two dimensions then allows ranking the different risks and 3) the design of a risk management plan detailing the actions to put any type of exploitation risk under proper control. The risk management will be implemented during the project with recommendations for the pre-exploitation phase in the final exploitation plan.

Task 8.4 Business Models and Plans for KER. Task leader: EXERGY with the support of all partners.

Development of business models: EXERGY will develop business models to support the continued exploitation and eventual commercialisation of each key project result (please see Impact section for more details). This task will be led in coordination with the dissemination/communication (Tasks 8.1/8.2) activities to ensure the coherence of the project's messages towards the outside world.

The task will involve outlining, refining and defining the business models through two actions:

Action 1. Business Model Development. A two-step process will be used to build an initial plan. Step 1) Understand, refine and define the innovations value proposition through the analysis of, and discussion with consortium partners and through literature review considering 1) Policy/Environmental/Social/Technological (PEST) issues, 2) competitive intensity analysis, 3) the need for alliances and collaboration and finally a Strengths/Weaknesses/Opportunities/Threats (SWOT) analysis. Step 2) Build economic models describing the cost and revenue projections for the innovations, this allows profit margins to be calculated and business sensitivities to be identified. This action will draw on information produced in Task8.2. Estimating revenue potential draws on the knowledge gathered in the development of the value proposition. The economic model will allow understanding of future cash flows and highlight the need for additional funding or external investment to reach the first sales.

Action 2. Business Model Validation. A two-step process will be used to validate the business model. Step 1) To ensure valuable feedback, careful stakeholder identification and recruitment is required. In collaboration with project partners, a targeted list based on stakeholder profiles will be drawn-up. The target list will ensure appropriate value chain coverage with an emphasis on the potential to become early innovation adopters. Stakeholders will be recruited through an initial contact, introduction of the value proposition and request for initial feedback, recruitment will draw on the stakeholder database developed for dissemination activities. Step 2) Feedback on the business model will be captured through interviews, questionnaires and, if relevant, visits to the experimental demonstration sites, depending on the targets' profiles. The feedback analysis will lead to a review and update of the value proposition and revenue/cost models within the business models.

Business Plan: The developed business models will be detailed in a business plan that will be the central part of the exploitation plan. Business Plan will be developed as a sustainability strategy outlining what should happen to the project outputs at the end of the project, and to explore how they can be sustained. The business plan will include client definition, a strategy for marketing, sales and operations financial models and go-to market strategies.

New business opportunities will be assesed during the drafting of the different business models and the business plan. The new value chains (three at least) will be analysed from a business approach along with the real opportunities of penetration in the market from MWS as raw material. The impact on the actual industrial sectors where PERCAL is envisaged to be introduced (See Section 2.1) will be determined and compared to current industries practises. This activity, in alignment with the exploitation activities, will set the most approapiate route for the results: licensing, marketing/direct sales or vending. The analysis of the new business opportunities will be performed taking into consideration these options for the consortium members as well as other biobased stakeholders which might be involved in the new value chains.

Task 8.5 Intellectual Property Management. Task leader: EXERGY with the support of all partners.
This task will undertake a regular review and assessment of the consortiums 'Freedom to Operate' in the areas of exploitable outputs. Reviewing 'Freedom to Operate' throughout the project will ensure that the consortium is safe to produce a product or use a production technology without infringing or violating third party intellectual property rights or highlight the need for mitigating actions by the consortium (eg. acquisition of license, which would impact the exploitation).

Furthermore, IPR management will include the mapping of existing patents and potentially overlapping IPR; and when the developing partner identifies possible novelty in a KER being developed, an assessment of the new knowledge generated during the project (outputs and other knowledge such as data); a proposition of optimal IPR protection options (e.g. patent, copyright, trademark, confidentiality), in line with the selected business model options (Task 8.4) and taking into account possible co-ownership; the proper implementation of the IPR protection measures during the project. The findings and recommendations for IPR protections will be included in the final exploitation plan (D8.3). EXERGY will lead the IPR management by providing guidance to the partners for their FTO and patentability search but each partner will be responsible of the right management of his result. When necessary, after the initial identification of a promising (patentable) idea, external professional support from a patent agent will be sought.

Task 8.6 Technology Transfer. Task Leader: EXERGY with the support of all partners.
Technology transfer will target the wide range of production companies in the interest areas of the project e.g. protein/ lipid separation and chemical building blocks for a wide range industrial applications. The activities planned to achieve a successful technology transfer are: a) Workshops/seminars oriented to different sectors (microbiology, green chemistry, engineering, bioprocesses, biorefinery, etc.). The two workshops will be organised in M24 and M36, in two different European countries (TBD), in cooperation with Task 8.2. The subject will be selected at M18 and M30, depending on the intermediate results; b) Training at companies' facilities. Technology transfer will be facilitated through the arrangement of 'short stay' training courses designed for scientists/Technicians to take place at partner companies' facilities; c) Technical Guidance Document: A 'Guide to the production and use of green chemical intermediates using MSW as a main raw material' will be prepared and published after completion of the project, in order to contribute to the technology transfer, dissemination and training. According to best industrial and environmental practices, all partners will co-operate in the elaboration of this Guide, which will contribute to an optimal profit of the new chemical products advantages. This guide will be developed by all Partners (D8.8) and will include the safety, regulatory, economic and environmental issues identified in WP8, and will be made public in the second Project Workshop (M36); d) Publishing scientific articles in journals and conferences (subject to the consortium's IP protection publication policy) and detailed in table 2.2.c. (section 2.2); e) Development of policy briefs and case studies targeting policy makers responsible for drafting EU or national legislation, regulations and guidelines affecting MSW based products valorisation, including sustainability guidelines and standards. This will also be fuelled by the results of task 8.5 in terms of influence of current policies on the market deployment of the project results (eg. barrier or gaps).

Task 8.7 Set up of PERCAL Stakeholders' Community of Interest. Task Leader: IMECAL with the support of all partners.
This task aims at tying up a critical mass of stakeholders to i) amplify the impacts of PERCAL at EU and international level by involving them in project activities performed in previous tasks ii) understand their specific market demands and properly address to their needs. This implies the execution of a stakeholder analysis (identify and prioritize who to mobilise and engage with) approaching key actors across all stages of the value chain. The analysis will be based on a 3-step process: i) Identify additinonal Stakeholders ii) Create a Stakeholder Map iii) Create a Stakeholder Management Strategy considering their knowledge, interests, positions, alliances, relationships, key roles and responsibilities. Target stakeholders will include chemical companies, MSW managers companies, policy makers, industrial associations, companies/reseach centres/universities that are leading projects with the same target (identified in task 7.1) and potential users of target chemical intermediates.
Measures for establishing and maintaining stakeholder liaisons include: 1) Bilateral and multilateral exchange of information, participation in roundtables, open days, action groups. 2) Open stakeholder consultation by means of the internet; submitting surveys and interviews in order to gather relevant information to feed into subsequent activities of this WP. The stakeholders' community will be approached by providing them all the available knowledge on PERCAL and other potential paralel routes of research identified during the project.

Participation per Partner	
Partner number and short name	**WP8 effort**
1 - IMECAL	

2 - AIMPLAS	
3 - ATB	
Partner number and short name	**WP8 effort**
4 - AUA	
5 - UGent	
6 - CENER	
7 - TBWR	
8 - VISUM	
10 - HAYAT	
12 - EXERGY	
Total	

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type**[15]	**Dissemination level**[16]	**Due Date (in months)**[17]
D8.1	Project web site	2 - AIMPLAS	Report	Public	3
D8.2	Initial Communication and dissemination Plan	2 - AIMPLAS	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D8.3	Draft communication activities & dissemination Plan updates	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	18
D8.4	IPR policy implementation in line with Exploitation Plan	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	6
D8.5	Report of the IPR management activities	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	36

D8.6	Final Exploitation Plan including Business Plans of key results	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	36
D8.7	Technology Transfer Activities, including training to SMEs, other activities and workshops from M1 to M18	12 - EXERGY	Report	Public	18

List of deliverables

Deliverable Number[14]	**Deliverable Title**	**Lead beneficiary**	**Type[15]**	**Dissemination level[16]**	**Due Date (in months)[17]**
D8.8	Guide to explore the conversion routes of MSW in high added value products.	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	34
D8.10	Technology Transfer Activities, including training to SMEs, other activities and workshops from M18 to M36	12 - EXERGY	Report	Public	36
D8.11	Final communication activities & dissemination Plan updates	12 - EXERGY	Report	Confidential, only for members of the consortium (including the Commission Services)	36

Description of deliverables

D8.1 Project web site.
D8.2 Initial Communication and dissemination Plan.
D8.3 Report on performed communication activities and dissemination Plan updates.
D8.4 IPR policy implementation in line with the Exploitation Plan.
D8.5 Report of the IPR management activities.
D8.6 Final Exploitation Plan including Business Plans of key results.
D8.7 Report of Technology Transfer Activities, including training to SMEs, other activities and workshops.
D8.8 Guide to explore the conversion routes of MSW in high added value products.
D8.10 Technology Transfer Activities, including training to SMEs, other activities and workshops from M18 to M36
D8.11 Final communication activities & dissemination Plan updates

D8.1 : Project web site [3]

Project web site

D8.2 : Initial Communication and dissemination Plan [6]

Initial Communication and dissemination Plan

D8.3 : Draft communication activities & dissemination Plan updates [18]

Draft communication activities & dissemination Plan updates

D8.4 : IPR policy implementation in line with Exploitation Plan [6]

IPR policy implementation in line with Exploitation Plan

D8.5 : Report of the IPR management activities [36]

Report of the IPR management activities

D8.6 : Final Exploitation Plan including Business Plans of key results [36]

Final Exploitation Plan including Business Plans of key results

D8.7 : Technology Transfer Activities, including training to SMEs, other activities and workshops from M1 to M18 [18]

Technology Transfer Activities, including training to SMEs, other activities and workshops from M1 to M18

D8.8 : Guide to explore the conversion routes of MSW in high added value products. [34]
Guide to explore the conversion routes of MSW in high added value products.

D8.10 : Technology Transfer Activities, including training to SMEs, other activities and workshops from M18 to M36 [36]

Technology Transfer Activities, including training to SMEs, other activities and workshops from M18 to M36

D8.11 : Final communication activities & dissemination Plan updates [36]

Final communication activities & dissemination Plan updates

Schedule of relevant Milestones

Milestone number[18]	**Milestone title**	**Lead beneficiary**	**Due Date (in months)**	**Means of verification**
MS18	Project website running.	2 - AIMPLAS	6	Website uploaded content & running.
MS19	Advanced exploitation strategy defined.	12 - EXERGY	18	Previous to issue the draft PEDR, the exploitable results are completely agreed, as well as co-ownership and preliminary business plans for each result outlined.

MS20	Final Business/ Exploitation & Communication Plans agreed.	12 - EXERGY	36	Agreement on final Business and Communication Plans.
MS21	Up to eight peer-reviewed papers and two patents ready to be submitted.	12 - EXERGY	36	Paper submitted to different journals and patents prepared to be submitted.
MS22	Two workshops successfully organized.	12 - EXERGY	36	Assistance of 50 firms in each one.

Work package number [9]	WP9	**Lead beneficiary** [10]	1 - IMECAL
Work package title	Project Management		
Start month	1	**End month**	36

Objectives

Overall project co-ordination and management. To optimize the resources and to ensure that all BBI JU requirements for communication and reporting are met. Specific objectives:

- Setting up of Project Management and Administration: Definition of Management Procedures; development ofprocedures, rules and templates for the consortium reporting activity, management platform through web-based project intranet.
- Management Co-ordination of the Project: Administrative and financial project management; legal representation ofthe project.
- Project Administration, control and support activities: Co-ordination and development activity reporting; Requestingand processing cost statements and cost certificates; Follow-up the progress of the project; Provide generic support at project level.
- Project Communication: Define, provide, set-up and maintain the necessary communication procedures and toolsbetween all the partners of the project to ensure an efficient communication flow. Communication with the BBI JU representatives.

Risk management: to carry out a continuous (ongoing) assessment of the progress of the project. To review the progress of the work in relation to the defined success measures and to report on this progress. Where appropriate, contingency plans will be identified and implemented.

Description of work and role of partners

WP9 - Project Management [Months: 1-36]
IMECAL, AIMPLAS, ATB, AUA, UGent, CENER, TBWR, VISUM, COVESTRO, EXERGY
Task 9.1 Management of the Resources. Task Leader: IMECAL Involved partners: All WP leaders.
The project will be managed on a daily basis by the lead partner IMECAL with the support of AIMPLAS. Full consortium project management meetings will be held every quarter to discuss technical progress and contractual issues (conference calls). Project management duties will include a dedicated project intranet which will act as a project management and communication tool and will include a password protected area for the sharing of meeting arrangements, minutes, reports and financial statements.
The Project Coordinator has the responsibility of managing the project cost effectively, including day to day management aspects, distribution of work within project partners, assessment of results, risk management, etc. Due to the number of partners involved (12) and the number of Member States (9), the management effort required should not be underestimated, requiring of an efficient and well-designed management structure. The project management structure, i.e. Project Coordinator, General Assembly (GA) and Executive Board (EB) is fully described in section 3.2. The General Assembly, representing the voice of the whole consortium, will form the highest decision making body. The EB will form the decision-making and arbitration body (on behalf of the GA) and will be responsible for the overall strategic direction of the project, in order to ensure that the project progresses properly and the risks are properly identified and managed. Intellectual/Industrial Property Rights (IPR) are managed by the IPR & Exploitation Board (IPREB), whose responsibilities are also described in section 3.2. In all these boards, the membership ensures that there is a strong industrial leadership in all Project activities. The TMC, formed by the WP leaders, who are in turn responsible for the coordination of all the activities performed by each participant within the work package and for on time delivery

of their specific tasks and the subsequent report of problems or special issues, will assess the technical progress of the Project reporting to the EB and the GA.
Management Co-ordination. The Project Co-Coordinator is IMECAL, supported by AIMPLAS, who will assume the administrative management of the project, assuring the project's progress. The Coordinator supports the Project Consortium in legal, administrative and financial issues. This co-ordination will lead the strategic, political, economic and operational of the project, articulated through the different Boards. AIMPLAS will thus be in charge of the preparation, organization, administration, minuting and follow up of the main project meetings (GA, EB, IPREB, TMC) . The tasks are fully detailed in section 3.2.2.
Project administration. Progress of each task will be assessed every three months to closely control the development activities of the partners in order to ensure that there is no deviation from the original plan or that corrective actions are taken in case of deviations. Financial control will be performed for the official reporting periods (requesting and processing the cost statements and the necessary cost certificates) as well as periodic Intermediate internal financial controls. It is foreseen to send to the BBI JU, an official progress report (M15) and a final report in M36. Details about the methods for monitoring and reporting progress, and documenting results, are provided in Section 3.2. An internal project web site will be created at the beginning of the project, in order to facilitate the monitoring and reporting of progress and documenting of results (see WP9). The website will also allow interactive internal communication between the partners involved in the different tasks. An overall description of all the Committees, their membership and procedures is provided in section 3.2.

Task 9.2 Risk management. Task Leader: IMECAL. Involved partners: TMC Members
Based on the initial risk identification reported in table 3.1.c., at the beginning of the project and for each work package, the risk indicators will be defined taking into account the overall project progress, and will be reported in MS23 and D9.1 Each WP leader will collect the risk indicators and will monitor technical progress at WP level along the project implementation. Progress of each task will be assessed every 3 months to control the developed activities of the partners in order to ensure that there is no deviation from the original plan or that corrective actions are taken when appropriate. The TMC and EB will review and assess the progress of each WP in relation to the identified risk indicators and assist the WP leaders. When appropriate, contingency plans and countermeasures will be developed and when approved by the EB, will be discussed with the BBI JU Scientific Officer. If approved, they will be implemented. The progress will be reported periodically, in the contractual reports (month 15, 36) and in the internal intermediate progress reports, where deviations from the work programme objectives (Deliverables and Milestones) are identified, they will be addressed and corrective actions/plans and countermeasures implemented to ensure that the Project reaches its planned outcomes.

Task 9.3. Communication with BBI JU. Task Leader: IMECAL.
The coordinator is the sole point of contact between the BBI JU and the project consortium. IMECAL will ensure that all milestones are met and that all reports are prepared in accordance with the contractual requirements. A Milestone report will be issued and sent to the BBI JU describing the Milestone completion and its implication in the project progress. The structure for monitoring and reporting progress will consist of a series of reports and meetings. Technical reports will be sent to the BBI JU to meet the reporting pattern expected. Documentation will be in English and in Microsoft Word format. All reports will be assembled by the Project Coordinator in a master file, which will be uploaded to the project intranet. Review procedures will be agreed and included in the Consortium Agreement to ensure that all the project deliverables reach a satisfactory quality standard (e.g. technical deliverables will be assessed by the TMC or certain appointed members and the Project Coordinator prior to submitting them to the EC).

Participation per Partner	
Partner number and short name	**WP9 effort**
1 - IMECAL	
2 - AIMPLAS	
3 - ATB	
4 - AUA	
5 - UGent	

6 - CENER	
7 - TBWR	
8 - VISUM	
11 - COVESTRO	
12 - EXERGY	
Total	

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D9.1	Risk indicators and Risk assessment	1 - IMECAL	Report	Confidential, only for members of the consortium (including the Commission Services)	9

Description of deliverables

D9.1 Risk indicators and Risk assessment

D9.1 : Risk indicators and Risk assessment [9]
Risk indicators and Risk assessment

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification
MS23	WPs risk indicators identified.	1 - IMECAL	3	Risk indicators for all WPs identified in order to start Risk Management procedures.

Work package number [9]	WP10	**Lead beneficiary** [10]	1 - IMECAL
Work package title	Ethics requirements		
Start month	1	**End month**	36

Objectives

The objective is to ensure compliance with the 'ethics requirements' set out in this work package.

Description of work and role of partners

WP10 - Ethics requirements [Months: 1-36]
IMECAL
This work package sets out the 'ethics requirements' that the project must comply with.

List of deliverables

Deliverable Number[14]	Deliverable Title	Lead beneficiary	Type[15]	Dissemination level[16]	Due Date (in months)[17]
D10.1	NEC - Requirement No. 1	1 - IMECAL	Ethics	Confidential, only for members of the consortium (including the Commission Services)	1

Description of deliverables

The 'ethics requirements' that the project must comply with are included as deliverables in this work package.

D10.1 : NEC - Requirement No. 1 [1]

Partner 10: Hayat Kimya San. A.S. is based in Turkey. As such, the applicants should confirm to the BBI that the ethical standards and guidelines of Horizon2020 will be rigorously applied, regardless of the country in which the research is carried out. It is recommended that the applicants should ensure that appropriate health and safety procedures conforming to relevant local/national guidelines/legislation are followed for staff involved in this project.

Schedule of relevant Milestones

Milestone number[18]	Milestone title	Lead beneficiary	Due Date (in months)	Means of verification

1.3.4. WT4 List of milestones

Milestone number[18]	Milestone title	WP number[9]	Lead beneficiary	Due Date (in months)[17]	Means of verification
MS1	Composition of MSW available	WP1	1 - IMECAL	14	Comparison with reported values in literature for similar feedstocks.
MS2	Improved enzymatic hydrolysis.	WP1	1 - IMECAL	16	Comparison of hydrolysis yields
MS3	Samples of hydrolysed MSW available for the different subsequent tasks	WP1	1 - IMECAL	18	Sufficient amounts of different fractions for lab tests within WP2, WP3 and WP4
MS4	LA with purity higher than 97% for L-LA .	WP2	3 - ATB	22	Application of chiral HPLC to estimate the optical purity of all LA samples
MS5	Yield at pilot plant facilities similar to other feedstocks already tested.	WP2	3 - ATB	30	Max. 10% losses compared to current process using pure lignocellulosic biomass feedstock.

MS6	Production of succinic acid in fed-batch and continuous processes	WP3	4 - AUA	18	Report on lab-scale experiments and comparison with preliminary experiments with the integrated system.
MS7	Extraction of succinic acid directly from the extraction cell at 99% purity.	WP3	4 - AUA	24	Report on the purity of succinic acid crystals extracted directly from the concentrated solution produced via membrane electrolysis.
MS8	A method for the production of amino acid and lipid surfactants from MSW by-products.	WP4	6 - CENER	32	Comparison of surface active properties with those described in literature.
MS9	Biosurfactants has fulfil the HAYAT internal standards.	WP4	10 - HAYAT	34	Comparison of bio-surfactant properties with commercial ones using DIN/EN/ISO standard.
MS10	Simultaneous esterification & products separation (reactive distillation) for ethyl lactate established.	WP5	7 - TBWR	34	Continuous mode reactive distillation established and validated at scale 3-5 L/h.
MS11	Renewable HMA has fulfil the end-user requirements or YPR internal standards.	WP5	2 - AIMPLAS	34	Comparison of HMA main properties with commercial EVA and renewable source HMA using the suitable EN/ ISO.

Milestone number[18]	**Milestone title**	**WP number[9]**	**Lead beneficiary**	**Due Date (in months)[17]**	**Means of verification**
MS12	Polyester-Polyols by conversion of SA with diamines have similar reactivity than using commercial bio-based SA (less than 20% time increase) and the PU formation obtained have at least 80% of performance than commercial ones.	WP5	11 - COVESTRO	34	Comparative analysis using internal and well-defined methodology of COVESTRO.
MS13	Main process parameters identified.	WP6	1 - IMECAL	18	Study of a versatile plant for bioethanol, LA and SA production.

MS14	Monitoring system built and tested.	WP6	8 - VISUM	30	The sensors and control system will be available for lab validation and suitable results will be reported.
MS15	Environmental performance of the new materials and processes according to standards and EU legislation in force.	WP7	2 - AIMPLAS	36	Check the environmental regulatory assessments that affect the product.
MS16	Final cost of intermediates and products at least 15% cheaper than current biobased market analogues.	WP7	12 - EXERGY	34	Final costs of the products taking into account all process steps.
MS17	The new biobased products and intermediates fulfil the related regulatory requirements.	WP7	1 - IMECAL	36	Check the regulatory assessments that affect the product.
MS18	Project website running.	WP8	2 - AIMPLAS	6	Website uploaded content & running.
MS19	Advanced exploitation strategy defined.	WP8	12 - EXERGY	18	Previous to issue the draft PEDR, the exploitable results are completely agreed, as well as co-ownership and preliminary business plans for each result outlined.
MS20	Final Business/ Exploitation & Communication Plans agreed.	WP8	12 - EXERGY	36	Agreement on final Business and Communication Plans.

Milestone number[18]	**Milestone title**	**WP number[9]**	**Lead beneficiary**	**Due Date (in months)[17]**	**Means of verification**
MS21	Up to eight peerreviewed papers and two patents ready to be submitted.	WP8	12 - EXERGY	36	Paper submitted to different journals and patents prepared to be submitted.
MS22	Two workshops successfully organized.	WP8	12 - EXERGY	36	Assistance of 50 firms in each one.
MS23	WPs risk indicators identified.	WP9	1 - IMECAL	3	Risk indicators for all WPs identified in order to start Risk Management procedures.

1.3.5. WT5 Critical Implementation risks and mitigation actions

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
1	Failure to accomplish deadlines and BBI JU procedures	WP1, WP10, WP2, WP3, WP4, WP5, WP6, WP7, WP8, WP9	Risk: Delays Mitigation: Experienced Coordinator that will keep continuously updated about new procedures.
2	IPR/innovation conflicts	WP8	Risk: lack of cooperation, delays Mitigation: a solid Consortium Agreement will be signed before accessing to the G.A, which will include clear rules, Background included/excluded, clear IPR ownership & consensus and conflict resolution mechanisms through the IPREB, external arbitration if necessary.
3	Partners not committed, under-performing, lack capabilities or leaving	WP9	Risk: Work not done, delays Mitigation: The Partners have a strong network of contacts that can replace such a partner and the RO have broad capabilities to fill some temporary gaps.
4	Insufficient cooperation among Partners	WP9	Risk: Work not done, delays Mitigation: frequent meetings to keep partners connected. Strong and proven Organizational structure to ensure workflow.
5	High variability in the feedstock composition (M, 60%)	WP1	This variability has been taken into account in the strategies proposed in WP1 and mitigated already by collecting feedstock from diverse but related provenances. Additionally, a representative feedstock with a reasonable content of all the fractions required for further upgrading can be selected. Based in the composition the enzymatic hydrolisys cocktail will be adapted.
6	Interferences of fractions for the recovery of others (M, 30%)	WP1	To interact with WP4 to optimize the procedures.
7	Production of toxic compounds for further fermentation as a result of pre-treatment (M, 30%)	WP1	To implement detoxify mechanisms, depending on the target compounds. Select less agressive pretreatments.
8	Undesired product formation (M, 40%)	WP1	To add industrial antibiotics if required (this is not possible if the objective is bacterial fermentation of the substrate, but for yeast transformation it is recommended). To add industrial inhibitors to undesired product formation. To improve the fermentation conditions for required product formation
9	Inability of microbial strains to efficiently utilize all the sugars in MSW feedstock	WP2, WP3	To do all operations with the highest cleaning and sterilization measures.

10	Low titters and/or productivities of the fermentation products on MSW (M, 50 %)	WP2, WP3	Iteration with WP1 to improve pre-treatment. If not sufficient, optimize the medium further, e.g., by supplementing the MSW feedstock with fractions rich in minerals and vitamins. Consider other host strains.

Risk number	Description of risk	WP Number	Proposed risk-mitigation measures
11	Fermentation results are not always reproducible due to variations in the composition of MSW (medium, 50%)	WP2	Identify optimum nutrient requirements and supplement fermentation media with commercial sources of nutrients when there is lack of specific nutrients from crude fermentation media
12	Low performance (yield, productivity, titer etc.) of screened cultures in waste hydrolysates (low to medium, 25%)	WP2	Identify the biowaste fractions that lead to the required fermentation efficiency and optimise fermentations using only these fractions. Use alternative strains.
13	The presence of inhibitors may reduce fermentation efficiency (low, 15 %)	WP2	Adapt additional steps of purification during upstream processing, select more tolerant strains or improve existing strains
14	The LA fermentation broth contains significant impurities originally coming from the MSW feedstock (such as protein/ polypeptides, amino acids, high amount of ashes, other organic acids such as acetic acid (high 50%, expected technical challenge)	WP2	The option to encounter this shortcomings are: (1) Adapt the downstream processing DSP to handle the the composition of the final fermentation broth by applying combinations of different DSP unit operations such as membranes (ultra and nano filtration), electro dialyses, ion exchange and recover a high quality LA product. (2) Evaluate and adopt the feedstock supply chain and first processing to insure impurities in feedstock are kept at an acceptable level for final end product; (3) Remove potential acetic acid in the fermentation broth by stripping; (4) Use the esterification as an actual cleaning process for LA to deliver a high quality end product to the market.
15	Sugar to succinic acid conversion yield is not improved with the integrated fermentation membrane electrolysis system (high 50%, expected technical challenge).	WP3	The expected increase in succinic acid concentration and productivity will suffice for the development of an efficient process
16	Other impurities are present in the succinic acid concentrated solution derived via membrane electrolysis (high, 30%)	WP3	A downstream protocol will be developed based on the impurities present in the concentrated solution

17	Excessive fouling of in situ membrane (Medium, 40%)	WP3	Different membranes and operational conditions will be tested, alongside operational regimes that can minimise fouling (e.g high cross-flow velocity, regular cleaning).
18	Negative impact of H2 on the fermenting organisms due to a shift in partial pressure and available CO2 (High, 50%)	WP3	H2 is expected to have a positive impact on the fermentation, but this remains a hypothesis to be tested. If the H2 strips too much CO2, an increased flow of CO2 may be required.
19	Low yield in the biosurfactants condensation reactions (Medium, 40%)	WP4	Different reaction approaches will be tested. Previous purification of some aminoacidos and lipids can be implemented.

Risk number	**Description of risk**	**WP Number**	**Proposed risk-mitigation measures**
20	Too complex mixture of biosurfactants (M, 50%)	WP4	Purification of the most interesting aminoacids and lipids in advance in order to perform the condensation reaction with pure species.
21	Impurities in the product LA can potentially interact or harm catalyst for reactive distillation process (low < 10%)	WP5	Different solid catalysts will be tested and evaluated
22	Esterification rate in the reactive distillation process is insufficient and final ethyl lactate product contains significant LA amounts. (Medium < 25%)	WP5	Process parameters need adaption, especially increase average resident time within the reactive part of the distillation column; change feed parameters; increase reactive zone in column; in worst case redesign/change geometry by increasing total height of column. Alternatively the process pathway may be extended by adding an additional distillation process to separate LA from ethyllactate to insure final end product.
23	Water amount within the reactor (coming from LA and ethanol raw material) plus water produced by esterification is getting to high to achieve high efficiencies (High, 50%, expected technical challenge)	WP5	The process needs to be specifically designed, and operated to handle the water in the distillation; options: (i) change the process parameters (temperature, feed rate…) decrease water content in raw materials (use only 96% ethanol and high concentrate LA product >80%); increase thermal input, change boiler; in worst case redesign column
24	Thermo-mechanical resistance of LA hotmelts is not enough for further processing Reduction in biopolymer properties, bad smell. (High, 30 %).	WP5	Different compatible antioxidants, lubricants and plasticisers will be tested to reduce the action of compounding extruder over the material. Use of low shear screw elements and minimize residence time thanks to the modular configuration of multi screw extruders.

25	Impurities in the product SA can potentially interact or harm catalyst for production of polyester polyols or other building blocks (H, 20%)	WP5	Covestro pilot plant have additional technologies to improve monomers purity, some of them will be studied/adapted to the new biobased SA. Alternative catalyzers will be proposed and evaluated.
26	Parameters of interest are difficult to study and monitor (M, 50%).	WP6	To this end a special effort in technology screening and preliminary tests will be done in order to select the most appropriate sensor technology.
27	Sensor development and testing taking a long time to achieve stable and satisfactory results (M, 40%).	WP6	Early stage sensor specification based on process requirements will be done and an iterative approach will be taken when starting test definition and sensor design in order to correct and foresee deviations.
28	Low versatility in the Process plant adaptation for Bioehanol, LA, SA production (L, 30%).	WP6	Study the best techno-economical case in each geographic situation having in account the biogenic fraction composition of each MSW.

Risk number	**Description of risk**	**WP Number**	**Proposed risk-mitigation measures**
29	PERCAL's products might be relatively expensive (M, 40%)	WP7	To revaluate and improve the production process to minimize energy consume or look for alternative and cheaper energy source. Look for alternative catalysts that improve yield or reduce energy consume.
30	LCA or Energy consume is higher than current process (H, 10%)	WP7	A comprehensive assessment methodology will be developed early in the project lifetime. If the target results cannot be achieved after the demonstration phase, strategies for further improvements will be developed and optimisation solutions will be applied. Modelling and simulation activities to be carried out after the demonstration phase will enable the consortium to choose the best and most efficient scenario. This will include the use of Aspen Energy Analyzer and exergy analysis as tools to optimise the energy consumption (heat integration) and to potentially reduce GHG and CO2 emissions.
31	Simulation software is insufficient to simulate, scale-up and model some parts of the process or even the entire process (M, 10%)	WP7	A deep research of the most accurate thermodynamic models for the simulation of the process will be carried out regarding each part of the process and the components involved in it. If not sufficient yet, a mathematic modelling will be developed based on the specification of the process and the experimental information by means of specialised software (e.g Matlab, Simulink…), especially where biologic processes are involved. The model will be then integrated into the AspenPlus or the required simulation software to have a unique final simulation flowsheet.

32	Lack of interest in the PERCAL outcomes (M, 50%)	WP8	Implementation of dissemination and exploitation activities focussing on the advantages of PERCAL benefits will be carried out.

1.3.6. WT6 Summary of project effort in person-months

	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WP8	WP9	WP10	Total Person/Months per Participant
1 - IMECAL											
2 - AIMPLAS											
3 - ATB											
4 - AUA											
5 - UGent											
6 - CENER											
7 - TBWR											
8 - VISUM											
9 - SOFTER											
10 - HAYAT											
11 - COVESTRO											
12 - EXERGY											
13 - YPR											
Total Person/Months											

1.3.7. WT7 Tentative schedule of project reviews

Review number [19]	Tentative timing	Planned venue of review	Comments, if any
RV1	18	venue tbc, probably BBI JU offices in Brussels	This review will be based on the documents submitted within the period in question: periodic technical report, financial statements, and all deliverables due in the reporting period plus possible other documents (e.g. needed for the interim impact evaluation). The reference document for the progress assessment is the GA and especially Annex I (Description of Action, DoA)

1. Project number

The project number has been assigned by the Commission as the unique identifier for your project. It cannot be changed. The project number **should appear on each page of the grant agreement preparation documents (part A and part B)** to prevent errors during its handling.

2. Project acronym

Use the project acronym as given in the submitted proposal. It can generally not be changed. The same acronym **should appear on each page of the grant agreement preparation documents (part A and part B)** to prevent errors during its handling.

3. Project title

Use the title (preferably no longer than 200 characters) as indicated in the submitted proposal. Minor corrections are possible if agreed during the preparation of the grant agreement.

4. Starting date

Unless a specific (fixed) starting date is duly justified and agreed upon during the preparation of the Grant Agreement, the project will start on the first day of the month following the entry into force of the Grant Agreement (NB : entry into force = signature by the JU). Please note that if a fixed starting date is used, you will be required to provide a written justification. **5. Duration**

Insert the duration of the project in full months.

6. Call (part) identifier

The Call (part) identifier is the reference number given in the call or part of the call you were addressing, as indicated in the publication of the call in the Official Journal of the European Union. You have to use the identifier given by the Commission in the letter inviting to prepare the grant agreement.

7. Abstract

8. Project Entry Month

The month at which the participant joined the consortium, month 1 marking the start date of the project, and all other start dates being relative to this start date.

9. Work Package number

Work package number: WP1, WP2, WP3, ..., WPn

10. Lead beneficiary

This must be one of the beneficiaries in the grant (not a third party) - Number of the beneficiary leading the work in this work package

11. Person-months per work package

The total number of person-months allocated to each work package.

12. Start month

Relative start date for the work in the specific work packages, month 1 marking the start date of the project, and all other start dates being relative to this start date.

13. End month

Relative end date, month 1 marking the start date of the project, and all end dates being relative to this start date.

14. Deliverable number

Deliverable numbers: D1 - Dn

15. Type

Please indicate the type of the deliverable using one of the following codes:

R Document, report
DEM Demonstrator, pilot, prototype DEC
Websites, patent fillings, videos, etc.
OTHER
ETHICS Ethics requirement
ORDP Open Research Data Pilot

16. Dissemination level

Please indicate the dissemination level using one of the following codes:

PU Public
CO Confidential, only for members of the consortium (including the Commission Services)
EU-RES Classified Information: RESTREINT UE (Commission Decision 2005/444/EC)
EU-CON Classified Information: CONFIDENTIEL UE (Commission Decision 2005/444/EC) EU-SEC Classified Information: SECRET UE (Commission Decision 2005/444/EC)

17. Delivery date for Deliverable

Month in which the deliverables will be available, month 1 marking the start date of the project, and all delivery dates being relative to this start date.

18. Milestone number

Milestone number:MS1, MS2, ..., MSn

19. Review number

Review number: RV1, RV2, ..., RVn

20. Installation Number

Number progressively the installations of a same infrastructure. An installation is a part of an infrastructure that could be used independently from the rest.

21. Installation country

Code of the country where the installation is located or IO if the access provider (the beneficiary or linked third party) is an international organization, an ERIC or a similar legal entity.

22. Type of access

VA if virtual access,
TA-uc if trans-national access with access costs declared on the basis of unit cost,
TA-ac if trans-national access with access costs declared as actual costs, and
TA-cb if trans-national access with access costs declared as a combination of actual costs and costs on the basis of unit cost.

23. Access costs

Cost of the access provided under the project. For virtual access fill only the second column. For trans-national access fill one of the two columns or both according to the way access costs are declared. Trans-national access costs on the basis of unit cost will result from the unit cost by the quantity of access to be provided.

4 MEMBERS OF THE CONSORTIUM

4.1 PARTICIPANTS (APPLICANTS)

PARTNER 1. Coordinator **Name**: Industrias Mecanicas Alcudia S.A. (IMECAL) **Web page**: www.imecal.com/perseo	**Type:** SME **Country:** Spain	

Brief description of the organization: IMECAL is a SME which was founded in 1979 as an engineering company in the metal sector, working on demand and manufacturing all type of metal-mechanical equipment for civil works, petro-refineries and petrochemical, transportation, refrigeration industry, industrial auxiliary motor and ceramics industry, among others. R & D IMECAL Department has allowed the company to progress in the technological field, having made significant partnerships with companies, universities and public research organizations. The R&D line of production technology for second generation bioethanol, has developed a biotechnological patented technology for obtaining bioethanol from Organic Municipal Waste. In 2007 IMECAL built a second generation bioethanol plant with a capacity to process 25 ton/day of organic fraction of MSW (PERSEO Bioethanol®). Since this year IMECAL has participated in different projects to test and improve PERSEO technology to different MSW raw materials with different companies. IMECAL is member of the Biofuels TP Platform-European Union, and participates actively in several national and international conferences on biofuels.

Role in the Project: Project coordinator. In WP1 they work in the selection and analysis of the most representative MSW. Develop in collaboration with NOVO the new enzyme cocktails to increase the amount of hydrolysed sugars. Development a biosurfactants from the protein and lipid fraction of MSW. Evaluate with CENER, EXERGY and VISUM the best technologies to integrate the new products (LA, SA and lipid/protein) in the PERSEO technology plant. Contribution to communication, dissemination, exploitation and other related activities.

Scientific team:

Ms Caterina Coll Lozano, PMP® (F), Chemical engineer by the Higher Technical School of Industrial Engineers of the Polytechnic University of Valencia. Project Management Profesional (PMP®) certificate by Project Management Institute (PMI). Executive MBA by Bussines & Marketing School ESIC. Master degree in environment coursed in the Department of Chemical and Nuclear Engineering from the Polytechnic University of Valencia. Technical Industrial engineer in Chemist by the Technical School of Industrial Engineering of the Polytechnic University of Valencia. Stay in the "School of Science" of "Institute of Technology Sligo" in Sligo (Ireland). Stay on the Biomass Unit of the Division of Renewable Energy from the department of Energy of CIEMAT. Professional experience of over 15 years in process engineering and environment, mainly developed in CEAM (Centro de Estudios Ambientales del Mediterraneo) as Head of laboratory and environmental projects, and in IMECAL as Plant Manager of second generation bioethanol pilot plant PERSEO, and R&D Manager. Over the last years he has coordinated and participated in several projects at national and international level and performed scientific and technical publications in various journals, books and conference contributions.

Dr. Ing. Marcos Latorre Sánchez (M), Chemical Engineer by the Higher Technical School of Industrial Engineers of the Polytechnic University of Valencia. Masters in Design for Sustainability from Cranfield University (England). PhD in Green Chemistry at Institute of Chemical Technology (Polytechnic University of Valencia - Spanish National Research Council). Stay at Max Planck Institute of Colloids and Interfaces (Germany). He has more than 6 years of experience in new materials development at laboratory and pilot scale and advanced characterization for renewable energy systems. He has participated in a number of R&D nationally and internationally funded projects aimed at the valorisation of biomass into higher valuable products. He is author of 10 papers in peer-reviewed journals and has participated in different international conferences.

Ms María Frau Martínez (F), Technical Agricultural Engineering by the Polytechnic University of Valencia. International Welding Specialist (IWS) by CESOL. Senior technician in occupational risk prevention in the fields of occupational safety (CIEGS, Polytechnic University of Valencia), Ergonomics and Applied Psychology (IMF) and Health at Work (IMF). Extensive experience in handling graphic design programs like Auto-Cad, Inventor, Solid Works and Lantek among others. Professional experience in process engineering in Bonduelle Iberica SAU (Frudesa) and currently as a mechanical and process engineering in IMECAL, where performs mechanical and industrial design projects.

Mr Jaime Mesa Arguello (M), Industrial Engineer by the Polytechnic University of Valencia. Technical specialist in engineering design with CAD Software, manufacturing and inspection of pressurized tanks. Professional experience in R&D projects development and process engineering at IMECAL, where performs mechanical and industrial design projects.

Mr Juan Manuel Ferri Palop (M), Telecommunications technical engineer (Electronic Systems specialty) by the Polytechnic University of Valencia and Fachhochschule Ulm - Hochschule für Technik (Germany). Technical specialist in electronic systems, computer design and modelling programs. Professional experience in R&D projects development, and process engineering at IMECAL SA., where performs mechanical, electrical and industrial design projects.

Related publications, patents or products:

- Method for Recovering Energy from the Organic Fraction of Solid Urban Waste and Associated Facility. Signes Nunez, Vicente, ; Ballesteros Perdices, Mercedes, ; Ballesteros Perdices, Ignacio, ; Manzanares Secades, Paloma, ; Martinez Garcia, Jose Maria, ; Negro Alvarez, Maria Jose, ; Oliva Dominguez, Jose Miguel, ; Castaneda Sanchez, Rafael, ; Coll Lozano, Caterina. Spanish patent nº 2303792, European Patent EPO 2112226, USA Patent 8399228 B2 and international PCT WIPO 099038.
- Ethanol production from the organic fraction obtained after thermal pretreatment of Municipal Solid Waste. Mercedes Ballesteros, Felicia Sáez, Ignacio Ballesteros, Paloma Manzanares, Maria Jose Negro, Jose Maria Martínez, Rafael Castañeda, Jose Miguel Oliva Dominguez. *Appl Biochem Biotechnol* (2010) 161:423–431.
- PERSEO Bioethanol®: Obtención de bioetanol y combustible orgánico a partir de RSU. Caterina Coll y Vicente Signes. INFOENVIRO, nº 72, p. 49-42, 2012.
- Sistema de comunicaciones para el control de una planta de biocombustibles. Josep F. Signes Palop, Víctor M. Sempere Payá, Caterina Coll Lozano. *Automática e Instrumentación*, nº 444, p. 40-42, 2012.
- Proyecto PERSEO. Innovación tecnológica para el bioetanol de segunda generación. Rafael Castañeda, Caterina Coll, Vicente Signes, Mercedes Ballesteros. *RETEMA*, nº 121, p.50-56, Especial biomasa 2007.

Related Projects:

Project acronym/description.	Scope	End Year	Website
LO2X: Supercritical water co-oxidation of urban sewage sludges and wastes	LIFE 12 ENV	2016	www.lo2x.com
BIOFORSU: Valorisation of the organic fraction of municipal solid wastes by conversion to bioethanol and combustible biomass	Regional	2013	Not Available
BIOFUEL: Development for a new integral waste treatment process with production of liquid biofuel	Spanish Project	2009	Not Available
ATENEA: Production of bioethanol from citrus wastes	Regional	2008	www.imecal.com/perseo
PERSEO: Production of bioethanol from organic municipal solid wastes	Spanish Project	2006	www.imecal.com/perseo

Technical equipment:

- 25.000 m^2 of metal-mechanic engineering facilities to transform metallic plates and sheets to a wide variety of industrial equipment: pipes, chemical reactors, distillation columns, etc. Machinery includes: Laser cutting, forming, machining, blasting, passivating, assembly and welding, etc.
- 2000 m^2 semi-industrial biotechnology plant (PERSEO Bioethanol®) with a capacity to process 25 ton/day of organic fraction of MSW. The plant includes the following units: Pre-treatment, Fermentation, SolidLiquid Separation, Distillation, Effluent Treatment, Auxiliary Systems, Final Product Storage, Control.
- Two physical-chemical laboratories included in the PERSEO semi-industrial plant, with different equipment: High pressure liquid chromatography, analytical balance, thermo-balance, centrifuge, microbiology fume extraction, pH meters, incubators with orbital shakers, solid grinders, oven, kiln, autoclave…

Pictures of bioethanol PERSEO semi-industrial plant and its laboratories

PARTNER 2. **Full name**: AIMPLAS-Asociación de Investigación de Materiales Plásticos y Conexas (AIMPLAS) **Web page**: www.aimplas.es	**Country:** Spain **Type:** RO	AIMPLAS PLASTICS TECHNOLOGY CENTRE

Brief description of the organization: AIMPLAS, Technological Institute of Plastics located in Valencia, is a private, non-profit Association with more than 500 associated companies created in 1990. AIMPLAS is formed by +120 highly skilled professional, more than 65% with a Masters, Engineering or equivalent degree in Chemistry, polymer engineering, materials engineering or equivalent, including 15 PhD.

AIMPLAS' fields of work are related to technological research and development on thermoplastic and thermosetting plastic materials & products, its transformation processes and their recyclability and sustainability. AIMPLAS generates new knowledge and technologies that can be transferred to companies in order to help them

to increase their effectiveness and competitiveness.

AIMPLAS has a broad expertise in the fields of recycling, compatibilization, reactive extrusion, biopolymers and renewable source materials, special assisted processing technologies and nanocomposites, gained during the last years in the frame of EU and national projects. Nowadays, AIMPLAS is involved in more than 25 European projects and has participated in 77 projects in FP5, FP6, FP7, LIFE+, CIP-EcoInnovation EU Programmes, among others, coordinating 27 of them. At National and Regional levels AIMPLAS participates in around 100 projects yearly.

AIMPLAS, as RO is focused to help companies in the plastic sector to develop new products and increase their competitiveness through innovation, has more than 20 pilot plants representing the most relevant polymer/plastics/composites production technologies present in the industry nowadays and has state-of-the art test facilities for chemical, optical, morphological, mechanical and physical characterisation. These pilot lines and laboratories are used by many customer's every year allowing them to test new materials, optimize production processes and launch new products to the market, supported by AIMPLAS technical staff, resulting in more than 5000 assays, 170 technical assessments and 120 skills training actions to more than 1500 clients per year. AIMPLAS has state-of-the-art 8500 m^2 facilities, including thermoplastics and thermoset pilot plants, analysis and testing laboratories (physical-mechanical, chemical, packaging, automotive and construction) and training areas.

Role in the Project: In WP5 Production of HMA from LA using reactive extrusion and Collaboration in Polyester polyols production. Leader of regulatory and safety analysis. Contribution to communication, dissemination, exploitation and other related activities.

Scientific team:

Dr. Amador García (M), PhD in Organic Chemistry by the University of Valencia, working in AIMPLAS as researcher in the chemical laboratory since 2009. He has been involved in several regional and FP7 European projects focused in nanotechnology areas, materials from renewable resources and coatings with enhanced properties, etc. (SUPIN, TRAINER, NMP INNOREX, RECAN, KBBE BRIGIT, KBBE BISIGODOS, BIOREFINE-2G). Before joining to AIMPLAS, he worked as researcher assistant in CIPF (Centro de Investigación Principe Felipe) and in the University of Valencia. During 2007 he was in the chemical group of Professor Dennis P. Curran (University of Pittsburgh) as visiting researcher. In 2010 he received the Plastic & Innovation award.

Dr. Alba María Ortiz Álvarez (F), European PhD in chemistry by the Universidad de Alcalá, Madrid (December 2012), working in AIMPLAS as PhD researcher in the chemical laboratory since 2013. She has been involved in several national and international research projects focused in different areas such as synthesis of new polymers, nanotechnology, materials from renewable resources and coatings with enhanced properties, and so on. Moreover, she has being involved in the technical work of on Food contact plastics as well as Pharmacopoeia. Before joining AIMPLAS, she worked at the Universidad de Alcalá, where she was involved in research projects. Her research focused on the design and synthesis of new enlarged-palladium complexes as recoverable catalysts by nanofiltration techniques, contributing in international peer-reviewed journals and in national and international conference papers in the field of catalysis, dendrimers and organometallic chemistry.

Ms. Raquel Llorens-Chiralt (F), Master in Material Engineering by the Polytechnic University of Valencia, working in AIMPLAS as Senior Researcher since 2005 in the Compounding Department, focused in new material formulations. She has been involved as main researcher in the technical execution of several FP7 Projects focused on nano-materials, biopolymers, flame retardants, blends, mineral and natural fibres composites and PVC. Before joining AIMPLAS she worked in different private industries (paints, ceramics and metallurgical).

Dr. Nuria Lopez (F), PhD in Materials by the Universidad Politécnica de Valencia, working as a researcher in the Extrusion Department at AIMPLAS since 2003. Extensive experience in polymer processing field, analysis of properties and development of new products. Technical advice, evaluation and optimization of materials using extrusion processing technology, new developments for private companies and close collaboration with other

Research Centres at Regional National and International level. Experience in sustainability developments, as the use of biodegradable and compostable materials and additives, materials from renewable sources, the use of natural fillers and studies of recyclability of materials.

Ms. María Llorens (F) graduated in Information Sciences from the Cardenal Herrera-CEU University. Then she holds a Master's Degree in Communication Management and a Master's Degree in Marketing Management from the Universitat Oberta de Catalunya. She has also a Master's Degree in Management Skills from the University of Valencia. She is currently working at AIMPLAS since 2004 and since 2011 she is the Head of the Marketing and Communication department, where she coordinates the relation with media, social networks, organization of events, digital marketing and graphic design of the company. As a result, AIMPLAS was awarded with the LIFE prize for the best textual communication for the project BREAD4PLA.

Mrs. Itziar Carracedo (F) has a University Degree in Chemical Engineering by Universidad Complutense de Madrid. She works in the International Projects Unit at AIMPLAS since 2007. She has been involved in international R&D projects funded under, the 6th (PEGASUS, FLOWFREE, INNOFOOT) and 7th (MAC-RTM, BUGWORKERS, PLA4FOOD, ESPRIT, NANOONSPECT) Framework Programmes, developing technical management and coordination tasks.

Publications:

- Signori, F., Boggioni, A., Righetti, M. C., Rondán, C. E., Bronco, S. and Ciardelli, F. (2015), Evidences of Transesterification, Chain Branching and Cross-Linking in a Biopolyester Commercial Blend upon Reaction with Dicumyl Peroxide in the Melt. Macromol. Mater. Eng., 300: 153–160.
- Campanella, A., Wool, R. P., Bah, M., Fita, S. and Abuobaid, A. (2013), Composites from northern red oak (*Quercus robur*) leaves and plant oil-based resins. J. Appl. Polym. Sci., 127: 18–26. doi: 10.1002/app.36663.
- R.M. González, M.P. Villanueva19 – Starch-based polymers for food packaging, Multifunctional and Nanoreinforced Polymers for Food Packaging,2011, Pages 527–570
- Coltelli, M.-B., Maggiore, I. D., Bertoldo, M., Signori, F., Bronco, S. and Ciardelli, F. (2008), Poly(lactic acid) properties as a consequence of poly(butylene adipate-*co*-terephthalate) blending and acetyl tributyl citrate plasticization. J. Appl. Polym. Sci., 110: 1250–1262. doi: 10.1002/app.28512
- Martí-Ferrer, F., Vilaplana, F., Ribes-Greus, A., Benedito-Borrás, A. and Sanz-Box, C. (2006), Flour rice husk as filler in block copolymer polypropylene: Effect of different coupling agents. J. Appl. Polym. Sci., 99: 1823–1831
- Fang Jm, Fowler Pa, Escrig C, Gonzalez R, Costa Ja, Chamudis L: Development of Biodegradable Laminate Films Derived From Naturally Occurring Carbohydrate Polymers. *Carbohydrate Polymers* 2005, 60:39-42.

Projects:

Project acronym/description.	Scope	End Year	Website
BUGWORKERS: New tailor-made PHB-based nanocomposites for high performance applications produced from environmentally friendly production routes.	FP7-NMP - 2009-SME	2013	www.bugworkersproject.eu/.
BRIGIT: New tailor-made biopolymers produced from sugars obtained from lignocellulosic residues for flame retardancy highly demanding applications.	FP7KBBE-SME	2017	www.brigitproject.eu/

INNOREX: Realising the continuous, highly precise, metalfree polymerisation of PLA using alternative energies for an innovative reactive extrusion.	FP7-NMP2012	2016	www.innorex.eu/
BIOREFINE 2G : Development of 2nd Generation Biorefineries Production of Dicarboxylic Acids and Biobased Polymers Derived Thereof	FP7-KBBE 2013	2017	www.biorefine2g.eu/
BISIGODOS: High value-added chemicals and BIoreSIns from alGae biorefineries produced from CO_2 provided by industrial emissions.	FP7 KBBE 2013	2017	www.bisigodos.eu

Technical equipment:

- Equipment for mechanical, thermal, rheological and physical tests accredited to perform plastic characterization using the EN/ISO standards
- Rheometer, Optical and electronic Microscopy, DMA, DSC, Thermogravimetry. Raman Spectrometer.
- Twin-screw co-rotating extruders with gravimetric feeders (to feed solid and liquid additives) and other multiscrew planetary extruder.
- Well-established kilo laboratory for the synthesis of polymers in multigrams to kilograms level. During kilolab synthesis, organic chemists and chemical engineering work together to reach the appropriate cost effectiveness of the polymer synthesis process, optimizing parameters as reaction time, temperature profile, amount of catalyst and post-purification steps.
- The kilolab is equipped with 1, 2, 5 and 15 l glass jacketed reactors able to reach temperatures of 190ºC and another 10 l steel reactor able to reach temperatures up to 250ºC. Moreover, high volume Buchi rotavap and high capacity crystallization/filtration systems are available. Other auxiliary equipment as vertical stirrers for viscous polymers, vacuum pumps, water traps, nitrogen line and vacuum oven are currently available.

Reactor and Buchi 20 l rotavap

AIMPLAS kilolab is also able to explore the polymer synthesis by means of controlled reactive extrusion process. In this context the Kilolab is fullequipped with HAAKE™ MiniLab II Micro Compounder which is ideal for reactive extrusion optimization process, compounding expensive materials or small scale materials such as nano-composites, bio-polymers or pharmaceuticals.

To perform all of these approaches, we take up several kind of strategies in polymer synthesis as:

- Ring opening polymerization (ROP) HAAKE™ MiniLab II Micro Compounder
- Radical polymerization
- Ionic polymeriation

PARTNER 3 **Name**: Leibniz Institute for Agricultural Engineering and Bioeconomy (ATB), Dept. Bioengineering **Web page**: www.atb-potsdam.de/en.html	**Type:** RTO **Country:** Germany	

Brief description of the organization: The Leibniz Institute for Agricultural Engineering and Bioeconomy (ATB) is a European research and development centre for agricultural engineering with an interdisciplinary field of activity at the nexus of biological and technical systems. ATB is funded by the Federal Government and the State of Brandenburg and receives additional third-party funds of around 35% of its budget (total 2015 budget >17.0 Mio).

ATB has multi-disciplinary research teams with currently about 250 staff members, 90 of them scientists and PhD students. In 2015 scientists published over 115 reviewed papers in international peer-reviewed journals.

The Institute's mission is to develop sustainable and highly innovative technologies for the resource efficient utilization of biological systems to produce food, raw materials and energy. Climate change and the incipient changes in the structure of agriculture represent particular challenges for agricultural engineering research.

ATB research activities are of interdisciplinary nature and follow a systems approach covering the entire spectrum from primary production and consumption of food, feed and non-food biomass including technology assessment.

The Institute's research work addresses topical scientific questions and contributes to the call for practical guidance and support in the domains of politics, industry, agriculture and horticulture. The institute owns outstanding infrastructure including pilot plants and experimental fields. Intensive collaboration with universities, other nonuniversity research institutions and enterprises on national and EU/international level contributes largely to elaborate solutions for the agricultural and food industry.

Role in the Project: Bioconversion of the hydrolysed carbohydrates from organic municipal solid waste to lactic acid (LA), which is a main activity of WP2. Collaboration within the other WP's and support in view of scale-up (piloting) and downstream processing sub-tasks.

Scientific team:

Dr. Joachim Venus (M), Senior Scientist "Industrial Biotechnology", holds a diploma on Biotechnology (graduated from Anhalt University of Applied Sciences Köthen/Germany 1984) and a PhD on Bioengineering (TU Dresden). He is employed permanently at the Dept. Bioengineering of ATB Potsdam (http://www.atbpotsdam.de/en/institute/about-us/team/portrait/portrait/joachim-venus.html).

Joachim Venus is Program Coordinator for the research program "Material and energetic use of biomass" and head of the research group bioconversion/fermentation of biogenic feedstocks/residues at the ATB. His work emphasized on the development of continuous processes for the production of basic chemicals - in particular lactic acid - from biomass. He has published a great deal of papers in well-known journals of those research fields. Moreover, he has taken part in more than 100 presentations and proceedings, since 2003, such as in "World Congress on Industrial Biotechnology and Bioprocessing", "International Conference on Environmental, Industrial and Applied Microbiology/ BioMicroWorld", "European Biomass Conference and Exhibition", "European Forum for Industrial Biotechnology/EFIB2010", "International Conference on Renewable Resources and

Biorefineries/RRB" or "Symposium on Biotechnology for Fuels and Chemicals".

At ATB Joachim Venus is in charge of numerous research projects being carried out in the multi-functional pilot plant for the biotechnological production of lactic acid from plant biomass. His expertise and research areas are

- White Biotechnology, Biorefineries, Scaling-up of Bioprocesses
- pre-treatment of biomass for microbial conversion processes, bioconversion of renewable resources
- kinetics of cell growth/product formation and modelling of fermentation processes
- development of continuous mode processes for the production of basic chemicals (e.g. lactic acid) and biomass
- operation of a pilot plant facility for the optimization of biotechnological processes

He is heading the national advisory board "Biotechnology of Renewable Resources" at DECHEMA. On international level Joachim Venus is being also a Management Committee member for Germany in the COST Action TD 1203 (Food waste valorisation for sustainable chemicals, materials & fuels, EUBis).

Dr. Donna Dietz (F), PhD in Bioprocess Engineering by Hamburg University of Technology in Hamburg, Germany. During the time in Hamburg, she worked as scientist in FP7 Project Propanergy – Integrated bioconversion of glycerol into value added products at pilot plant scale. Since 2012 she is working as senior researcher in ATB. Dr.Dietz's expertise includes microbial fermentation of biobased raw material to value added products from lab to pilot scale.

Ms Kerstin Mehlmann (F), Diploma in biotechnology by the Beuth University of Applied Sciences (Germany). She has more than 15 years of experience in fermentations at laboratory and pilot scale and advanced characterization of several substrates. Over the last years she has participated in many projects at national and international level. Experience in microbiology works and analytical methods, HPLC in particular.

Related publications, patents or products:

- Turon, X., Venus, J., Arshadi, M., Koutinas, M., Lin, C., Koutinas, A. (2014) Food Waste and Byproduct Valorization through Bio-processing: Opportunities and Challenges. BioResources [Online] 9: 4, 5774-5777
- Pleissner, D.; Venus, J.: Agricultural residues as feedstocks for lactic acid fermentation. - ACS Symposium Series, Vol. 1186 "Green Technologies for the Environment" (2014) Chapter 13, pp 247–263, Chapter DOI: 10.1021/bk-2014-1186.ch013, ISBN13: 9780841230187/eISBN: 9780841230194
- Kwan, T.H.; Pleissner, D.; Lau, K.Y.; Venus, J.; Pommeret, A.; Lin, C.S.K.: Techno-economic analysis of a food waste valorization process via microalgae cultivation and co-production of plasticizer, lactic acid and animal feed from algal biomass and food waste. Bioresource Technology 198 (2015) 292–299. http://dx.doi.org/10.1016/j.biortech.2015.09.003
- Pleissner, D.; Venus, J.: Utilization of protein-rich residues in biotechnological processes. Applied Microbiology and Biotechnology 100(2016) Issue 5, 2133-2140, http://link.springer.com/article/10.1007/s00253-015-7278-6
- Pleissner, D.; Neu, A.-K.; Mehlmann, K.; Schneider, R.; Puerta-Quintero, G.I.; Venus, J.: Fermentative lactic acid production from coffee pulp hydrolysate using *Bacillus coagulans* at laboratory and pilot scales. Bioresource Technology 218(2016) 167–173, http://dx.doi.org/10.1016/j.biortech.2016.06.078

Related Projects: - A.N.Bach Institute of Biochemistry of Russian Academy of Sciences, INBI, Moskau „Enzymes for lignocellulosic feedstocks degradation and production of sugars (C6, C5) and value added products", BMBF / DLR (Russia/Germany) – 2010-2013

- BREAD4PLA: Demonstration-plant project to produce poly-lactic acid (PLA) biopolymer from waste products of bakery industry, EU LIFE10/ENV/ES/479 – 2011-2014
- Food waste valorisation for sustainable chemicals, materials & fuels (EUBis,) – MC Member Germany, COSTAction TD1203 – 2012-2016
- Process optimization and investigations for the valorization of residues from the dairy industry (LACTOplus, Project-No: 848572), Program "Production of the future", Austrian Research Promotion Agency (FFG) – 2016-
2017-

- Thermophilic lactic acid production utilizing rice bran in continuous cultivation (BranLact, Bilateral project with Beijing University of Chemical Technology/BUCT, German Call Bioeconomy International 2015) – 2016-2019

Technical equipment:

The ATB has a powerful research infrastructure. We can provide services for external research partners that include primary research data, bacterial strain collections, analytical instruments and methods, as well as application laboratories. ATB operates facilities for biotechnological and physico-chemical conversion processes to produce materials (lactic acid, a base chemical for bioplastics, and biochar) and energy (biogas) pursuing biorefinery and cascade concepts for the utilization of biomass.

R&D facilities description (including analytical suite at ATB)

- Upstream processing of raw materials & fermentation feedstocks (Equipment for pre-treatment, filtration, purification…; Micro-/Ultra-/Nanofiltration; Reverse Osmosis; Ceramic membranes; Hollow fiber membranes)
- Fermentation units from lab to pilot scale (1 litre up to 1000 litres) including continuous mode devices coupled with cell retention
- Downstream processing of raw lactates & lactic acid (Softening, Electrodialysis, Ion exchange cations/anions, Decolorization, Evaporation) which could be used/adapted for other bio-based products, i.e. organic acids
- Analytical methods for the main parameters of bioprocesses (e.g. cell numbers, dry weight biomass, strain characterization, HPLC for sugars and acids, IC, AAS, element analyzer, GC, ICP, Bio-LC, fibre analytics…)
- Pilot plant for lactic acid (own facility at ATB site for the demonstration of the entire process chain from the raw material(s) via bioconversion/fermentation up to final product) http://www.atbpotsdam.de/en/institute/about-us/research-infrastructure/pilot-plant-lactic-acid.html

From plant raw or waste materials lactic acid, a basic chemical for bioplastics, is produced in a continuous fermentation process. The practical scale of the mentioned pilot plant enables scale-up and optimization of the entire process: from testing various feedstocks to conversion and the production of product samples with specific characteristics. The aim is to make the process on basis of a wide range of raw materials stable and economically competitive.

PARTNER 4 **Name**: Agricultural University of Athens **Web page**: www.aua.gr/bioprocesses	**Type:** HE **Country:** Greece	

Brief description of the organization:

AUA serves agricultural sciences since 1920 and is involved in various research and educational projects including over 1500 research projects since 2002 mainly funded by the EC or the National Strategic Reference Framework. During the period 2010 - 2015, AUA has been involved in 40 European projects, four as the coordinator and in thirty six as partner in the corresponding consortia. The overall EU funding for AUA has been over 8 million euros. The staff members of AUA have established numerous collaborations with European/International researchers, with regular student exchange through the European Projects, bilateral actions with international universities or on-going long term collaborations.

The Department of Food Science and Human Nutrition (DFSHN) of AUA promotes the existing knowledge on Food Science and Human Nutrition through research and development programs conducted in collaboration with other Greek and foreign Universities and research institutes as well as with research and development departments of small and large Greek food industries. The Department consists of seven laboratories. The Laboratory of Food Engineering from the DFSHN is well equipped with adequate infrastructure to support a number of research projects funded by national and international research institutions. There are 3 post-doctoral researchers, 13 PhD students and 10 MSc students working in research topics related to valorization of renewable resources (including agri-industrial waste streams), industrial biotechnology, biorefinery development, platform chemicals production (including succinic acid), LCA and techno-economic studies and bioprocess design. AUA research team has received more than €1.500.000 since 2011 in the aforementioned research topics.

The group has more than 30 publications since 2013 on relevant topics. Since 2011, AUA has been involved in 3 national, 3 bilateral (collaborating with institutions in Spain, Brazil and Germany), 1 FP7, 1 Life+ research projects, and 1 project supported by the private sector (Petrobras, Brazil) focusing on biorefinery development from renewable resources and waste streams for the production of various products (including platform chemicals and succinic acid in particular). An MSc course entitled "*Food Bioprocesses and Biorefineries*" is organized by DFSHN-AUA covering topics in refining of renewable resources, industrial biotechnology and biorefinery engineering and design (18 month duration with 6 month dissertation). Dr Ioannis Kookos from the Chemical Engineering Department of the University of Patras is also member of the AUA team with expertise in process systems engineering, bioprocess design, and techno-economic evaluation.

Role in the Project: AUA will participate in PERCAL in the following topics: 1) Bioprocess development and optimisation for the production of succinic acid using MSW hydrolysates; 2) Collaboration with UGENT in the development of an integrated bioprocess based on production of succinic acid via fermentation coupled with extraction of organic acids by electrolysis; 3) Development and optimization of downstream separation of succinic acid via an electrodialysis based process in collaboration with ATB; 4) Process design, techno-economic evaluation and LCA of the developed bioprocesses.

Scientific team:

Dr Apostolis Koutinas (male, Assistant Professor, DFSHN-AUA) is Chemical Engineer with PhD in biochemical engineering and 18 year experience in industrial biotechnology, biorefinery development and design of bioprocesses/biorefineries. He is currently supervising two post-doctoral researchers and three PhD students on succinic acid production, downstream separation and bioprocess optimization including life cycle analysis and techno-economic evaluation. He has 75 papers in peer-reviewed scientific journals (*h-index 21, non-self-citations*), 13 book chapters, 1 edited book and more than 90 international conferences and workshops. He has participated in 15 research projects since 2011 and serves as Associate Editor in *Biochemical Engineering Journal* (Elsevier). He is currently Guest Editor in a special issue on "*Advances on Biorefinery Engineering*" in the Biochemical Engineering Journal.

Dr Seraphim Papanikolaou (male, Associate Professor, DFSHN-AUA) is Agronomist with 2 MSc Diplomas and a PhD in fermentation technology. He has expertise in fermentation technology and bacterial metabolism. He has participated in 20 projects, published 85 papers in peer-reviewed journals (*h-index 35, non-self-citations*) and serves on the editorial board of 5 international journals.

Dr Ioannis Kookos (male, Associate Professor, Department of Chemical Engineering, University of Patras) is Chemical Engineer with MSc and PhD in Process Systems Engineering. He has expertise in the development of mathematical programming formulations for biorefineries including bioprocesses. He has 50 papers in peerreviewed scientific journals with expertise in (bio)process design, techno-economic evaluation, modeling and control.

Mrs Chrysanthi Pateraki (female, PhD student) is Biologist with MSc in wine making and currently a PhD student at AUA to graduate at the beginning of 2017. She has carried out research on optimization of succinic acid production focusing on bioprocessing aspects and elucidation of microbial metabolism using spent sulphite liquor and other renewable resources as feedstock. She has closely collaborated with UGENT in the development of the integrated succinic acid production system based on electrolysis membrane extraction coupled with fermentation.

Dr Vasiliki Kchrimanidou (female, post-doctoral researcher) is an Agronomist who acquired her Diploma from the DFSHN. She concluded her PhD at AUA in 2016 and has significant expertise in industrial biotechnology and biorefinery development

Related publications, patents or products:

1) Alexandri M, Papapostolou H, Vlysidis A, Gardeli C, Komaitis M, Papanikolaou S, Koutinas AA, 2016. Extraction of phenolic compounds and succinic acid production from spent sulphite liquor. *Journal of Chemical Technology and Biotechnology*, DOI: 10.1002/jctb.4880
2) Koutinas AA, Yepez B, Kopsahelis N, Freire DMG, Machado de Castro A, Papanikolaou S, Kookos IK. 2016. Techno-economic evaluation of 2,3-butanediol production from renewable resources. *Bioresource Technology*, 204: 55-64.
3) Alexandri M, Papapostolou H, Komaitis M, Stragier L, Verstraete W, Danezis GP, Georgiou CA, Papanikolaou S, Koutinas AA. 2016. Evaluation of an integrated biorefinery based on fractionation of spent sulphite liquor for the production of an antioxidant-rich extract, lignosulphonates and succinic acid. *Bioresource Technology*, 214: 504–513
4) Pateraki C, Ladakis D, Stragier L, Verstraete W, Kookos I, Papanikolaou S, Koutinas A, 2016. Pretreatment of spent sulphite liquor via ultrafiltration and nanofiltration for bio-based succinic acid production. *Journal of Biotechnology*, 233: 95–105.
5) Koutinas AA, Vlysidis A, Pleissner D, Kopsahelis N, Garcia IL, Kookos IK, Papanikolaou S, Kwan TH, Lin CSK. 2014. Valorization of industrial waste and by-product streams via fermentation for the production of chemicals and biopolymers. *Chemical Society Reviews* 43:2587-2627.

Related Projects:

Project acronym/description.	Scope	End Year	Website
FOODPRINT: Development of an integrated strategy for reducing the carbon footprint in the food industry sector	Life+ 2013	2017	www.foodprint.gr
Production of enzymes and building blocks (1,3propanediol, 2,3-butanediol and fumaric acid) for petrochemical production from industrial residues	Industrially funded R&D	2017	-
BRIGIT: New tailor-made biopolymers produced from sugars obtained from lignocellulosic residues for flame retardancy highly demanding applications.	FP7KBBE-SME	2016	www.brigit-project.eu/
BIO4OIL: New bioprocess for microbial oil from crude glycerol and cellulosic sugars	GreeceGermany bilateral project	2015	-
Nutri-Fuel: Development of novel bioprocesses for the production of biofuels from food industry waste streams	National funded project	2015	-

Technical equipment:

- The DFSHN-AUA has the adequate infrastructure to carry out submerged fermentations, qualitative and quantitative analysis of renewable resources, microbiological techniques and techno-economic evaluation.
- 7 bioreactors (1-7 L working volume), 8 shakers, 2 GC-FID, 1 GC-MS, 2 UV-Vis spectrophotometers, 1 microplate reader, 3 HPLC units with RI, UV and diode-array detectors, 1 FPLC, 5 incubators, 2 rotary evaporators, 8 balances with accuracy of 0.0001 g and 0.01 g, laboratory freezers (-40 °C), 7 water baths, freeze dryer, 3 centrifuges, 3 laminar flow cabinet, 3 microscopes and various generic equipment for microbiological techniques.
- one semi pilot scale spray dryer, UF/RO membrane system, double effect rising film evaporator and high pressure homogeniser.

PARTNER 5 **Name**: Center for Microbial Ecology and Technology **(**UGENT). **Web page**: http://cmet.ugent.be	**Type:** HE **Country:** Belgium	

Brief description of the organization:

The **Center for Microbial Ecology and Technology** (CMET), formerly **LabMET (**www.cmet.ugent.be) is part of the Faculty of Bioscience Engineering at Ghent University (UGent). CMET houses a group of 80 fulltime personnel of which ~25 at a postdoctoral level. CMET is specialized in the fields of applied microbial ecology, microbial electrocatalysis, host microbe interactions, treatment and resource recovery of water and waste and biomaterials and nanotechnology. The principal research strategy at CMET aims at understanding phenomena of microbial ecology in order to manage the vast metabolic capabilities of complex microbial communities to solve practical problems. The group is also involved in a series of environmental biotechnology projects ranging from nitrogen recovery processes, anaerobic digestion, (trace) metal recovery and so on. CMET has always been a leading research group in the field of anaerobic digestion (AD). This is reflected in the number of publications on ADs (> 100) and several spin-off companies (e.g. Organic Waste Systems-OWS, EPAS, BIOTIM) and the coordinating role CMET takes in e.g. ManureEcoMine (FP7, 2013-2016). Since 15 years CMET also houses a (bio)electrochemistry group, with extensive experience in membrane electrolysis e.g. for sulphide and nitrogen recovery as well as production of caustic from waste streams. This experience puts CMET in an excellent position to perform the tasks within PERCAL. Furthermore, CMET has extensive experience in participation in EU collaborative projects as well as acting as coordinator e.g. for ManureEcomine.

Role in the Project: UGENT is responsible for the development of a membrane electrolysis system coupled to the AUA succinic acid fermenter. The process will be optimized in terms of operational parameters, and in a second phase be implemented at pilot scale.

Scientific team:

Prof. Dr. Ir. Korneel Rabaey (M) is full professor and Head of the Department of Biochemical and Microbial Technology at Ghent University. Until mid 2011, he was senior lecturer and incoming director/founder of the Centre for Microbial Electrosynthesis at The University of Queensland where he recently received an Honorary Professorship. He is internationally recognized for his work in the field of bioelectrochemical systems, particularly with respect to novel process development. He has published over 150 papers in this field (>100 A1/international peer reviewed, > 12000 citations, H-index 46) for which he was recently designated as “ISI Highly Cited Researcher”. Prof. Rabaey has driven technology development, by better integrating electrochemical concepts in the microbial or environmental engineering driven field.

Mr. Stephen Andersen is a graduating PhD student at UGENT, and was the key developer of the membrane electrolysis approach for succinic acid extraction.

Related publications, patents or products:

- Desloover, J., De Vrieze, J., Van de Vijver, M., Mortelmans, J., Rozendal, R. and Rabaey, K. 2015 Electrochemical Nutrient Recovery Enables Ammonia Toxicity Control and Biogas Desulfurization in Anaerobic Digestion. *Environmental Science & Technology* 49(2), 948-955.
- Andersen, S.J., Candry, P., Basadre, T., Khor, W.C., Roume, H., Hernandez-Sanabria, E., Coma, M. and Rabaey, K. (2015) Electrolytic extraction drives volatile fatty acid chain elongation through lactic acid and replaces chemical pH control in thin stillage fermentation. Biotechnology for Biofuels 8.
- Logan, B.E. and Rabaey, K. 2012 Conversion of wastes into bioelectricity and chemicals by using microbial electrochemical technologies. *Science* 337(6095): 686-690.
- Rabaey, K. and Rozendal, R.A. 2010 Microbial electrosynthesis: revisiting the electrical route for bioproduction. *Nature Reviews Microbiology* 8: 706-716.
- Andersen, S. J., T. Hennebel, S. Gildemyn, M. Coma, J. Desloover, J. Berton, J. Tsukamoto, C. Stevens, and K. Rabaey. 2014. Electrolytic Membrane Extraction Enables Production of Fine Chemicals from Biorefinery Sidestreams. Environmental Science & Technology 48:7135-7142.

Related Projects:

Project acronym/description.	Scope	End Year	Website
ELECTROTALK – impact of electricity on microbial metabolism, creation of interfaces.	ERC-StG	2017	www.electrotalk.org
MRP Ghent Bio-Economy – fermentation of biorefinery sidestreams	BOF-UGent	2016	
ProEthanol – fermentation of vinasses for production of carboxylic acids	EU FP7	2014	www.proethanol.eu
Scale up of carboxylic acid production and extraction from thin stillage	IOF	2017	
Microelectronics for Microbiology – development of intelligent steering	FWO	2019	

Technical equipment:

UGENT facilities are divided between analytical, microbial and molecular laboratories, temperature controlled labs, host interaction laboratories and a technology hall were several lab-scale reactors are available from millilitres to litres scale. The group is equipped with a broad range of analysis tools used in the determination of physicochemical and biochemical parameters (e.g. GC, HPLC, IC, AAS), microbial parameters (e.g. laminar flows, anaerobic chambers, flow cytometers, multiwall spectrophotometers), as well as the molecular characterization, identification and quantification of microbial species or groups in diverse communities. Moreover, UGENT has a broad knowledge and experience in reactor technology (e.g. SBR, CSTR, UASB, RBC, MFC, MEC) including also 5 fully automated, 5-L fermenter systems. Also due to its high activity in electrochemical research, UGENT is provided with different tools for electrochemical characterization (e.g. ~60 potentiostats, rotating disc electrodes and a wide variety of test cells).

PARTNER 6. **Full name**: FUNDACION CENER-CIEMAT **Web page**: www.cener.com	**Country:** Spain **Type:** RTO	CENER ADItech

Brief description of the organization:

The National Renewable Energy Centre (CENER) is a technology centre specialised in applied research, and the development and promotion of renewable energies. It is highly rated and has acknowledged national and international prestige. CENER, as a foundation, is an organisation destined to support renewable energies sector competitiveness in order to help companies to reach their R&D needs: It offers high added value services and activities and transferring technology by means of a cutting-edge technological infrastructure, with the most modern laboratories and facilities in Europe.

CENER´s Biomass Energy Department is composed of a multidisciplinary staff, over 20 people, some of them with more than ten years' experience in the Bioenergy Sector and specific knowledge in biology, chemistry, agronomy and chemical engineering. There is also a full equipped laboratory with specific areas dedicated to: chemical processes, microbiology, fermentation processes and analytical techniques and the Second Generation Biofuel Centre (CB2G) which is a research infrastructure devoted to advanced biofuels and bioproducts process development. Since its foundation in 2002, it has been developing R&D activities in the bioenergy sector. These activities range from the evaluation of biomass potentials to conversion using different routes (biochemical and thermochemical) and also including the lifecycle analysis of all these processes.

Other relevant information:

- CENER's Biomass Department is an associated member of the BIC Initiative.
- CENER's Biomass Department is a member of the EERA Bioenergy Joint Programme Steering Committee

Role in the Project: CENER participates in WP1 in characterization of MSW samples and fermentation byproducts; in collaboration with IMECAL, Validation of MSW as fermentation feedstock. In WP4, CENER will be leading the WP4 and production of new biosurfactants from amino acids and lipids. Contributes to WP7 (LCA and techno-economic assessment) and WP9 (Business plan, exploitation, dissemination and communication activities).

Scientific team:

Dr Javier Gil (M) took his PhD in Science (Chemical Engineering program) in 1999. He is the Director of the Biomass Department of CENER (National Renewable Energy Centre of Spain) since September 2015 coordinating the activities in several bioenergy fields: resources evaluation, sustainability studies, biofuels, bioproducts and thermochemical applications (torrefaction and gasification). Since 2002 to 2015 he also worked at CENER, prior he has occupied several R&D positions in Fraunhofer UMSICHT Institute and University of Zaragoza working in projects focused on biomass gasification and tar cleaning. In all these R&D organisations he has had the opportunity of coordinating and participating in several RTD - Biomass related projects, both

National and European funded

Ms Irantzu Alegria (F) holds a MSc in Biology by the University of Navarra. As Researcher of the Biomass Department since 2002, she has been working in second generation and advanced biofuels and biorefineries. Her main expertise is in biochemical conversion of biomass into biofuels and/or bio-based products, completed with a wide knowledge of analytics and biological conversion. She has participated in RTD projects focused on biochemical conversion of biomass into bioenergy and/or bio-based products. She also actively participated in the design of a pilot plant for biochemical conversion of biomass located in Aoiz (Navarra). She coordinates the technical work and monitoring the work from several EU funded (H2020, Interreg, etc.) as well as national and regional projects.

Ms Ibai Funcia (M) has a Degree in Chemistry and an Interuniversity Master's degree in Project Management. Since July 2003 has been working in CENER's Biomass Energy Department as a Biomass Researcher. Main research lines are thermo-chemical route: biomass torrefaction, biomass combustion (ash behaviour at high temperature) and biomass gasification (tar cleaning, sampling and analysis), biochemical route: biomass pretreatment, enzymatic hydrolysis and fermentation, biomass and Bio-fuel characterization. Chromatography methods development. Laboratory quality and assurance procedures.

Ms Mercedes Munárriz (F) holds a MSc in Biology by the University of Navarra. As Researcher of the Biomass Department since 2003, she works on the Development and Coordination of R&D Projects related to biodiesel production from microalgae, microalgae cultivation as well as in upscaling of biochemical processes. She has **Ms**

Ms Laura García (F) holds a MSc in Biology by the University of Navarra, and a master degree in Biomedical Investigation. From September 2009 to September 2010, she was research fellow at the Microbiology Dpt. in the University of Navarra Clinic. As Researcher of the Biomass Department since 2011, she works on the Development and Coordination of R&D Projects related to bioethanol production from residues and lignocellulosic biomass, syngas fermentation, as well as in microbiological processes for obtaining compounds of interest. She has developed his professional career both in laboratory and pilot plant, specializing in processes of enzymatic hydrolysis, fermentation as well as characterization of biomass.

Five relevant publications, patents or products:

- Alegría, I.; Garcia, L.; Ortigosa, I; Del Campo, I.; Sánchez, D.; Munárriz, M.; Iruretagoyena, J. Development of a biorefinery concept design for the Navarre region. 23rd European Biomass Conference. Vienna. June 2015.
- Toledano. A. Alegria, I, Labidi, J. Biorefining of olive tree (Olea europea) pruning. Biomass & Bioenergy. 2013.59 (503-511)
- I. del Campo; I. Ortigosa; I. Alegria; I. Echeverría. Experimental study of using butanol as an additive for reducing the vapor pressure of ethanol-gasoline blends. XIX ISAF- International Symposium of Alcohol Fuels. Verona. Italy.(October 2011).
- Funcia, I. Alegria, F.J. Iruretagoyena, M. Munárriz, M. Piera, F.M. Reyes-Sosa and I. Echeverría. Validation of a high throughput analysis of microalgae carbohydrates for bioethanol production "18th European Biomass Conference- Visual presentation. Lyon ,May 2010
- I. del Campo, I. Alegría, M. Zazpe, M. Echeverría and I. Echeverría. "Diluted acid hydrolysis pretreatment of agro-food wastes for bioethanol production". Industrial Crops and Products Vol 26 (2006), 214-221

Five Relevant Projects:

Project acronym/description.	Scope	End Year	Website

BUTANexT. highly efficient production process to convert sustainable feedstocks in the next generation of biobutanol through and enzymatic hydrolysis and fermentation process	H2020	2018	www.butanext.eu
BIOrescue. develop and demonstrate a new innovative biorefinery concept based on the cascading use of spent mushroom substrate (SMS) supplemented by wheat straw and other seasonal underutilised lignocellulosic feedstocks.	H2020	2019	www.iorescue.eu
AMBITION. develop a long-term joint European Community Research and Innovation Agenda on the integration of biofuels production and surplus grid electricity valorisation.	H2020	2019	Not available yet. http://cordis.europa.eu/project/rcn/206587_en.html
CYCLAG "A network of technology centers to develop an algae-based biorefinery". develop and validate the technological processes to obtain biodiesel from cultivating microalgae	Interreg	2019	hhttp://www.cyclalg.com/en/presentacion/
BIOPEST: Research study on fermentation processes for Biopesticides production at lab and pilot scale.	Regional project funded by the Navarra Government	2016	Not availabe

Technical equipment:

CENER owns a pilot plant for the production of second generation and advanced biofuels and bioproducts. This facility is called CB2G (Second Generation Biofuel Center). The general objective of this infraestructure is to develop and validate processes and components through R&D projects focused on pilot scale as an intermediate step towards industrial scale-up. The CB2G include several Process Development Units (PDUs) as pre-treatment, torrefaction, gasification and biochemical processes units. Some of CB2G capabilities include flexibility to handle a wide range of biomass feedstocks, ability to incorporate and test equipment from partners or third-parties into whole conversion processes and changing configurations and flexible scaling-up: from pilot to semi-industrial. Therefore, by means of this facility, CENER not only intends to produce biofuels but also other biobased products that could be interesting for the development of the biorefinery concept.

PARTNER 7 **Name**: Tbw research GesmbH (TBWR) **Web page**: www.tbwresearch.org/	**Type:** SME **Country:** Austria	

Brief description of the organization: Tbw research (TBWR) is a R&D SME based in Vienna, founded in Nov. 2013 as a development spin off company of the IC-group, a highly experienced technical and engineering consulting company active in many fields mostly international with more than 400 employees. TBWR`s focus is on applied research only especially in cooperation with industry partners. Thematic fields of R&D activities are in general mobility, energy & renewables, construction and water management. Within the energy& renewables field TBWR is deeply involved in developing integrated processes for generating value added products and energy - biorefineries. There are several ongoing biorefinery R&D projects to develop valorisation pathways for various feedstocks (such as waste of food processing, grass, Lucerne or plants extracts for various product applications).
Lactic acid has been a focus of several projects of TBWR's key staff such as developing a downstream processing
(from lab upto pilot scale) for recovering LA from complex solutions such as grass silage juice. Furthermore TBWR has inhouse access to a reactive distillation apparatus at pilot's scale which was designed and integrated in cooperation with its sister company CES- Clean energy solutions. There is an ongoing research initiative to develop a simultaneous esterification and separation process for Ethyl lactate by means of a reactive distillation device. This pilot equipment and background expertise will also be available for the PERCAL project to test and develop a customized processing for the organic acids produced. TBWR also coordinates a national biorefinery network to enable and optimize interaction between academia and industry actors. Key staff of TBWR represent Austria within the international expert group of the IEA Bioenergy Task 42 - Biorefining.

Role in the Project: Involved in downstream processing of fermentation broth to lactic acid with partner ATB; Development and Assessment of a reactive distillation process for producing Ethyl Lactate from the two PERCAL intermediates LA and Ethanol.

Scientific team:

Michael G. MANDL, DI (M), Senior Researcher at TBWR and leader of the Biorefining group; Renewable Resources Engineering Graduate at University of Life Science and Renewable Resources (BOKU) Vienna. M. Mandl is key scientific staff and project manager of various national R&D projects. He has a comprehensive track record and expertise in applied research for more than 17 years. His specific field in biorefining is in green biorefinery e.g. separation of lactic acid and amino acids from grass silage juice. He joined the TBWR team in Aug. 2015, after he had worked for JOANNEUM RESEARCH for more than 17 years in the applied R&D. Currently he is also involved in modelling and upscaling a reactive distillation process for ethyl lactate and validating process with pilot plant performance data. M. Mandl is Austrian National Leam Leader of the IEA Bioenergy Task 42- Biorefinig and has been active within the IEA expert group form more than 6 years so far.

Dr Patrick J. Walsh BSc. MBA. D.Phil. (M) has worked in many countries, Scotland, Northern Ireland, USA, New Zealand, Australia and latterly the Republic of Ireland. His career covers the academic world (Canterbury University, NZ and the GMIT and NUIG in the RoI) as well as research institutes Foretry Commission Northern Research Station, Scion NZ). Recently he has entered private research and development by founding a company to commercialise some of his unpublished work. He is, at present, contracted by TBW to work on Biorefinery research and development. Dr Walsh has published widely on subjects dealing with a variety of subjects including Forest Health and Protectio, Viruses as species specific biocontrol, Bacterial control Methods, the use of plant extracts e. g; Terpenes and limonene. He represented Ireland at the IEA Task 42 Biorefining from 20072013. He has published over 100 scientific reports and papers and presented at conferences in many countries.

Ms. Stefanie Wong (F), Junior Researcher at TBWR; MSc degree in Biology, University of Vienna. Joined TBWR Energy and renewables in 2016.

NN – PhD (M/F) in Chemical Engineering, to be deployed to PERCAL Project

Related publications, patents or products:

- TBWR/ CES is currently defining a IPR for reactive distillation and separation process for ethyl lactate
- S.H. Sharma, M. Mandl, Green Biorefining in Sustainable Bioenergy Production, Edited by Lijun Wang, CRC Press 2014, Pages 535–564, Print ISBN: 978-1-4665-0552-0, eBook ISBN: 978-1-4665-0556-8, DOI: 10.1201/b16764-30
- De Jong, E, Higson, A, Walsh, P. and Wellisch (2012) Bio-based Chemicals: Value Added Products from Biorefineries. International Energy Agency
- De Jong, E, Higson, A, Walsh, P. and Wellisch (2012) Product developments in the bio-based chemicals arena. Biofuels, Bioprod. Bioref. Accepted June 2012
- M.G. Mandl (2010) Status if green biorefining in Europe, *Biofpr*-Biofuels, Bioprod. Bioref., DOI:10.1002/bbb.219

Related Projects:

Project acronym/description.	Scope	End Year	Website
Green Safer Solvent – Esterification of lactic acid and ethanol to ethyl lactate by means of reactive separation techniques.	FFG Austria	2016	
Demonstration Green Biorefinery Utzenaich plant for the recovery of lactic acid and amino acids from grass silage by means of innovative separation techniques combined with an anaerobic digester.	FFG Austria	2012	//nachhaltigwirtschaften.at/de/fdz /projekten/demonstrationsanlagegruene-bioraffinerieutzenaich.php

GREEN BIOREFINERY - Technical, economic and ecological optimisation of value chains by the introduction and efficient use of sustainable raw materials.	SUSPRISE joint call	2010	//nachhaltigwirtschaften.at/en/fdz/projects/susprise/greenbiorefinery-technical-economicand-ecological-optimisation-ofvalue-chains-by-the-introductionand-efficient-use-of-sustainableraw-materials.php

Technical equipment:

- Pilot scale reactive distillation plant for the production and separation of ethyl lactate, established by national funds form the Austrian Research promotion Agency, Grant
- Strategic cooperations for analytic services with several university departments

PARTNER 8 **Full Name:** Visum Ltd.(Ex. IRIS Advanced Engineerging Limited) (VISUM) /Ireland **Web page**: www.seeingnewdata.com	**Country:** Ireland **Type:** SME	

VISUM is an optical engineering company from Dublin that specializes in the development and commercialization of Near Infrared Spectroscopy (NIRS)-based portable and inline analyzers and measurement systems for process monitoring and quality control applications in the food, pharmaceutical and chemical applications. The company was established in 2014 and currently employs 9 staff from its facilities in Nexus UCD (technology park at the University College Dublin campus). In 2016, the company launched the VISUMPalm (a handheld NIR measurement instrument) and has secured first sales. An integration platform (VisumNIR) for the inline version is currently under development in cooperation with early adopter clients from the food processing industry (dairy, meat and olive oil) and pharmaceutical industries. VISUM Ltd. is an SME Champion-awarded a Phase 1 and a subsequent Phase 2 Grant under H2020's SME Instrument.

The optical engineering and smart systems teams at VISUM Ltd. have specialized expertise in sensor integration, optical engineering, chemometrics, data analytics, cloud infrastructures and data management systems.

Role in the Project:

VISUM will lead WP6 where they will develop and test novel advanced sensors for monitoring selected process parameters and will design and evaluate the feasibility of a plant level integration of the sensors with a control system. They will also collaborate in the study of the relevant parameters to be analysed in order to select the most suitable sensor technologies.

Additionally, VISUM will contribute to WP8 (LCA and techno-economic assessment) and WP9 (Business plan, exploitation, dissemination and communication activities).

Scientific team:

Dr. Timothy Kehoe. *(male)* PhD in Engineering, University College Cork, Ireland- developed an IR photometric measurement system for thermal diffusivity in anisotropic materials. Masters in Optoelectronics, Queens University Belfast. Designed an Echelle spectrometer while based at Andor Technology Ltd. Worked at Tyndall National Institute, Cork, and Catalan Institute of Nanotechnology, Barcelona, developing diffractive optical and acoustooptical metrology techniques for nanotechnology applications, in particular for characterision of nanoscale structures fabricated by nanoimprint lithography and directed self-assembly. Other research activities: quantification of ordering in self-assembled structures by image analysis techniques and optical sensors using hydrogel materials. Tim is the VISUMPalm Product Manager, as well as the Technical Lead at VISUM Ltd. and he brings vast optical integration and device instrumentation expertise.

Dr. Severine Philippe *(female)* graduated with a 1st class Hons. B.Sc. in Applied Sciences (Physics and Physics Technology) from Dublin Institute of Technology, Ireland, in 2002 and received her PhD in Physics from Trinity College Dublin in 2007, for the study of polarisation effects in semiconductor optical amplifiers (SOA). She went on to work on the development and characterization of low noise SOAs, as well as THz generation by multimode lasers and their detection using four wave mixing in SOA, both in the Semiconductor Photonics Group, Trinity College Dublin, and the School of Electronic Engineering, Dublin City University. She previously worked as a R&D technician for the plastics industry and in quality control for electronic appliances. She joined VISUM in 2014 as an optical engineer and is currently involved in projects including various microscopy techniques, planar waveguide technologies and MIR spectroscopy.

Nikolina Mikulcic *(female)* User Interface Designer. Nikolina comes from web design and development background with more than 15 years of experience in this field and she is an expert on UI/UX design strategy and front-end development (CSS/HTML). She studied graphic design and she started to learn web design due to her

deeply interested for digital world. Since joining VISUM in September 2016, she has been creating a new User Interface for the VISUM product lines. Prior to VISUM, she has an extensive experience in the online world, having worked in companies such as EuroArt93 – digital agency, BlueBOX IT, Biroval. She also was a founder of design studio In Love With Blue and she worked as Project Manager, UI/UX designer and front-end developer. She was leader numerous digital projects for different clients as Erste Bank, Unicredit Bank, Telenor Bank, Komb Bank, ZAMP etc.

Chris Dermody *(Male)* Software Developer. Bachelor's degree in Product Design from Dublin Institute of Technology. Chris excels in creative problem-solving, using the latest technologies to transform ideas into real opportunities that provide exponential value to business operations. With a keen eye for user experience, Chris builds robust, responsive applications that stand the test of time. Since joining VISUM, Chris has been hard at work developing both front and backend features that push the VISUM product lines forward. Prior to VISUM, Chris enjoyed working across the full spectrum of business, founding startups and overseeing product development throughout the lifecycle of many successful businesses.

Ekaitz Hernandez Troyas *(Male)* Fullstack Developer at Visum. Bachelor Degree in Computing Science (5 years). Ekaitz comes from startup background focusing in web developing. He has 4 years of experience in the web industry. He started to work in Dublin for a small startup related with geolocation industry. He had the opportunity to create his own startup related with the cosmetic industry. This experience has given him the opportunity to learn cutting edge technologies as well to learn other facets of a company like business development or marketing. Also he worked as a freelancer implementing solutions with the Squarespace CMS. Ekaitz always wants to face new challenges and learn new things that are out his core competency.

Méabh Garrick *(female)* Optical Engineer. M.Sc Photonics, Friedrich Schiller Universität, Jena, Germany, B.Sc Physics with Medical Physics, National University of Ireland, Galway she worked as a research scientist at the Fraunhofer IOF in Jena, Germany after completing my Master Thesis at the institute. I specialised in the area of spectrally resolved laser-induced damage testing using a high-power tunable OPO laser system as well as angle resolved scattering (ARS) from surfaces and the unique combination of both techniques. My work was presented at the OSA Interference Coatings conference (June 2016).She speak Irish, German (and English). Motivated, teamorientated person with a flair for problem solving. She was awarded a scholarship to study at the Abbe School of Photonics, Friedrich Schiller Universität, Jena where she spent almost four years.

Five relevant publications, patents or products:

1. Development and commercialisation of the VISUM instruments- NIR spectroscopy devices- in-pipe, handheld
2. Publication in relation to Innovation Management: O. Mc Nerney, A. Stachowicz, P. Czuprynski, E. Philpott, D. Kolman, D. Martin, E. Herrmann (2010). Strategic Guide to Successful Use and Dissemination of the Results of Research and Development Projects. 1st edition, Steinbeis-Edition Stuttgart 2010. ISBN 978-3-941417-274
3. On 08/02/2016, VISUM was awarded the Seal of Excellence form the European Commission

Five Relevant Projects:

Project acronym/description.	Scope	End Year	Website
PREVIEW- PREdictiVe system to recommend Injection mold sEtup in Wireless sensor networks. (Grant Agreement number: 636892). VISUM Ltd. is responsible for the development of the Data Acquisition System and in in developing the Signal Conditioning, its integration with the ARM main board, as well as implementing a prototype.	H2020	2017	www.preview-project.eu/

Phase 1 SME Instrument: An IR-spectroscopy-based platform for the rapid integration and customisation of affordable, real-time monitoring solutions for the food and pharmaceutical industry (Grant Agreement number: 684767).	H2020	2017	-
Phase 2 SME Instrument: Cloud-enabled IR spectroscopy platform for the rapid integration and customisation of affordable, real-time chemical monitoring solutions for food and pharmaceutical industries (Grant Agreement number: 731043**)**	H2020	-	-

Technical equipment:

Spectroscopic Laboratory: equipped with several instruments, namely, an FTIR-ATR (Interspectrum), two UV-VIS (PerkinElmer and Ocean Optics), one Extended-SWIR (StellarNet) and one NIR (Mightex) spectrophotometers as well as a SWIR Hyperspectral imager (Specim-Xenics). The associated staff are trained and skilled in chemometrics modelling by using a commercial suite (SOLO+MIA from Eigenvector) as well as ad hoc non-commercial chemometric tools (Matlab and Python toolboxes).

Electronics, mechanical and optical workshops: equipped with up-to-date hardware and tools: signal generators, wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-milling machines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others. ZEEMAX software for the design of optical systems and design setup by means of advanced optical components and facilities (optical tables, windows, lenses, lasers and light sources and related "opticalware").

PARTNER 9 **Name**: SO.F.TER. SPA (SOFTER) **Web page**: www.softergroup.com Partner from Month 1 until Month 7.	**Type:** IND (Large Enterprise) **Country:** Italy	So.F.TeR.

Brief description of the organization:

SO.F.TER. SpA (SOFTER) is part of SO.F.TER. GROUP (www.softergroup.com) a privately held multinational company active in the development and production of thermoplastic materials for over thirty-five years. Production facilities are located in Italy, Mexico, Brazil and USA.

SOFTER is an Italian large enterprise and one of the major independent compounders of thermoplastics in Europe. SOFTER has a great expertise in material design and production of polymers-based materials and is considered the ideal technical partner in co-design activities of final products thanks to its high operational flexibility and to its tailor-made approach to the design of materials.

The company currently supplies leading European manufacturing companies active in sectors such as automotive, building, household appliances, E&E, consumer goods, interior design goods, mechanical industry, sport&leisure items and footwear.

SOFTER employs over 270 persons and is certified according to ISO 9001 and ISO 14001. Its turnover in 2015 was about 220 million euros. In its more than thirty-five-years activity as developer and producer of thermoplastics and hot-melts SOFTER has gained a strong knowledge in the areas of:

- material formulation and design
- continuous melt-mixing/compounding processes
- polymerization processes by polycondensation
- thermal, rheological, chemical and physical-mechanical characterization of thermoplastic materials

SOFTER has an established know-how about polymerization, production and characterization of co-polyesters, polyamides and co-polyester-based hot melts. This expertise will consent to support the partners in the project.

Role in the Project: In the framework of the project SOFTER will act as industrial discussion partner in the valorisation route aimed at the production of lactic-acid-based hot-melts. The company will have a guiding/advising role and will follow the project activities, evaluating results and giving opinions/suggestions/feedback in an industrial perspective.

Scientific team:

M.Sc. Raffaella BATTELLI (F) (**Contact person**), M.Sc. degree in Chemistry at University of Bologna and 2nd level European Master in Nanotechnologies of Polymeric Materials at University of Perugia, has been working at Softer's since 2010 as a researcher and project manager and has been involved in the development of thermoplastics for engineering applications. She is in charge of the management of international cooperation research projects related to the development & production of thermoplastic materials. In this project she will coordinate SOFTER's technical team and she will be active also as a researcher.

M.Sc. Michele PONTEVIVO (M), M.Sc. degree in Chemistry at University of Florence in 1981, has worked for Montedison and is co-inventor of three patents about high performance fluorinated polymers. He has been working at SOFTER's since 2004 in several different leading positions up to the present role of Technology Manager and Factory Manager. In the project he will be the person in charge for general and technical management.

Mr. Denis Grechi (M), Secondary school diploma as Industrial Expert in Phisics, has been working in the plastic sector since 2006 covering different positions (shift supervisor of the polyester polymerization process, analytical operator and coordinator of the CQ laboratory) up to the present role of researcher. He has a background in testing and analysing polymers and thermoplastic materials and in developing new co-polyester polymers and hot-melts.

Related publications, patents or products:

- In the last eight years SOFTER has developed the technology and the expertise for the **development and production of different bio-based thermoplastic materials for multiple applications**, with a focus on their technological processability and industrial implementation.
- SOFTER has developed the **technology for converting PHA into high performance biobased thermoplastic materials:** "Durable polyhydroxyalkanoate compositions" Ref: WO 2013124361 A1 "
- SO.F.TER. GROUP is the third largest co-polyester producer in Europe. SO.F.TER. GROUP is one of the top producers of thermoplastics in Europe and its products portfolio is one of the widest in Europe.
- SOFTER has been producing for more than thirty years a **wide array of elastomers** (i.e. thermoplastic elastomeric materials) and today is one of the top players in the European market with respect to both production volumes and variety of its products portfolio. Elastomers marketed worldwide under trend names FORPRENE®, PIBIFLEX®, LAPRENE®, SOFPRENE® are currently used for the production of parts supplied to several manufacturing sectors such as the automotive, household appliances, electronics, building, footwear, interior design goods and sport&leisure items industry
- The company has been developing, producing and marketing for more than fifteen years **engineering thermoplastic materials for high performance applications,** in sectors such as automotive, household appliances, E&E, building, interior design goods and mechanical industry (e.g. POLYFOR®, TECNOPRENE®, PIBITER® and NYLFOR® technical grades).

Related Projects:

Project acronym/description.	Scope	End Year	Website
TRASH2CASH, «Designed high-value products from zerovalue waste textiles and fibres via design driven technologies»,	H2020 NMP18	2018	//trash2cashproject.eu/
BioGrapPA «Bioplastics from the grapes wastes to the package»,	EIT Climate	2016	
GREEN INDUSTRIAL SYMBIOSIS, value chain of valorization of the biomasses from agro-industrial waste and on its conversion into high-added value materials	Italian project	2015	
MINANO "New high-quality mined nanomaterials mass produced for plastic and wood-plastic nanocomposites,	FP7-NMP	2013	http://minano.vtt.fi/
LOTOTUBO "Development of a new generation of piping for construction sector based on nanomaterials, new formulations and new processes in order to replicate water-repellence and dirt-resistance of lotus leaf	Italian project	2014	

Technical equipment:

SOFTER's Technology Centre includes Laboratories and Material Prototyping Facilities dedicated to R&D activities. Laboratories are equipped with instruments to perform a wide range of mechanical, thermal and rheological analysis of thermoplastic materials. The Material Prototyping Facilities are equipped with continuous and batch melt-mixers and with pilot chemical batch reactors that are used for the preparation of experimental samples and for scaling-up activities. A CAE service for material flow simulation, mold virtual prototyping and predictive structural analysis is also part of the Centre.

PARTNER 10 **Name:** Hayat Kimya Sanayi Anonim Sirketi (HAYAT) **Web Page**: www.hayat.com.tr	**Type:** IND **Country:** Turkey	

Brief description of the organization:

Hayat Kimya was established in 1937 as a textile trading company. Company has entered the FMCG industry in 1987. Today, with 5200 employees Hayat Kimya is one of the largest FMCG (household cleaning & personal care, baby–care & feminine hygiene, and paper tissue & towels) manufacturers headquartered in Turkey.

Product portfolio of Hayat Kimya can be classified in three main groups:

i) Household laundry detergent/surface cleaning products, ii) Hygiene products, iii) Tissue products,

Company have several global manufacturing sites including the locations in Bulgaria, Ukraine, Russia, Iran, Algeria, Egypt, Nigeria and Turkey. Hayat Hygiene Group is the fifth largest player in global diaper industry and works with 8.6 billion baby diaper unit/year production capacity. Hayat also have an overseas distribution network, and thus they have sales operations in more than 100 countries in five continents.With its Papia and Familia brands, as well as its non-home brand Focus and economic brand Teno, the company is the 7th largest tissue producer in Europe, and the largest in the Middle East and the Balkans. Hayat's cellulose and packaging paper plant in BosniaHerzegovina serves both the local market and international markets. Bingo, the favorite detergent of Turkish women, is produced in the Algerian market under the brand TEST.

Even though Hayat Kimya is originally an FMCG manufacturer, also produce several key raw materials and some mid products at production sites including Turkey (e.g. SLES and LABSA for detergent industry and flexible packaging films, nonwoven material for diapers and sanitary napkins) in commercial quantities that are indispensable to the detergent and hygiene industry.

All of the products of Hayat Kimya are certified by ISO EN 9001. Hayat Kimya has also the certificates of OHSAS 18001, ISO 14001, ISO 27001 and ISO50001.

Hayat Kimya has 5411 sqm facility in Turkey for R&D activities. More than 70 researchers are working in R&D team including 13 PhD and 18 MSc. The main objective of the R&D Center is to develop environment-friendly products in the FMCG segment.

R&D spending in 2015 is nearly 10 million dollar. In 2015 Hayat R&D center was selected the best R&D Center in chemical sector by ministry of science, industry and technology of the Republic of Turkey.

Role in the Project: They will be involved as an end-user in the project for the PERCAL project outputs in three different areas: as a solvent replacement in printing, as a surfactant replacement in surface cleaners, and as a white oil replacement in cardboard hotmelts. HAYAT also participates in the activities described in WP7 and WP8 related to environmental, economical, safety, regulatory, Business plan, IPR, dissemination and technological transfer.

Scientific team:

Dr Fatih Mehmet Erguney, PhD, R&D Manager (M). Fatih Mehmet Erguney is responsible for the product and raw material research and development of nonwoven and flexible packaging at Hayat R&D Center. He is also responsible for innovation management. After receiving his BSc degree in Chemical Engineering at the Bogazici University in 2000, Istanbul, he started working for a chemical company as a Process Development Engineer. After two years, he decided to pursue his PhD degree and joined the University of Akron, USA in 2003 to study Polymer Science. Afterwards, started working for a research oriented SME, called Sampas Nanotechnology. Dr. Erguney also has a lot of experience in project management. He is managing 20+ projects at any time at Hayat R&D Center.

Dr. Ipar Nimet Uzun, PhD, Specialist Researcher (F). Ipar Nimet Uzun studied Chemical Engineering at Yildiz Technical University (1998-2002). She carried out her MSc studies (2002-2005). After she got her MSc degree in 2005 she started her PhD studies (2005-2011) which are based on preparation of drug containing polymeric microspheres for controlled drug release using supercritical fluid technology (supercritical assisted and/or antisolvent processing). She also worked as a research and teaching assistant (2002-2011) at the Chemical Engineering Department of Yildiz Technical University.

Dr. Sebnem Dogan, PhD, Specialist Researcher (F). She becomes her BSc degree with honours in Chemical Engineering at the Kocaeli University in 2007, Kocaeli, while also during her BSc degree she participate academical research studies and straight after the university she began her academic carrier as Research Assistant at the Kocaeli University. She started her PhD in 2009, its aim was to prepare polymer blends PLA / TPU with shape memory for a potential usage for biomedical applications. From 2007-2014, her academic carrier as Research Assistant in different universities has resulted with 6 projects funded by different national and international fundings, 10 publications in popular science journals and about 25 conference proceedings in all over the world. She joined Hayat authorized R&D Center as a research scientist in 2014 to create huge amount of benefits for the future with those new materials.

Publications: No relevant publications.

Projects: No previous relevant projects.

Technical equipment:

- Rheometer, Brookfield Viscometer, Tensiometer, High Resolution Optical Microscopy, Microtome
- FTIR, GC, GPC, HPLC, GC-MS, XRF, Raman-Microscopy, Cary Eclipse Fluorometer
- Filter Pressure Test, Lab Compounder, Drainage Tester, PFI Mill, Microfluidics M-110EH-30, CT SCanner

PARTNER 11 **Name**: Covestro Deutschland AG (COVESTRO) **Web page**: www.covestro.com	**Type:** IND **Country:** Germany	covestro

Brief description of the organization:

Covestro is a world-leading supplier of high-performance polymer materials for key industries. Business activities are focused on the manufacture of high-tech polymer materials and the development of innovative solutions for products used in many areas of daily life. The main segments served are the automotive, electrical and electronics, construction and the sports and leisure industries.

Covestro, formerly Bayer MaterialScience, is headquartered in Leverkusen, Germany which is one of its biggest sites. It has around 30 sites throughout Europe, Asia and America, eight of which are world-scale production facilities. With sales of about €12.1bn in 2015 and 15,800 employees, Covestro is among the world's largest polymer companies.

As a global leader in polyurethanes, Covestro develops and produces the components for rigid and flexible foams as well as for solid materials. Diverse applications range from mattresses and car seats to insulating materials for refrigeration appliances and buildings.

Covestro is also the leader in the development and production of polycarbonates. Lightweight, tough and easy to shape, these high-performance plastics can be used in a wide variety of ways, and have large-scale use in industries such as automotive and construction.

As a world-leading supplier of raw materials for high-value coatings, adhesives and specialty chemicals, Covestro supplies these high-value niche products to protect and enhance the appearance of anything from cars, via furniture and flooring, applications in the sports and leisure sector through to cosmetic, textile and medical products.

Covestro seeks sustainable solutions for the major challenges of our times such as climate change, resource scarcity, population change and urbanization – providing the backdrop for favorable industry dynamics and opportunities for further growth.

Role in the Project: Succinic acid can be a potential bio-based substitute of conventionally applied fossil-based adipic acid or other diacids. Covestro could exemplary use succinic acid in polyester-based polyols, resins and polyurethane dispersions (PUD). Within this project, Covestro will give meaningful insights into the quality requirements for succinic acid and the correlation between product quality and final applicability. Furthermore, Covestro will analyze and test the provided succinic acid in various applications.

Scientific team:

Mr. Thomas Görgen (M) is heading the Alliance Management function, a part of Covestro Innovation Management. He coordinates the external R&D activities including the publicly funded projects. Thomas is a polymer chemist by training and experienced in H2020 due to his former appointment as German National Contact Point for FP6 (2002-2005)

Dr. Natalie Rahmen (F), graduated in Bioengineering at RWTH Aachen is responsible for scouting of bioprocesses relevant to Covestro

Dr. Raul Pires (M) is responsible for the New Technologies function within the Business Unit Coatings, Adhesives, Specialties. He will analyze and test the provided succinic acid in various applications.

Related publications, patents or products: Succininc Acid:

http://press.covestro.com/news.nsf/id/covestro-and-reverdia-develop-bio-based-tpu
www.plasticstoday.com/covestro-and-reverdia-partner-development-bio-based-tpu/24790357423530

http://news.bio-based.eu/reverdia-covestro-in-succinic-acid-based-tpu/

www.chemanager-online.com/en/news-opinions/markets-companies/covestro-and-reverdia-develop-bio-basedtpu-foam

Other Biobased products:

http://press.covestro.com/news.nsf/id/Bayer-MaterialScience-develops-first-coating-hardener-from-biomass

www.wageningenur.nl/en/newsarticle/New-insights-in-furanic-intermediates-and-aromatic-derivatives-frombiomass.htm

Related Projects:

Project acronym/Title	Scope	End Year	Website
ThermoBIK- Bio-Thermoaktivierbare Klebstoffe	Biobased adhesives	2017	http://foerderportal.bund.de/foekat/jsp/SucheAction.do?actionMode=view&fkz=22018212

Technical equipment:

- Succinic acid will be used as a building block for polyols, resins and polyurethane dispersions (PUD). Equipment for polymerization is available. Furthermore, we have test facilities i.e. for urethane formation of polyols with isocyanates.
- All kinds of polymer analytic is available (GPC, NMR, IR, RAMAN, DMTA,…)

PARTNER 12 **Name**: Exergy Ltd (EXERGY) **Web page**: http://www.exergy.uk.com/	**Type:** SME **Country:** United Kingdom	exergy

Brief description of the organization: Exergy is one of the UK's leading renewable energy companies with the expertise to provide services, across the globe, to develop, engineer, construct and operate projects that contribute to our goal of a low carbon, sustainable future. We have been in the industry for over 10 years, contributing to the development of the global renewable energy market. We have helped to move the sustainable energy debate from the margin to the mainstream. Operating across a range of innovative low-carbon energy technologies, our capabilities cover everything from project inception to energy generation and operation. Sustainability lies at the core of our values; balancing commercial, environmental and social concerns to address the world's energy problems is what makes us different.

We are unique in our ability to offer vertically integrated skills in a range of sustainable energy technologies. Exergy can be involved at every stage: from project design and development, through planning and financing, engineering and construction to operation. Exergy ensures schemes are designed and built to generate energy in the most economical way and with minimal environmental impact.

Exergy is also member of SPIRE and BIC.

Role in the Project: Exergy will be in charge of the socio-environmental analysis (S-LCA) and LCC as well as

will lead the WP8 where Exergy will undertake the market analysis and the development of the business plans.

Scientific team:

Dr. Fernando Centeno, (M): degree in Physics, Master Degree and PhD in Renewable Energies and Energy Efficiency. Mr. Centeno has wide experience on the engineering and consultancy field related to the building construction & energy sector. Has been involved in two FP7 coordinations focused on the renewable energy sector. As Head of the R&D Group, he has participated in more than 38 international R&D project (i.e. H2020, FP7, IEE, Life+, Eco-innovation, Eureka, Eurostars, among others).

MSc.Eng Jose Luis Molto, (M): holds a master in Chemical Engineering by Rey Juan Carlos University. He has a strong background in simulation, optimisation and modelling of chemical and biochemical processes (Aspen HYSYS, Aspen PLUS and SuperPro) along with the feasibility and economical assessments of them. He is an active member of the European Federation of Chemical Engineers (EFCE) since 2013 and he is associate member of IChemE since 2016.

MSc.Eng Rocio Roldan, (F): is Chemical Engineer by Complutense University of Madrid. Her expertise area is related to the modelling and simulation of chemical and biochemical processes and unitary operations, especially focused on the AspenOne software handling and mathematic modelling. She has been involved in projects related to the valorisation of biomass into valuable products and related to circular economy. She is an associate member of IChemE.

MSc.Eng Sreenivaasa R. Pamidi, (M): is a bachelor in Mechanical and Energy Engineering from VIT, India and holds a master's degree in MSc and PhD Renewable Energy and Environmental Modelling from University of Dundee, Scotland. He has worked on several European research projects which have focussed on his main area of expertise in Energy Harvesting Infrastructure, solar Photovoltaics, solar thermal, Energy Efficiency Optimisation, and Modelling in the Area of Renewable Energies.

Related publications, patents or products:

i) Simulation of chemical pre-treatment process and Optimisation and scale up of the system based on simulation. Liquid and gas Fischer-Tropsch fuel production from olive industry waste: fuel from waste. Exergy Ltd, report for the European Commission. 2015.

ii) Sustainable techno-economic solutions for the agricultural value chain Scale-up and design of an optimised algae process for biodiesel and biogas production. Molto Marin, Jose L. Chemical Engineering, Thesis - Madrid, 2015.

iii) Scale-up and design of an optimised biomass to liquid process (BtL) for diesel production from agroforestal wastes. Roldan Aguayo, Rocio. Chemical Engineering, Thesis. Madrid, 2016

iv) Reducing the impact of forest plant production - design of a stand-alone PV-hybrid system for powering an innovative forestry incubator. Sreenivaasa R Pamidi, Marco Hernández Velasco. (2014)

Related Projects:

Project acronym/description.	Scope	End Year	Website
INNOBITE – "Transforming urban and agricultural residues into high performance biomaterials for green construction".	(FP7 ENV NO. 308465)	2015	www.innobite.eu
FFW - "Liquid and gas Fischer-Tropsch fuel production from olive industry waste: fuel from waste"	(FP7 ENV, No. 308733)	2016	www.fuelfromwaste.eu
AGROCYCLE - "Sustainable techno-economic solutions for the agricultural value chain"	H2020 No. 690142	2019	-
AgriMax - "Agri and food waste valorisation co-ops based on flexible multi-feedstocks biorefinery processing technologies for new high added value applications"	(BBI JU – VC3.D5 2015)	2020	Not available yet
CloseWEEE - "Integrated solutions for pre-processing electronic equipment, closing the loop of post-consumer high-grade plastics, and advanced recovery of critical raw materials antimony and graphite"	(H2020 WASTE No 641747)	2018	www.closeweee.eu

Technical equipment: The main office of Exergy is located in Coventry, United Kingdom, within the Technology Park where around 70 different business companies are placed. Exergy utilises industry-leading software for the simulation and optimisation of chemical processes (AspenOne software engineering), along with the specialised software for the measure and evaluation of the potential environmental impacts linked to processes and products SimaPro and GaBi.

PARTNER 9 **Name**: Yparex BV (YPR) **Web page**: www.yparex.com From Month 8 on.	**Type:** SME **Country:** Netherlands	YPAREX Extrudable adhesive resins

Brief description of the organization:

Yparex BV is part of The Compound Company (http://www.thecompoundcompany.com) a privately held multinational company active in the development and production of thermoplastic adhesives for over thirty years. Production locations located in The Netherlands and Indonesia.

Yparex is a Dutch small enterprise and independent niche producer of extrudable adhesives in Europe. Yparex has a great expertise in material design and production of polymers-based extrudable adhesive materials by grafting and is considered the ideal technical partner in co-design solutions in plastic products thanks to its high operational flexibility and to its tailor-made approach to the design of materials.

The company currently supplies leading European manufacturing companies active in sectors such as automotive, building & construction, medical and packaging

Yparex BV employs 6 persons and is certified according to ISO 9001:2015. In its more than thirty-years activity as developer and producer of thermoplastics adhesives, Yparex BV has gained a strong knowledge in the areas of:

- Grafting of polyolefins with maleic anhydride and other monomers by reactive extrusion.
- thermal, rheological, chemical and physical-mechanical characterization of thermoplastic materials

Yparex has an established know-how about development, grafting and characterization of polyolefins grafted with Maleic anhydride. This expertise will consent to support the partners in the project.

Role in the Project: In the framework of the project Yparex will act as industrial discussion partner in the valorisation route aimed at the production of lactic-acid-based hot-melts. The company will have a role try to graft the PLA with MAH and will follow the project activities, evaluating results and giving opinions/suggestions/feedback in an industrial perspective.

Scientific team:

B.Sc. Frank HUIJNEN (M) (**Contact person**), B.Sc. degree in Chemistry at Zuyd Highschool Heerlen (1987) and a B.Sc. degree in process technology (Zuyd Highschool, 1996) and Polymer technology (Hogeschool West Brabant (1992). Started working for Yparex in dec 2015. Started his carreer at 1988 at DSM NV in polymer testing and development. In 2006 he started working for Intertek as a polymer consultant, with automotive as special attention area. At Yparex he is responsible for Business Development and involved in several international cooperation research projects related to the development & production of thermoplastic adhesives. In this project he will coordinate Yparex's technical team.

M.Sc. Peter Siers (M), M.Sc. in Chemistry at University Twente (2011). Started working for Yparex in 2011 and is responsible R&D manager with knowledge of grafting polyolefins. He will support the project with his grafting know how.

Related publications, patents or products:

- For the 3-layer encapsulant film, Yparex has developed a special grafted POE grade and applied for a patent (Title: 'Electronic Device Module') January 21st 2016: PCT/EP2016/051219.
- The company was the first one to bring a sustainable extrudable adhesive to the market, called Yparex RENEW.

Related Projects:

Project acronym/description.	Scope	End Year	Website
SuMMiT, « Smart large lightweight long life Multifunctional PV Module Technology for large Power Installations and Distributed Energy Generation »,		2018	// summit-project.eu /

Technical equipment:

Yparex's Technology Centre includes Laboratories and Material Prototyping Facilities dedicated to R&D activities. Laboratories are equipped with instruments to perform a limited range of basic mechanical, thermal and rheological analysis of thermoplastic materials. The Material Prototyping Facilities are equipped with a reactive extruder, compounder and injection molding machine to prepare small scale samples and for scaling-up activities.

4.2 THIRD PARTIES INVOLVED IN THE PROJECT (INCLUDING USE OF THIRD PARTY RESOURCES)

There are no third parties involved in the development of PERCAL project as shown in Table 4.2 from below.

There are two beneficiaries IMECAL and CENER subcontracting several activities in the project as described in the from below.

The subcontracted activities will also respect the article 13 of the Grant Agreement and the Best Value for Money, BIOPOLIS company will be therefore subcontracted by their specialisation and experience in the biotechnology sector and in particular in the tasks detailed below:

Table 4.2: Subcontracting and third parties breakdown

Partner		Cost (€)	WP	Task	Description
Subcontracting	1.IMECAL	62,000	WP1	Task 1.3. Characterization of MSW samples	Laboratory analyses to determine the chemical composition of selected substrates
			WP1	Task 1.4. Pre-treatment and enzymatic hydrolysis improvement from PERSEO bioethanol® process	Laboratory analyses for the evaluation of pre-treatment and Saccharification processes
			WP1	Task 1.5. Validation of fermentation processes	Laboratory analyses to validate the fermentation processes for the production of target bioproducts (ethanol, LA, SA).
			WP6	Task 6.1. Study of parameters of the pretreatment and fermentation process for the production of bioethanol, lactic & succinic acid and the byproducts stream.	Study for the selection and optimization of the relevant fermentative parameters for the production of bioproducts at industrial scale. Analysis and control of parameters.

			WP6	Task 6.4. Integration and adaptation of processes to a biotechnological plant configuration	Study of the integration of fermentative parameters within the appropriate working ranges in the demo plant configuration. Analysis of parameters.
Third parties		0	N/A		
Total		**62,000**			
Subcontracting	.AIMPLAS	0	N/A		
Third parties		0	N/A		
Total		**0**			
Subcontracting	.ATB 3	0	N/A		
Third parties		0	N/A		
Total					
Subcontracting	.AUA 4	0	N/A		
Third parties		0	N/A		
Total					
Subcontracting	5.UGENT	0	N/A		
Third parties		0	N/A		
Total					
Subcontracting	6· CENER	48,000	WP3	Task 3.1.Evaluation of MSW as fermentation feedstock	Screening and selection of bacterial strain for succinic acid production and first stage of fermentation media optimization assays at lab
Partner		**Cost (€)**	**WP**	**Task**	**Description**
					scale according to selected strain.
			WP4	Task 4.4.Synthesis of the amino-lipid biosurfactant	Consultancy and support for the selection of analytical techniques to evaluate the activity of biosurfactant samples produced.
Third parties		0	N/A		
Total		48,000			
Subcontracting		0	N/A		

Third parties	7. TBWR	0	N/A		
Total		**0**			
Subcontracting	8. VISUM	0	N/A		
Third parties		0	N/A		
Total		**0**			
Subcontracting	9. SOFTER	0	N/A		
Third parties		0	N/A		
Total		**0**			
Subcontracting		0	N/A		
Third parties		0	N/A		
Total		**0**			
Subcontracting	11. COVESTRO	0	N/A		
Third parties		0	N/A		
Total		**0**			
Subcontracting	12. EXERGY	0	N/A		
Third parties		0	N/A		
Total		**0**			
Subcontracting	13. YPR	0	N/A		
Third parties		0	N/A		
Total		**0**			

5: ETHICS AND SECURITY

5.1 ETHICS

There are no foreseen negative ethical issues associated with the implementation of the project results.

Appropriate health and safety procedures conforming to relevant local/national guidelines/legislation are followed for staff involved in this project.

PERCAL concerns the use of Municipal Solid Waste (MSW) as feedstock to develop intermediate chemical products of industrial interest. These include lactic acid, ethyl lactate solvents, succinic acid and biosurfactants.

Ethical issues are raised as part of the participation of partner from Turkey: Hayat Kimya San A.S. This applicant has confirmed to the BBI that the ethical standards and guidelines of Horizon2020 will be rigorously applied, regardless of the country in which the research is carried out.

The specific authorisations for the import/export of materials will be provided in due time after being provided by the relevant authorities. Partner HAYAT will be involved in the validation of the final products therefore they will not receive any intermediate product nor MSW.

Companies have the responsibility of collecting information on the properties and the uses of substances that they manufacture or import materials

Regarding the protection of human health and the environment from the risks that can be posed by chemicals, all the substances used in the project will be evaluated to ensure that no risks are involved in their use. And in case anyone could have any hazard property, the specific substance will be sent to the destinatary accompanied by a document describing the risk management measures to be used to make safe use of it. For partners established outside the EU this requirement will apply as for the rest of partners, in this particular project.

There will be no experimentation on animals, with human cloning, genetic modification or human embryos or stems cells as a part of the project.

5.2 SECURITY.

Please indicate if your project will involve:

- Activities or results raising security issues: NO.
- 'EU-classified information' as background or results: NO.

Although the self-assessment identified no issues, we want to summarise here how certain aspects of the project that could lead to ethical concerns are addressed in the work plan:

- Regulatory and safety-related activities have been incorporated from the outset of the project with the intention of gathering data to fuel in a further safe and compliant development phase. In terms of product, safety aspects will deal with water and waste regulations similar than other chemical industries, where disposal of solid and liquid by-products must be safety for workers and environment and best practices and security measures must be a priority to avoid uncontrolled discharges to the environment and fires. In terms of possible harm to the environment, the use of a LCA approach, which identifies environmental as well as health impact, can help identify where the maximum impact could occur providing the opportunity for process or material modifications to minimise these impacts.
- Finally, the appropriate health and safety procedures conforming to relevant local/national guidelines/legislation will be followed by staff involved in this project. In this sense, all partners follow recognized hygiene and safety rules, including chemical regulations such as REACH, and, recommendations of good practices for workers safety will be given by ARCHA when checking the pilot processes in WP7. An Ethical & societal assessment covering those as well as further aspects is also included in WP7.

All in all, the developed bio based products from biogenic faction of MSW are in fact expected to be safer for both the consumer and environment (products with less toxicity) and ethically more acceptable than their standard (synthetic) counterparts or even than some of their bio-based relatives, often agriculture land competing (sugars and oils), which ethics are questionable, leading to fulfilling the will of consumer to make responsible choice not jeoparzing our future generation's ability to access resources and to a clean environment.

ANNEX I- PERSEO® BIOETHANOL Technology

Bioethanol production. Following the pre-treatment, the PERSEO Bioethanol® Technology developed by IMECAL and CIEMAT for producing bioethanol will be demonstrated on a real-life environment with different organic fraction of municipal solid waste (OFMSW). **The demonstration plant is equipped with all the reactors, fermenters, and separation and distillation units needed for transforming the OFMSW to bioethanol. The typical ethanol yields obtained vary from 100 – 130 liters of bioethanol per tonne of organic waste in dry bases**. The bioethanol production involves several steps:

A. The raw material is subjected to a pre-treatment which aims to eliminate the inserts contained in the organic matter (stones, plastics, metals, glass, etc.).

B. A **thermochemical dilute acid hydrolysis** of the raw material will be done in order to produce a solubilizing of the cellulose and hemicellulose contained in the MSW.

C. After the feedstock is hydrolysed, a **simultaneous saccharification and fermentation process** is done. The cellulose saccharification and fermentation of glucose to ethanol is produced in the same fermenter, adding the enzymes, yeast and nutrients suitable for efficient conversion of sugars to ethanol.

D. The fermentation sludge obtained is subjected to a **distillation process** in order to concentrate the ethanol produced in the fermentation of the organic fraction of MSW.

E. **Dehydration process** is carried out of the mixture by passing through a molecular sieve to achieve dehydration required for ethanol (> 99.5%) as a final product.

Bioethanol production

State of the art: PERCAL will use an innovative patented process based on a thermochemical pre-treatment and a Simultaneous Sacharification and Fermentation (PERSEO Bioethanol® technology) to produce biofuels from municipal solid waste.

Figure A1. The PERSEO Bioethanol® Technology plant

Second generation bioethanol can be produced by two main production routes, the thermochemical and biochemical.

In thermal process, bioethanol can be produced via gasification of the lignocellulosic feedstock at a high pressure and in absence of inert gases and further conversion of the syngas into bioethanol. In the biochemical pathways, the lignocellulosic feedstock is firstly pre-treated in order to make the cellulosic component more accessible to the cellulases for the subsequent enzymatic saccharification step. During the pre-treatment a significant portion of hemicelluloses is converted into monosaccharides, mostly xylose (a C5 sugar). C6 sugars – mainly glucose is released after the enzymatic hydrolysis step and is then fermented with or without the C5 sugars depending on the design options. Several technology options are under development including separate hydrolysis and fermentation (SHF) or simultaneous saccharification and co-fermentation (SSCF) of C5 and C6 sugars with or without onsite production of the cellulases.

The main challenges of the biochemical route concern the integration of the system which requires a coordinated effort in improving each step to achieve an overall higher efficiency and a decreased production cost of bioethanol. In that sense, a particular attention should be paid to the feedstock cost. One of the strong assumptions in developing second generation bioethanol is the possibility to process low-cost feedstock. The concept of the process for enzymatic hydrolysis of cellulose and simultaneous fermentation, so-called Simultaneous Saccharification and Fermentation (SSF) is a promising technology. It was introduced by Gulf Oil Company, US and the University of Arkansas (Gauss et al. 1976; Huff et al. 1976) and various reports on bioethanol production have mentioned the superiority of ethanol yield and productivity using the SSF process[1,2].The sequence of steps for the SSF is virtually the same as for the separate process, except that saccharification and fermentation steps are combined in one vessel, however, the presence of yeast or bacteria along with enzymes minimizes the sugar accumulation in the vessel, consequently higher rates, yields and concentrations of ethanol are possible using SSF at lower enzyme loading. Additionally, the presence of ethanol makes the mixture less vulnerable to contamination by unwanted microorganisms[3,45].

Linked National and International R&D Activities.

[1] Söderström J, Galbe M, Zacchi G. Separate versus simultaneous saccharification and fermentation of two-step pretreated softwood for ethanol production. J Wood Chem Technol. 2005;25:187–202.

[2] Neves, M. A. D., Kimura, T., Shimizu, N., & Shiiba, K. (2006). Production of alcohol by simultaneous saccharification and fermentation of low-grade wheat flour. Brazilian Archives of Biology and Technology, 49(3), 481-490.

[3] Roble ND, Ogbonna JC, Tanaka H. A novel circulating loop bioreactor with cells immobilized in loofa (Luffa cylindrica) sponge for the bioconversion of raw cassava starch to ethanol. Appl Microbiol Biotechnol. 2003 Feb;60(6):671-8. Epub 2002 Dec 19.

[4] T. Montesinos et al., Production of alcohol from raw wheat flour by Amyloglucosidase and Saccharomyces cerevisiae, Enzyme and Microbial Technology,

[5] , 2000, 362-370

With regards to the production of bioethanol from MSW, there are some examples of industrial and pilot plants but none of these projects has validated a whole process able to transform the Stabilized Organic Fraction (SOF) available in MSW into not only biofuels (bioethanol and biomethane) but also other value-added co-products such as biofertilisers. There are a few industrial plants for conversion of MSW to bioethanol. The first international waste to

Project (Start/End date)	**Main Observations/ Advantages of BIOWASTE compared to these projects**
Fiberight (2105 - In operation)	ANDRITZ's technology utilizes a steam heating concept to continuously preheat and cook the feedstock **at elevated temperatures**. BIOWASTE employs an innovative pre-treatment process at low temperature in order to avoid dioxins formation and reduce the production costs
Enerkem (2014 - In operation)	Their technology is based on gasification, a thermochemical process at high temperature. **Gasification presents several disadvantages in comparison with fermentation**: refuelling is often required for continuous running of the systemand, in addition it handles with residues such as ash or tarry condensates.
Etanolix (2015 - In operation)	The ethanol plant is fully integrated into the functions and logistics of St1 oil refinery in Gothenburg and has an annual production capacity of 5 million litres of advanced bioethanol.
Abengoa (Babilafuente) (2013 - In operation)	Demonstration plant which has a capacity to treat 25,000 tons of municipal solid waste (MSW) from which it will obtain up to 1.5 million litres of bioethanol for use as fuel.
Bionolix (2013 - In operation)	The products of the Bionolix process are ethanol and solid biofuel which is used to produce energy part of which is used in the plant and waste water.
Coskata (2011 - In operation)	Semi-Commercial Facility Thermochemical / Biotechnical process: Pre-treatment, gasification to syngas, fermentation and separation. Gasification involves high temperature and pollutants in contrast with the BIOWASTE biological/chemical process.
IMECAL S.A. (PERSEO Bioethanol®) (2007 - In operation)	Demonstration plant based in biotechnological process to treat the organic fraction of MSW. Has a capacity to treat 25 t/d in fermentation to produce bioethanol and a solid fuel.

bioethanol demonstration plant was built by IMECAL – CIEMAT in 2007, based in the biotechnological process PERSEO® Bioethanol. The **PERSEO** project in Valencia (Spain) started to obtain bioethanol from the organic fraction of the municipal solid waste. Another **demonstration plant was inaugurated by Abengoa in 2013**. Located in Spain, the organic matter is pre-treated to produce organic fibre, which is subsequently converted into bioethanol by using a fermentation and enzymatic hydrolysis of SOF from MSW. However, their strategy is not focused on the study of the most appropriate pre- treatment. Since pre-treatment has been viewed as one of the most expensive processing steps in cellulosic biomass-to-fermentable sugars conversion.

In Sweden, **St1 Biofuels Oy has completed a waste-to-ethanol project (Etanolix)** which recycles feedstock such as biowaste and process residue from local bakeries and bread from shops that is past its sell-by date into ethanol for transport fuel. In Finland, there is another plant **(Bionolix) that produces ethanol from biowaste from shops and households**.

Outside Europe, the **first industrial-scale** municipal solid waste to biofuel facility was opened in Canada in 2014 by **Enerkem**. Despite being the first industrial scale waste-to-biofuels facility of its kind to turn household garbage into biofuels and biochemical, **their technology is based on gasification** and subsequent catalytic synthesis, differing from the novel configuration of BIOWASTE where bio-ethanol will be produced more efficiently through alcoholic fermentation. Another example is Coskata a semi-Commercial Facility in Illinois (USA) which produces ethanol from both biomass feedstock and natural gas. In July 2014, ANDRITZ received an order to supply services **for Fiberight's cellulosic ethanol plant** in Iowa, USA. ANDRITZ Group´s biorefinery is focused on the transformation of municipal solid waste (MSW) and other organic materials into cellulosic ethanol as the primary product. **Linked plants**

ANNEX II. Support letters.

novozymes

Letter of Interest and Support

Dear Ms. Caterina Coll, Chief Operating Officer of IMECAL Company.

I undersigned Johan Mogensen, acting as Vice President of Bioenergy and Feed Application Research at Novozymes, am pleased to declare the interest and support of our company to the project proposal entitled "***Chemical building blocks from versatile MSW biorefinery***" (Project acronym: PERCAL)" that you intend to submit to the H2020 call "BBI-2016-R08 Emerging technologies for conversion of the organic content of Municipal Solid Waste and improving waste-to-chemicals value chain" and coordinated by the company IMECAL.

Novozymes is a world leader in bio-innovation and specialised in industrial enzymes and microorganisms. As an enzyme supplier for various biorefining processes we would be interested in the results of the project in order to evaluate future collaborations with PERCAL's partners. Furthermore, Novozymes is interested to be involved in the PERCAL Advisory Group in order to give them more information about the materials and the processes we are developing, in regular meetings with the consortium or parts of the consortium.

Bagsværd, 30th of August 2016

Rethink Tomorrow

FORTISCHEM

Letter of Interest and Support

Dear Ms. Caterina Coll, Chief Operating Officer of IMECAL Company.

I undersigned Ing. **Ľubomír Kukučka,** acting as the **CEO and Member of the Board of Directors** of **FORTISCHEM a. s.,** am pleased to declare the interest and support of our Company to the project proposal entitled "***Chemical building blocks from versatile MSW biorefinery***" (Project acronym: PERCAL)" that you intend to submit to the H2020 call "BBI-2016-R08 Emerging technologies for conversion of the organic content of Municipal Solid Waste and improving waste-to-chemicals value chain".

FORTISCHEM a. s. as a chemical company is interested in the results of the project in order to evaluate future collaborations or commercial agreements with PERCAL's partners.

FORTISCHEM a. s. expects to attend communication and dissemination events, as well as contributing with advice and suggestions on industrial and market requirements during the project.

Nováky, 08.08.2016

Ing. Ľubomír Kukučka
CEO and Member of the Board of Directors

CONTACT INFORMATION

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ARKEMA
INNOVATIVE CHEMISTRY

Colombes, August 31st 2016

Jean-Luc DUBOIS
ARKEMA France
Corporate R&D,
420 Rue d'Estienne d'Orves
92705 Colombes, France

Ms. Caterina Coll
Chief Operating Officer of IMECAL Company

OBJET : PERCAL – letter of support

Dear Ms. Caterina Coll, Chief Operating Officer of IMECAL Company.

I undersigned **Jean-Luc DUBOIS,** acting as the **Scientific Director** of ARKEMA FRANCE, am pleased to declare the interest and support of our company to the project proposal entitled "***Chemical building blocks from versatile MSW biorefinery***" (Project acronym: PERCAL)" that you intend to submit to the H2020 call "BBI-2016-R08 Emerging technologies for conversion of the organic content of Municipal Solid Waste and improving waste-to-chemicals value chain" and coordinated by the company IMECAL

ARKEMA as a group operates already 5 biorefineries in the world, 4 of them using vegetable oils and fatty acids as starting materials. ARKEMA generates about 1 G€ of sales from products derived from renewable resources, and is looking for new and alternative sourcing of key starting materials for future productions. Arkema is a producer of several polymeric materials including polyamides, polyesters, adhesives, coating resins… and is interested in the results of the project in order to evaluate future collaborations with PERCAL's partners. Furthermore, **ARKEMA** is interested to be involved in the PERCAL Advisory Group in order to give them more information about the materials and the processes we are developing, in regular meetings with the consortium or parts of the consortium.

Colombes, August 31st 2016

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